

stat_longterm_measurement_design

Specialization: v1

TL;DR. Measurement design is the argument of the budget-normalized two-source score bound, so the actionable population criterion is the observable design-value gap, not predictive fit. A full-catalogue pilot consistently estimates that criterion and supports selected-design inference under the declared learner conditions. Conversely, both unadjusted observational prediction risk $R_d(P)$ and the standard treatment-aware risk $R_d^A(P)$ can select the same regular but arbitrarily inefficient proxy system. On the binary witness, $R_{d_e}^A = 585/1024$, $R_{d_p}^A < 3/10$, and all four source-score or IKM-normalized selector ratios retain the sharp leading divergence $1/(726\epsilon^2)$. A split-sample saturated finite-cell pilot consistently implements the treatment-aware rule and inherits this divergence in probability.

1 Project justification

Gap. The fixed-system long-term data-combination theory and surrogate-efficiency literature take the measured short-term system and source ratio as given. Published comparisons of a given surrogate set with or without surrogates do not furnish a guarantee for selecting among cost-constrained measured subsets. The missing question remains even for the two standard population squared-risk objectives, with and without treatment in the prediction regression.

Niche. The note treats the measured subset, sequential role assignment, and budget split as design variables while keeping identification and estimation conceptually separate. It gives exact finite-catalogue selection constants, a full-catalogue pilot rule, post-selection inference under explicit selected-path conditions, and a regular binary witness that isolates efficiency failure rather than identification failure.

Fill. The design-value gap is an observable, computable selection criterion and the independent full-catalogue pilot yields oracle regret and feasible allocation. The negative side now covers both $R_d(P)$ and $R_d^A(P)$: both uniquely select d_p , whereas d_e is arm-normalized and IKM-normalized profile-optimal. Their fixed-law constants are exactly enumerable but have no finite uniform guarantee; the sharp center envelope diverges as $1/(726\epsilon^2)$. A held-out saturated finite-cell estimator of $R_d^A(P)$ selects d_p with probability tending to one along fixed- χ triangular paths, so the impossibility is operational rather than an oracle-only artifact.

2 Related work

Imbens–Kallus–Mao–Wang provide identification, doubly robust estimation, and fixed-system efficiency theory for a given sequential proxy system and limiting source ratio. Surrogate-index and surrogate-efficiency results likewise compare information from a fixed surrogate set rather than optimize cost-constrained subsets and role assignments. Existing optimal-measurement work in linear front-door, Gaussian instrumental-variable, or fixed-variable two-phase designs addresses different design spaces. The present raw squared-risk and treatment-aware squared-risk rules are generic standard diagnostics; no named published paper is claimed to use either as a proxy-selection workflow. Carneiro–Lee–Wilhelm appears only in the explicitly secondary residual-risk-per-budget shape, not as an attributed selection rule or guarantee.

3 Setup

Target (estimand / structural parameter): The causal estimand is $\tau = \mathbb{E}_P[Y(1) - Y(0) \mid G = O]$. The observable statistical target selected by the main theorem is $\theta_{\hat{d}_m}(P) = \mu_{\hat{d}_m,1} - \mu_{\hat{d}_m,0}$ on $\mathfrak{P}_{\mathcal{D}}^{\text{obs}}$; it equals τ only on $\mathfrak{P}_{\mathcal{D}}^{\text{id}}$, which contains the binary witness region but not all of $\mathfrak{P}_{\mathcal{D}}$. Primary functional / estimator / diagnostic: The main statistical functional is the selected observed bridge contrast $\theta_{\hat{d}_m}(P)$ on $\mathfrak{P}_{\mathcal{D}}^{\text{obs}}$. Its explicit selected-design cross-fitted estimator, studentizer, limit law, and Wald interval are given by Definition def:selected-wald-variance and Theorem thm:postselection-wald, conditional on the declared selected-path learner, multiplier, pilot, and variance conditions. Proposition prop:postselection-wald-ate-corollary converts this to inference for τ on $\mathfrak{P}_{\mathcal{D}}^{\text{id}}$. Absolute fixed-system causal efficiency uses the IKM-normalized moments and the separately cited canonical-gradient conditions..

- K — `catalogue_size`; $\mathbb{N}$; index

Definition. The finite number of candidate short-term outcomes.

- $\mathcal{J}$ — `index_set`; $\{1, \dots, K\}$; index

Definition. The catalogue index set $\mathcal{J} = \{1, \dots, K\}$.

- a — `treatment_level`; $\{0, 1\}$; index

Definition. A generic treatment level.

- g — `sample_label`; $\{E, O\}$; index

Definition. A generic experimental or observational sample label.

- G — `sample_indicator`; $\{E, O\}$; datum

Definition. The indicator of membership in the experimental sample E or observational long-term-follow-up sample O .

- X — `baseline_covariate`; $\mathcal{X}$; datum

Definition. Observed pretreatment covariates.

- U — `latent_confounder`; $\mathcal{U}_0$; latent

Definition. The persistent unobserved confounder in the observational population.

- A — `treatment`; $\{0, 1\}$; datum

Definition. The realized binary treatment.

- T — `signed_treatment`; $\{-1, 1\}$; derived datum

Definition. The signed treatment $T = 2A - 1$.

- s — `binary_proxy_signal`; $(0, 1)$; index

Definition. A generic signal strength for a binary proxy of U .

- W_s — `binary_proxy_family`; $\{-1, 1\}$; witness proxy

Definition. A binary proxy whose conditional mass is $\Pr(W_s = w \mid U = u) = (1 + suw)/2$ for $u, w \in \{-1, 1\}$.

- $Y(a)$ — `potential_long_term_outcome`; L_2 ; potential outcome

Definition. The long-term potential outcome under treatment a .

- Y — `realized_long_term_outcome`; $\mathbb{R}$; datum

Definition. The realized long-term outcome, observed only when $G = O$.

- $Z(a)$ — `potential_short_term_catalogue`; $\prod_{j \in \mathcal{J}} \mathcal{Z}_j$; potential outcome

Definition. The full vector $Z(a) = (Z_j(a) : j \in \mathcal{J})$ of candidate short-term potential outcomes under treatment a , ordered by measurement time.

- Z — `realized_short_term_catalogue`; $\prod_{j \in \mathcal{J}} \mathcal{Z}_j$; datum

Definition. The realized full-catalogue vector $Z = Z(A)$.

- P — `full_catalogue_law`; $\mathcal{P}(G, X, U, A, Y(0), Y(1), Z(0), Z(1))$; law

Definition. The full-catalogue superpopulation law.

- c_j — `measurement_cost`; $(0, \infty)$; cost

Definition. The known per-unit cost of measuring catalogue outcome Z_j .

- `b.E` — `base_experimental_cost`; $(0, \infty)$; `cost`

Definition. The known base cost of one experimental unit before catalogue measurements.

- `b.O` — `base_followup_cost`; (b_E, ∞) ; `cost`

Definition. The known base cost of one observational long-term-follow-up unit, including measurement of Y .

- `\kappa` — `per_unit_measurement_cap`; $(0, \infty)$; `budget`

Definition. The cap on the catalogue-measurement cost incurred for any main-study unit.

- `\mathcal{D}` — `admissible_design_family`; finite nonempty set; `decision_set`

Definition. The ex-ante-known finite family of pairs $d = (M_d, r_d)$, where $M_d \subseteq \mathcal{J}$, $\sum_{j \in M_d} c_j \leq \kappa$, and $r_d : M_d \rightarrow \{1, 2, 3\}$ assigns every measured outcome to one sequential role.

- `d` — `measurement_design`; $\mathcal{D}$; `decision`

Definition. A generic admissible measured-subset and role-assignment pair.

- `d'` — `measurement_design`; $\mathcal{D}$; `decision`

Definition. A second generic admissible measurement design used in pairwise comparisons.

- `M.d` — `measured_subset`; $2^{\mathcal{J}}$; `decision_component`

Definition. The catalogue subset measured under design d .

- `r.d` — `role_assignment`; $M_d \rightarrow \{1, 2, 3\}$; `decision_component`

Definition. The sequential-role assignment under design d .

- `S_{1d}` — `early_short_term_potential_and_realized_vector`; `product space`; `proxy`

Definition. The role-one potential subvector $S_{1d}(a) = (Z_j(a) : j \in M_d, r_d(j) = 1)$ and its realized version $S_{1d} = S_{1d}(A)$.

- `S_{2d}` — `middle_short_term_potential_and_realized_vector`; `product space`; `sequential_separator`

Definition. The role-two potential subvector $S_{2d}(a) = (Z_j(a) : j \in M_d, r_d(j) = 2)$ and its realized version $S_{2d} = S_{2d}(A)$.

- `S_{3d}` — `late_short_term_potential_and_realized_vector`; `product space`; `proxy`

Definition. The role-three potential subvector $S_{3d}(a) = (Z_j(a) : j \in M_d, r_d(j) = 3)$ and its realized version $S_{3d} = S_{3d}(A)$.

- `p_{g,a}` — `treatment_probability`; $(0, 1)$; `nuisance`

Definition. The sample-specific probability $p_{g,a} = P(A = a \mid G = g)$.

- `h.d` — `outcome_bridge`; $L_2(S_{3d}, S_{2d}, A, X)$; `nuisance`

Definition. The unique square-integrable solution h_d of

$$\mathbb{E}_P[Y \mid S_{2d}, S_{1d}, A, X, G = O] = \mathbb{E}_P[h_d(S_{3d}, S_{2d}, A, X) \mid S_{2d}, S_{1d}, A, X, G = O]$$

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- `q.d` — `selection_bridge`; $L_2(S_{2d}, S_{1d}, A, X)$; `nuisance`

Definition. A design-specific square-integrable candidate selection bridge.

- `T.d` — `conditional_expectation_operator`; $L_2(S_{3d}, S_{2d}, A, X) \rightarrow L_2(S_{2d}, S_{1d}, A, X)$; `operator`

Definition. The operator $(T_d f)(S_{2d}, S_{1d}, A, X) = \mathbb{E}_P[f(S_{3d}, S_{2d}, A, X) \mid S_{2d}, S_{1d}, A, X, G = O]$.

- T_d^* — adjoint_conditional_expectation_operator; $L_2(S_{2d}, S_{1d}, A, X) \rightarrow L_2(S_{3d}, S_{2d}, A, X)$; operator

Definition. The adjoint operator $(T_d^*u)(S_{3d}, S_{2d}, A, X) = \mathbb{E}_P[u(S_{2d}, S_{1d}, A, X) \mid S_{3d}, S_{2d}, A, X, G = O]$.

- $\|\cdot\|_{2,0,h,d}$ — strong_outcome_bridge_norm; seminorm; error_metric

Definition. For an outcome-bridge perturbation f , $\|f\|_{2,0,h,d} = \{\mathbb{E}_P[f(S_{3d}, S_{2d}, A, X)^2 \mid G = O]\}^{1/2}$.

- $\|\cdot\|_{2,0,q,d}$ — strong_selection_bridge_norm; seminorm; error_metric

Definition. For a selection-bridge perturbation u , $\|u\|_{2,0,q,d} = \{\mathbb{E}_P[u(S_{2d}, S_{1d}, A, X)^2 \mid G = O]\}^{1/2}$.

- $\|\cdot\|_{w,h,d}$ — weak_outcome_bridge_seminorm; seminorm; error_metric

Definition. The weak outcome-bridge seminorm $\|u\|_{w,h,d} = \|T_d^*u\|_{2,0,q,d}$, which measures violation of the design-specific observed outcome-bridge equation.

- $\|\cdot\|_{w,q,d}$ — weak_selection_bridge_seminorm; seminorm; error_metric

Definition. The weak selection-bridge seminorm $\|u\|_{w,q,d} = \|T_d^*u\|_{2,0,h,d}$, which measures violation of the design-specific observed selection-bridge equation.

- $\mu_{d,a}$ — bridge_mean; $\mathbb{R}$; nuisance

Definition. The experimental bridge mean $\mu_{d,a} = \mathbb{E}_P[h_d(S_{3d}, S_{2d}, a, X) \mid A = a, G = E]$.

- τ — causal_estimand; $\mathbb{R}$; estimand

Definition. The observational-population long-term average treatment effect $\tau = \mathbb{E}_P[Y(1) - Y(0) \mid G = O]$.

- $\phi_{E,d}$ — experimental_gradient_component; $L_2^0(P(\cdot \mid G = E))$; influence_component

Definition. The experimental score $\phi_{E,d} = \sum_{a=0}^1 (-1)^{1-a} \mathbf{1}\{A = a\} p_{E,a}^{-1} \{h_d(S_{3d}, S_{2d}, a, X) - \mu_{d,a}\}$.

- $\phi_{O,d}$ — observational_gradient_component; $L_2^0(P(\cdot \mid G = O))$; influence_component

Definition. The observational score

$$\phi_{O,d} = \sum_{a=0}^1 (-1)^{1-a} \mathbf{1}\{A = a\} p_{O,a}^{-1} q_d(S_{2d}, S_{1d}, a, X) \{Y - h_d(S_{3d}, S_{2d}, a, X)\}$$

- $V_{E,d}$ — experimental_information_component; $[0, \infty]$; efficiency_component

Definition. The variance component $V_{E,d} = \mathbb{E}_P[\phi_{E,d}^2 \mid G = E]$.

- $V_{O,d}$ — observational_information_component; $[0, \infty]$; efficiency_component

Definition. The variance component $V_{O,d} = \mathbb{E}_P[\phi_{O,d}^2 \mid G = O]$.

- $c_{E,d}$ — experimental_unit_cost; $(0, \infty)$; cost

Definition. The per-unit cost $c_{E,d} = b_E + \sum_{j \in M_d} c_j$.

- $c_{O,d}$ — followup_unit_cost; $(0, \infty)$; cost

Definition. The per-unit cost $c_{O,d} = b_O + \sum_{j \in M_d} c_j$.

- B — main_study_budget; $(0, \infty)$; asymptotic_index

Definition. The total monetary budget of the main study.

- n_E — fixed_experimental_sample_size; $\mathbb{N}$; sample_size

Definition. A generic fixed experimental sample size in the anchor reduction.

- n_0 — fixed_observational_sample_size; $\mathbb{N}$; sample_size

Definition. A generic fixed observational sample size in the anchor reduction.

- α — experimental_budget_share; $(0, 1)$; allocation

Definition. The fraction of B spent on experimental units.

- $n_{\{E, B\}}(d, \alpha)$ — experimental_sample_size; $\mathbb{N}$; sample_size

Definition. The feasible count $n_{E, B}(d, \alpha) = \lfloor \alpha B / c_{E, d} \rfloor$.

- $n_{\{O, B\}}(d, \alpha)$ — observational_sample_size; $\mathbb{N}$; sample_size

Definition. The feasible count $n_{O, B}(d, \alpha) = \lfloor (1 - \alpha)B / c_{O, d} \rfloor$.

- $\mathfrak{r}_{\{d, B\}}(\alpha; P)$ — integer_rounding_residual; $[0, \infty)$; rounding_error

Definition. The floor-count residual

$\mathfrak{r}_{d, B}(\alpha; P) = B\{V_{E, d}/n_{E, B}(d, \alpha) + V_{O, d}/n_{O, B}(d, \alpha)\} - \{c_{E, d}V_{E, d}/\alpha + c_{O, d}V_{O, d}/(1 - \alpha)\}$ whenever both integer counts are positive.

- $\mathcal{V}_d(\alpha; P)$ — budget_normalized_efficiency_bound; $(0, \infty]$; focal_object

Definition. The design-value profile $\mathcal{V}_d(\alpha; P) = c_{E, d}V_{E, d}/\alpha + c_{O, d}V_{O, d}/(1 - \alpha)$ for regular d , and $+\infty$ otherwise.

- $\alpha_d^*(P)$ — profiled_budget_share; $(0, 1)$; optimal_allocation

Definition. The optimizer $\alpha_d^*(P) = \sqrt{c_{E, d}V_{E, d}} / (\sqrt{c_{E, d}V_{E, d}} + \sqrt{c_{O, d}V_{O, d}})$.

- $\mathcal{V}_d^*(P)$ — profiled_design_value; $(0, \infty]$; focal_object

Definition. The profiled value $\mathcal{V}_d^*(P) = \inf_{0 < \alpha < 1} \mathcal{V}_d(\alpha; P)$.

- $D^*(P)$ — oracle_measurement_design; $2^{\mathcal{D}}$; optimal_design

Definition. The oracle set $D^*(P) = \arg \min_{d \in \mathcal{D}} \mathcal{V}_d^*(P)$, retaining every minimizer under ties.

- $\gamma(P)$ — oracle_separation; $(0, \infty]$; selection_margin

Definition. The scalar separation $\gamma(P) = \min_{d \notin D^*(P)} \{\mathcal{V}_d^*(P) - \min_{d' \in D^*(P)} \mathcal{V}_{d'}^*(P)\}$, with $\gamma(P) = +\infty$ when $D^*(P) = \mathcal{D}$.

- $\Delta_P(d, d')$ — design_value_gap; $\mathbb{R}$; selection_criterion

Definition. The pairwise gap $\Delta_P(d, d') = \mathcal{V}_d^*(P) - \mathcal{V}_{d'}^*(P)$.

- $R_d(P)$ — predictive_risk; $[0, \infty)$; benchmark_objective

Definition. The best observational long-term prediction risk $R_d(P) = \inf_{f \in L_2(X, Z_{M_d})} \mathbb{E}_P[(Y - f(X, Z_{M_d}))^2 \mid G = O]$.

- $n_{\{\mathrm{CLW}, B\}}(d)$ — cost_feasible_prediction_sample_size; $(0, \infty)$; benchmark_sample_size

Definition. The continuous jointly budget-feasible sample size $n_{\mathrm{CLW}, B}(d) = B/(c_{E, d} + c_{O, d})$ for the Carneiro–Lee–Wilhelm-shaped comparator, which charges one experimental and one long-term-follow-up unit with all measurements in M_d per analysis pair.

- $\Psi_{\{\mathrm{CLW}, d\}}(P; B)$ — secondary_cost_weighted_residual_risk_diagnostic; $[0, \infty)$; secondary_benchmark_objective

Definition. The secondary cost-weighted diagnostic $\Psi_{\mathrm{CLW}, d}(P; B) = (c_{E, d} + c_{O, d})R_d(P)/B$; the subscript records shape only and does not attribute a published workflow.

- $\mathcal{D}_{\{\mathrm{pred}\}}(P)$ — raw_prediction_risk_optimizer_set; $2^{\mathcal{D}}$; benchmark_decision

Definition. The minimizer set $\mathcal{D}_{\mathrm{pred}}(P) = \arg \min_{d \in \mathcal{D}} R_d(P)$, retaining every tie.

- $\rho_{\{\mathrm{pred}\}}(P)$ — raw_prediction_risk_approximation_ratio; $[1, \infty]$; diagnostic

Definition. The best-tie profiled design-value ratio of the raw population prediction-risk selector.

- m — pilot_sample_size; $\mathbb{N}$; asymptotic_index

Definition. The full-catalogue pilot sample size.

- η — test_level; $(0, 1)$; confidence_level

Definition. The nominal noncoverage probability for the post-selection Wald interval.

- $\mathcal{P}_m$ — full_catalogue_pilot; sample; pilot_data

Definition. An independent pilot of size m observing $G, X, A, Z, Y \mathbf{1}\{G = O\}$, with every component of Z measured.

- $\widehat{V}_{E,d}$ — pilot_experimental_variance_estimator; $[0, \infty)$; estimator

Definition. The cross-fitted pilot second-moment estimator of $V_{E,d}$.

- $\widehat{V}_{O,d}$ — pilot_observational_variance_estimator; $[0, \infty)$; estimator

Definition. The cross-fitted pilot second-moment estimator of $V_{O,d}$.

- $\widehat{\mathcal{V}}_{d,m}^*$ — pilot_design_value_estimator; $[0, \infty]$; estimator

Definition. The cross-fitted plug-in version of $\mathcal{V}_d^*(P)$ obtained by fitting h_d, q_d and the two score variances on $\mathcal{P}_m$, then inserting the known costs.

- $\widehat{d}_m$ — plugin_measurement_design; $\mathcal{D}$; estimator

Definition. A deterministic-tie-broken selector $\widehat{d}_m \in \arg \min_{d \in \mathcal{D}} \widehat{\mathcal{V}}_{d,m}^*$.

- $\underline{v}_m$ — variance_truncation; $(0, \infty)$; regularization

Definition. A deterministic sequence satisfying $\underline{v}_m \downarrow 0$, used only to keep the pilot allocation in the interior in finite samples.

- $\widehat{\alpha}_{d,m}$ — pilot_allocation_rule; $(0, 1)$; estimator

Definition. The pilot-feasible share

$$\widehat{\alpha}_{d,m} = \sqrt{c_{E,d}(\widehat{V}_{E,d} \vee \underline{v}_m)} / [\sqrt{c_{E,d}(\widehat{V}_{E,d} \vee \underline{v}_m)} + \sqrt{c_{O,d}(\widehat{V}_{O,d} \vee \underline{v}_m)}]$$

- $\widehat{\alpha}_m$ — selected_pilot_allocation; $(0, 1)$; estimator

Definition. The implemented main-study share $\widehat{\alpha}_m = \widehat{\alpha}_{\widehat{d}_m, m}$.

- $N_{E,B}$ — implemented_experimental_count; $\mathbb{N}$; sample_size

Definition. The pilot-feasible integer count $N_{E,B} = \lfloor \widehat{\alpha}_m B / c_{E, \widehat{d}_m} \rfloor$.

- $N_{O,B}$ — implemented_observational_count; $\mathbb{N}$; sample_size

Definition. The pilot-feasible integer count $N_{O,B} = \lfloor (1 - \widehat{\alpha}_m) B / c_{O, \widehat{d}_m} \rfloor$.

- ε_m — uniform_pilot_error; $[0, \infty]$; error

Definition. The error $\varepsilon_m = \max_{d \in \mathcal{D}} |\widehat{\mathcal{V}}_{d,m}^* - \mathcal{V}_d^*(P)|$.

- L — crossfit_fold_count; $\mathbb{N}$; index

Definition. A fixed number of main-study cross-fitting folds.

- $\widehat{h}_{\{d,B\}}(-\ell)$ — crossfit_outcome_bridge_estimator; $L_2(S_{3d}, S_{2d}, A, X)$; nuisance_estimator

Definition. The outcome-bridge estimator for design d trained off main-study fold $\ell \in \{1, \dots, L\}$.

- $\widehat{q}_{\{d,B\}}(-\ell)$ — crossfit_selection_bridge_estimator; $L_2(S_{2d}, S_{1d}, A, X)$; nuisance_estimator

Definition. The selection-bridge estimator for design d trained off main-study fold $\ell \in \{1, \dots, L\}$.

- $\delta_{\{h,B\}}$ — outcome_bridge_weak_rate; $(0, \infty)$; rate

Definition. A deterministic uniform weak-seminorm rate for the main-study outcome bridges.

- $\rho_{\{h,B\}}$ — outcome_bridge_strong_rate; $(0, \infty)$; rate

Definition. A deterministic uniform strong-norm rate for the main-study outcome bridges.

- $\delta_{\{q,B\}}$ — selection_bridge_weak_rate; $(0, \infty)$; rate

Definition. A deterministic uniform weak-seminorm rate for the main-study selection bridges.

- $\rho_{\{q,B\}}$ — selection_bridge_strong_rate; $(0, \infty)$; rate

Definition. A deterministic uniform strong-norm rate for the main-study selection bridges.

- $r_{\{V,m\}}$ — pilot_variance_rate; $(0, \infty)$; rate

Definition. A deterministic rate for uniform pilot estimation of the two source-specific gradient variances.

- $r_{\{\widetilde{V}, B\}}$ — main_study_variance_rate; $(0, \infty)$; rate

Definition. A deterministic rate for uniform main-study estimation of the two source-specific gradient variances.

- ϵ — weak_proxy_signal; $(0, 10^{-3})$; perturbation_index

Definition. The signal strength of the early proxy in the prediction-selected witness design.

- $d_{\{\mathrm{p}\}}$ — prediction_selected_design; $\mathcal{D}$; witness_design

Definition. The witness design using the weak early proxy and the strong late proxy.

- $d_{\{\mathrm{e}\}}$ — efficiency_selected_design; $\mathcal{D}$; witness_design

Definition. The witness design using two moderately informative proxies.

- $\mathsf{H}_{\{\mathrm{comp}\}}$ — raw_prediction_risk_comparison_handle; method; open_question_handle

Definition. The quadratic-form comparison handle for the generic raw population prediction-risk rule.

- $z_{\{1-\eta/2\}}$ — standard_normal_quantile; $\mathbb{R}$; critical_value

Definition. The $1 - \eta/2$ quantile of the standard normal distribution.

- $\psi_{\{E,d\}}^{\mathrm{IKM}}$ — ikmw_experimental_gradient_component; $L_2^0(P(\cdot | G = E))$; influence_component

Definition. The Imbens–Kallus–Mao–Wang-normalized experimental score $\psi_{E,d}^{\mathrm{IKM}} = (1 - p_{E,1})\phi_{E,d}$.

- $\psi_{\{O,d\}}^{\mathrm{IKM}}$ — ikmw_observational_gradient_component; $L_2^0(P(\cdot | G = O))$; influence_component

Definition. The Imbens–Kallus–Mao–Wang-normalized observational score $\psi_{O,d}^{\mathrm{IKM}} = (1 - p_{O,1})\phi_{O,d}$.

- $V_{\{E,d\}}^{\{\mathrm{IKM}\}}$ — `ikmw_experimental_information_component`; $[0, \infty]$; `efficiency_component`
Definition. The IKM-normalized experimental source moment $V_{E,d}^{\mathrm{IKM}} = \mathbb{E}_P[(\psi_{E,d}^{\mathrm{IKM}})^2 \mid G = E] = (1 - p_{E,1})^2 V_{E,d}$.
- $V_{\{O,d\}}^{\{\mathrm{IKM}\}}$ — `ikmw_observational_information_component`; $[0, \infty]$; `efficiency_component`
Definition. The IKM-normalized observational source moment $V_{O,d}^{\mathrm{IKM}} = \mathbb{E}_P[(\psi_{O,d}^{\mathrm{IKM}})^2 \mid G = O] = (1 - p_{O,1})^2 V_{O,d}$.
- $\mathcal{V}_{\{d, \mathrm{IKM}\}}^*(P)$ — `ikmw_profiled_design_value`; $(0, \infty]$; `focal_object`
Definition. The profiled IKM-normalized value
$$\mathcal{V}_{d,\mathrm{IKM}}^*(P) = \left(\sqrt{c_{E,d} V_{E,d}^{\mathrm{IKM}}} + \sqrt{c_{O,d} V_{O,d}^{\mathrm{IKM}}} \right)^2$$
.
- $D_{\{\mathrm{IKM}\}}^*(P)$ — `ikmw_oracle_measurement_design`; $2^{\mathcal{D}}$; `optimal_design`
Definition. The IKM-normalized oracle set $D_{\mathrm{IKM}}^*(P) = \arg \min_{d \in \mathcal{D}} \mathcal{V}_{d,\mathrm{IKM}}^*(P)$, retaining all ties.
- $\rho_{\{\mathrm{pred}\}}^{\{\mathrm{IKM}\}}(P)$ — `ikmw_raw_prediction_risk_ratio`; $[1, \infty]$; `diagnostic`
Definition. The best-tie ratio of the raw population prediction-risk selector computed with the IKM-normalized profile.
- $\theta_d(P)$ — `observed_bridge_contrast`; $\mathbb{R}$; `estimand_functional`
Definition. The identified designwise observed-data bridge contrast $\theta_d(P) = \mu_{d,1} - \mu_{d,0}$. It equals the causal target only under an additional common-identification condition.
- $\sigma_{\{d, \mathrm{IKM}\}}^2$ — `ikmw_local_efficiency_bound`; $[0, \infty)$; `efficiency_bound`
Definition. The Imbens–Kallus–Mao–Wang fixed-system local efficiency lower bound for design d and causal target τ .
- n — `combined_sample_size`; $\mathbb{N}$; `sample_size`
Definition. The combined fixed-system sample size $n = n_E + n_O$.
- P_n° — `artificial_two_source_law`; `probability_law`; `comparison_law`
Definition. The artificial law with source masses n_E/n and n_O/n and source-conditional distributions inherited from P .
- λ — `source_sample_ratio`; $(0, \infty)$; `asymptotic_ratio`
Definition. The limiting fixed-system ratio n_E/n_O .
- L_0 — `fixed_design_fold_count`; $\{2, 3, \dots\}$; `index`
Definition. The fixed number of observational-only cross-fitting folds in the published fixed-design comparison.
- S_1 — `fixed_system_early_outcome`; `product space`; `proxy`
Definition. The early short-term outcome in the fixed-system IKM comparison.
- S_2 — `fixed_system_middle_outcome`; `product space`; `sequential_separator`
Definition. The middle short-term outcome in the fixed-system IKM comparison.
- S_3 — `fixed_system_late_outcome`; `product space`; `proxy`
Definition. The late short-term outcome in the fixed-system IKM comparison.
- h_0 — `fixed_system_outcome_bridge`; $L_2(S_3, S_2, A, X)$; `nuisance`
Definition. The true outcome bridge in the cited fixed-system theorem.

- q_0 — fixed_system_selection_bridge; $L_2(S_2, S_1, A, X)$; nuisance

Definition. The true selection bridge in the cited fixed-system theorem.

- μ_0 — fixed_system_bridge_mean; map from $\{0, 1\}$ to $\mathbb{R}$; nuisance

Definition. The treatment-indexed bridge mean in the cited fixed-system score.

- T_0 — fixed_system_conditional_expectation_operator; $L_2(S_3, S_2, A, X) \rightarrow L_2(S_2, S_1, A, X)$; operator

Definition. The conditional-expectation operator in the cited fixed-system theorem.

- T_0^* — fixed_system_adjoint_operator; $L_2(S_2, S_1, A, X) \rightarrow L_2(S_3, S_2, A, X)$; operator

Definition. The adjoint of T_0 in the cited fixed-system theorem.

- $\widehat{h}_{(-\ell)}$ — fixed_system_crossfit_outcome_bridge; $L_2(S_3, S_2, A, X)$; nuisance_estimator

Definition. The outcome-bridge fit trained outside fixed-system fold ℓ .

- $\widehat{q}_{(-\ell)}$ — fixed_system_crossfit_selection_bridge; $L_2(S_2, S_1, A, X)$; nuisance_estimator

Definition. The selection-bridge fit trained outside fixed-system fold ℓ .

- $\delta_{h,n}$ — fixed_system_outcome_weak_rate; $(0, \infty)$; rate

Definition. The weak outcome-bridge rate in the cited fixed-design result.

- $\rho_{h,n}$ — fixed_system_outcome_strong_rate; $(0, \infty)$; rate

Definition. The strong outcome-bridge rate in the cited fixed-design result.

- $\delta_{q,n}$ — fixed_system_selection_weak_rate; $(0, \infty)$; rate

Definition. The weak selection-bridge rate in the cited fixed-design result.

- $\rho_{q,n}$ — fixed_system_selection_strong_rate; $(0, \infty)$; rate

Definition. The strong selection-bridge rate in the cited fixed-design result.

- $V_E^{\{\mathrm{IKM}\}}$ — fixed_system_ikmw_experimental_moment; $[0, \infty)$; efficiency_component

Definition. The experimental score second moment printed in IKM Theorem 5 for a fixed short-term system.

- $V_O^{\{\mathrm{IKM}\}}$ — fixed_system_ikmw_observational_moment; $[0, \infty)$; efficiency_component

Definition. The observational score second moment printed in IKM Theorem 5 for a fixed short-term system.

- $\widehat{\tau}_{\{\mathrm{DR}\}}$ — ikmw_fixed_system_estimator; $\mathbb{R}$; estimator

Definition. The doubly robust estimator in Imbens--Kallus--Mao--Wang Theorem 5.

- $\mathcal{I}_{E,B}$ — experimental_index_set; finite index set; index_set

Definition. The main-study experimental indices, of cardinality $N_{E,B}$.

- $\mathcal{I}_{O,B}$ — observational_index_set; finite index set; index_set

Definition. The main-study observational indices, of cardinality $N_{O,B}$.

- $\mathcal{I}_{O,B,\ell}$ — observational_fold_index_set; finite index set; index_set

Definition. Observational evaluation fold ℓ in a deterministic balanced partition of $\mathcal{I}_{O,B}$.

- `e_{B,a}` — `experimental_arm_count`; $\mathbb{N}_0$; `sample_size`

Definition. The number of experimental main-study observations assigned treatment a .

- `o_{B,\ell,a}` — `observational_fold_arm_count`; $\mathbb{N}_0$; `sample_size`

Definition. The number of observations in observational fold ℓ with treatment a .

- `\widehat{p}_{E,a}` — `empirical_experimental_arm_probability`; $[0, 1]$; `estimator`

Definition. The empirical experimental arm proportion $e_{B,a}/N_{E,B}$, assigned zero if its source count vanishes.

- `\widehat{p}_{0,\ell,a}` — `empirical_observational_fold_arm_probability`; $[0, 1]$; `estimator`

Definition. The empirical arm proportion within observational fold ℓ , assigned zero if its fold size vanishes.

- `\widehat{\mu}_{d,a}^{(-\ell)}` — `foldwise_experimental_bridge_mean`; $\mathbb{R}$; `estimator`

Definition. The experimental arm mean of the fold- ℓ outcome-bridge fit, with zero-denominator value zero.

- `\widehat{\mu}_{B,d}(a)` — `crossfit_bridge_mean_estimator`; $\mathbb{R}$; `estimator`

Definition. The average of experimental bridge predictions and observational bridge residual corrections for treatment a , totalized at zero denominators.

- `\widehat{\tau}_B(d,\alpha)` — `design_allocation_bridge_contrast_estimator`; $\mathbb{R}$; `estimator`

Definition. The cross-fitted contrast $\widehat{\mu}_{B,d}(1) - \widehat{\mu}_{B,d}(0)$ at design d and allocation α , under the declared total convention.

- `\widehat{\phi}_{E,d,B,i}` — `experimental_plugin_score`; $\mathbb{R}$; `estimated_influence_component`

Definition. The exact arm-normalized fold-averaged experimental plug-in score for observation i .

- `\widehat{\phi}_{0,d,B,i}` — `observational_plugin_score`; $\mathbb{R}$; `estimated_influence_component`

Definition. The exact arm-normalized observational plug-in score for observation i , evaluated with its fold-external fits.

- `\widetilde{V}_{g,d,B}` — `total_main_study_score_variance_estimator`; $[0, \infty)$; `variance_estimator`

Definition. The within-source empirical second moment of the exact arm-normalized plug-in score $\widehat{\phi}_{g,d,B,i}$, with value zero when its source count or a required arm denominator is zero.

- `\mathsf{D}_B` — `positive_denominator_event`; `event`; `event`

Definition. The event that both selected source counts and every experimental arm and observational fold-arm count are positive.

- `\widehat{s}_B^2` — `total_post_selection_variance_estimator`; $[0, \infty)$; `variance_estimator`

Definition. The total variance estimator

$$\mathbf{1}\{D_B\} B\{\widetilde{V}_{E,\widehat{d}_m,B}/N_{E,B} + \widetilde{V}_{O,\widehat{d}_m,B}/N_{O,B}\}$$

, with all zero-denominator fractions assigned zero.

- `\widehat{s}_B` — `total_post_selection_standard_error`; $[0, \infty)$; `standard_error`

Definition. The nonnegative square root $(\widehat{s}_B^2)^{1/2}$, hence zero outside D_B .

- `\mathsf{G}_{\{\mathrm{tot}\}}` — `total_wald_event`; `event`; `event`

Definition. The event $D_B \cap \{\widehat{s}_B^2 > 0\}$ on which the ordinary studentized statistic is evaluated.

- `\widehat\tau_B` — `total_post_selection_bridge_estimator`; $\mathbb{R}$; `estimator`

Definition. On G_{tot} , the explicit cross-fitted estimator $\widehat{\tau}_B(\widehat{d}_m, \widehat{\alpha}_m)$; on its complement, zero. It targets $\theta_{\widehat{d}_m}(P)$ on the full regular class and τ under common identification.

- `I.B` — `total_postselection_wald_interval`; random closed interval in $\mathbb{R}$; `confidence_interval`

Definition. On G_{tot} , the interval $[\widehat{\tau}_B - z_{1-\eta/2}\widehat{s}_B/\sqrt{B}, \widehat{\tau}_B + z_{1-\eta/2}\widehat{s}_B/\sqrt{B}]$; on the complement, $[0, 0]$.

- `s_{0,B}` — `oracle_selected_standard_error`; $[0, \infty)$; `oracle_standard_error`

Definition. The nonnegative square root of $B\{V_{E,\widehat{d}_m}/N_{E,B} + V_{O,\widehat{d}_m}/N_{O,B}\}$ on positive source counts and zero otherwise.

- `\mathsf{T.B}` — `total_studentized_statistic`; $\mathbb{R}$; `test_statistic`

Definition. On G_{tot} , the statistic $\sqrt{B}\{\widehat{\tau}_B - \theta_{\widehat{d}_m}(P)\}/\widehat{s}_B$, and zero otherwise.

- `\mathcal{C}\backslash\kappa` — `induced_catalogue_cost_manifold`; subset of $(0, \infty)^{K+2}$; `parameter_region`

Definition. The cap-feasible primitive cost vectors $(b_E, b_O, c_1, \dots, c_K)$ inducing every $c_{E,d}$ and $c_{O,d}$ through the catalogue subset formulas.

- `\tau_0` — `ikmw_fixed_system_target`; $\mathbb{R}$; `estimand`

Definition. The fixed-system causal target in the cited IKM limit theorem.

- `\|\cdot\|_2` — `fixed_system_l2_norm`; norm; `error_metric`

Definition. The $L_2(P_n^\circ)$ norm in the cited fixed-design comparison.

- `e.B` — `deterministic_path_experimental_count`; $\mathbb{N}_0$; `sample_size`

Definition. The floor count $\lfloor \alpha_B B / c_{E,d_B} \rfloor$ on a deterministic design-allocation path.

- `o.B` — `deterministic_path_observational_count`; $\mathbb{N}_0$; `sample_size`

Definition. The floor count $\lfloor (1 - \alpha_B) B / c_{O,d_B} \rfloor$ on a deterministic design-allocation path.

- `n.B` — `deterministic_path_total_count`; $\mathbb{N}_0$; `sample_size`

Definition. The total deterministic-path count $n_B = e_B + o_B$.

- `P_{B,d,\alpha}^{\circ\circ}` — `deterministic_path_two_source_law`; probability law; `comparison_law`

Definition. The artificial law with source masses proportional to e_B, o_B and source-conditional laws inherited from P along design d and share α .

- `\|\cdot\|_{2,B}^{\circ\circ}` — `deterministic_path_l2_norm`; norm; `error_metric`

Definition. The $L_2(P_{B,d,\alpha}^\circ)$ norm used only for functions having declared versions on both sources or for terms explicitly zeroed outside $G = O$; it is not applied to an observational equivalence class lacking a declared experimental-source version.

- `\chi` — `prescribed_cost_disadvantage_factor`; $(1, \infty)$; `index`

Definition. The arbitrary prescribed finite paired-cost ratio of the prediction-selected witness design to the efficient witness design.

- `y_{\chi}` — `late_proxy_signal`; $(0, 1)$; `witness_proxy_signal`

Definition. The late-proxy signal $y_\chi = \sqrt{1 - 3/(10\chi)}$ in the prediction-selected center design.

- `\mathcal{M}_{\{\mathrm{bin}\}}(\chi)` — `binary_constrained_finite_cell_model`; set of finite-support laws; `witness_model`

Definition. The χ -indexed constrained finite-cell family defined in Definition def:binary-witness, with relative topology induced by the complete latent-cell vector.

- $\mathfrak{P}_{\mathcal{D}}^{\text{obs}}$ — observable_bridge_law_class; set of observed two-source laws; model_class

Definition. The observable bridge-law class in Definition def:observable-bridge-law-class.

- $\mathfrak{P}_{\mathcal{D}}$ — regular_law_class; set of laws; model_class

Definition. The stronger causal full-catalogue subclass of $\mathfrak{P}_{\mathcal{D}}^{\text{obs}}$, containing the latent exchangeability, external-validity, sequential-outcome, and completeness restrictions for every design.

- $\mathcal{U}_{\text{bin}}$ — regular_witness_region; $2^{\mathfrak{P}_{\mathcal{D}}}$; witness_region

Definition. The relatively open finite-cell perturbation region around the binary witness path specified in Definition def:binary-region.

- $\mathcal{U}_{\text{bin}}^{\text{IKM}}$ — ikmw_regular_witness_region; $2^{\mathfrak{P}_{\mathcal{D}}}$; witness_region

Definition. The relatively open subregion of $\mathcal{U}_{\text{bin}}$ on which the strict IKM-normalized optimizer ordering is preserved.

- P_{ϵ} — finite_support_witness_law; $\mathfrak{P}_{\mathcal{D}}$; witness_law

Definition. The χ -indexed binary finite-support center law specified in Definition def:binary-witness.

- r_{ϵ} — binary_stability_radius; $(0, \infty)$; radius

Definition. The regularity and strict-optimizer stability radius furnished by the binary witness lemma at P_{ϵ} .

- $r_{\epsilon}^{\text{opt}}$ — binary_optimizer_stability_radius; $(0, \infty)$; radius

Definition. The alias $r_{\epsilon}^{\text{opt}} = r_{\epsilon}$ used by the corrected binary-region definition.

- $r_{\epsilon}^{\text{IKM}}$ — binary_ikmw_stability_radius; $(0, \infty)$; radius

Definition. The regularity, common-identification, and strict IKM-normalized optimizer-stability radius furnished at P_{ϵ} for the prescribed χ .

- $P_{\chi, \epsilon}$ — finite_support_witness_law; $\mathfrak{P}_{\mathcal{D}}$; witness_law

Definition. The binary finite-support center law indexed jointly by the prescribed cost-disadvantage factor χ and early-proxy signal ϵ ; for fixed χ , it is the law denoted P_{ϵ} in Definition def:binary-witness.

- $\mathfrak{P}_{\mathcal{D}}^{\text{id}}$ — common_identification_class; subset of $\mathfrak{P}_{\mathcal{D}}$; model_class

Definition. The causal subclass on which $\theta_d(P) = \tau$ for every admissible design.

- $R_d^A(P)$ — treatment_aware_predictive_risk; $[0, \infty)$; benchmark_objective

Definition. The best observational long-term squared-error prediction risk using treatment, baseline covariates, and the measured catalogue subset.

- $\mathcal{D}_{\text{pred}}^A(P)$ — treatment_aware_prediction_optimizer_set; $2^{\mathfrak{P}_{\mathcal{D}}}$; benchmark_decision

Definition. The tie-preserving minimizer set of $R_d^A(P)$ over the admissible family.

- $\rho_{\text{pred}}^A(P)$ — treatment_aware_prediction_source_score_ratio; $[1, \infty]$; diagnostic

Definition. The best-tie arm-normalized profiled design-value ratio of the treatment-aware population prediction-risk selector.

- $\rho_{\text{pred}}^{\text{IKM}}(P)$ — treatment_aware_prediction_ikmw_ratio; $[1, \infty]$; diagnostic

Definition. The best-tie IKM-normalized profiled design-value ratio of the treatment-aware population prediction-risk selector.

- $\widehat{f}_{d,m}^A$ — `saturated_treatment_aware_pilot_regression`; finite-cell regression map; estimator

Definition. The training-half saturated observational cell-mean predictor of Y from A, X, Z_{M_d} in the full-catalogue pilot.

- $\widehat{R}_{d,m}^A$ — `heldout_treatment_aware_risk_estimator`; $[0, \infty)$; estimator

Definition. The validation-half empirical squared loss of $\widehat{f}_{d,m}^A$ on observational full-catalogue pilot records.

- $\widehat{d}_m^A$ — `empirical_treatment_aware_measurement_design`; $\mathcal{D}$; estimator

Definition. The deterministically tie-broken minimizer of $\widehat{R}_{d,m}^A$ over the admissible family.

4 Assumptions

Assumption 1 (`ass:longterm-consistency`). $Y = Y(A)$ almost surely.

Standard (*Long-term-outcome consistency*, [1]).

Assumption 2 (`ass:shortterm-consistency`). $Z = Z(A)$ almost surely.

Standard (*Short-term-outcome consistency*, [1]).

Assumption 3 (`ass:observational-latent-exchangeability`). $(Y(a), Z(a)) \perp A \mid U, X, G = O$ for each $a \in \{0, 1\}$.

Standard (*Observational latent exchangeability*, [1]).

Assumption 4 (`ass:observational-overlap`). $0 < P(A = 1 \mid U, X, G = O) < 1$ almost surely.

Standard (*Latent observational overlap*, [1]).

Assumption 5 (`ass:experimental-randomization`). $(Y(a), Z(a), U, X) \perp A \mid G = E$ for each $a \in \{0, 1\}$.

Standard (*Complete experimental randomization*, [1]).

Assumption 6 (`ass:experimental-overlap`). $0 < P(A = 1 \mid G = E) < 1$.

Standard (*Experimental treatment overlap*, [1]).

Assumption 7 (`ass:external-validity`). For every $d \in \mathcal{D}$ and $a \in \{0, 1\}$,

$$(S_{1d}(a), S_{2d}(a), S_{3d}(a), U, X) \perp G.$$

Standard (*Full short-term external validity across samples*, [1]).

Assumption 8 (`ass:source-support`). $p_P(U, X \mid A = a, G = E)/p_P(U, X \mid A = a, G = O) < \infty$ almost surely for each $a \in \{0, 1\}$.

Standard (*Cross-sample support overlap*, [1]).

Assumption 9 (`ass:sequential-outcomes`). $(Y(a), S_{3d}(a)) \perp S_{1d}(a) \mid S_{2d}(a), U, X, G = O$ for every $d \in \mathcal{D}$ and $a \in \{0, 1\}$.

Standard (*Designwise sequential-outcome restriction*, [1]).

Assumption 10 (`ass:late-proxy-completeness`). For every $d \in \mathcal{D}$, $a \in \{0, 1\}$, and square-integrable v , $\mathbb{E}_P[v(U) \mid S_{3d}, S_{2d}, A = a, X, G = O] = 0$ almost surely implies $v(U) = 0$ almost surely.

Standard (*Late-proxy completeness*, [1]).

Assumption 11 (`ass:early-proxy-completeness`). For every $d \in \mathcal{D}$, $a \in \{0, 1\}$, and square-integrable v , $\mathbb{E}_P[v(U) \mid S_{1d}, S_{2d}, A = a, X, G = O] = 0$ almost surely implies $v(U) = 0$ almost surely.

Standard (*Early-proxy completeness*, [1]).

Assumption 12 (`ass:selection-bridge-existence`). For every $d \in \mathcal{D}$, q_d is a square-integrable solution of $p_P(S_{3d}, S_{2d}, X \mid A, G = E)/p_P(S_{3d}, S_{2d}, X \mid A, G = O) = (T_d^* q_d)(S_{3d}, S_{2d}, A, X)$.

Standard (*Observed selection-bridge existence*, [1]).

Assumption 13 (ass:operator-bijectivity). T_d is bijective for every $d \in \mathcal{D}$.

Standard (Designwise bridge-operator bijectivity, [1]).

Assumption 14 (ass:nondegenerate-components).

$$0 < \min_{d \in \mathcal{D}} \min\{V_{E,d}, V_{O,d}\} \leq \max_{d \in \mathcal{D}} \max\{V_{E,d}, V_{O,d}\} < \infty$$

Standard (Finite nondegenerate influence-component variances, [1]).

Assumption 15 (ass:independent-pilot). $\mathcal{P}_m$ is independent of the main-study observations used to construct $\hat{\tau}_B$.

Standard (Pilot and analysis sample splitting, [12]).

Assumption 16 (ass:uniform-outcome-weak-rate). For every deterministic sequence $d_B \in \mathcal{D}$, $\alpha_B \in (0, 1)$ with $|\alpha_B - \alpha_{d_B}^*(P)| \rightarrow 0$, and fixed L ,

$$\max_{1 \leq \ell \leq L} \|T_{d_B}(\hat{h}_{d_B,B}^{(-\ell)} - h_{d_B})\|_{2,O,q,d_B} = O_P(\delta_{h,B}).$$

Assumption 17 (ass:uniform-outcome-strong-rate). For every deterministic sequence $d_B \in \mathcal{D}$, $\alpha_B \in (0, 1)$ with $|\alpha_B - \alpha_{d_B}^*(P)| \rightarrow 0$, and fixed L ,

$$\max_{1 \leq \ell \leq L} \|\hat{h}_{d_B,B}^{(-\ell)} - h_{d_B}\|_{2,B}^\circ = O_P(\rho_{h,B}).$$

Assumption 18 (ass:uniform-selection-weak-rate). For every deterministic sequence $d_B \in \mathcal{D}$, $\alpha_B \in (0, 1)$ with $|\alpha_B - \alpha_{d_B}^*(P)| \rightarrow 0$, and fixed L ,

$$\max_{1 \leq \ell \leq L} \|T_{d_B}^*(\hat{q}_{d_B,B}^{(-\ell)} - q_{d_B})\|_{2,B}^\circ = O_P(\delta_{q,B}).$$

Assumption 19 (ass:uniform-selection-strong-rate). For every deterministic sequence $d_B \in \mathcal{D}$, $\alpha_B \in (0, 1)$ with $|\alpha_B - \alpha_{d_B}^*(P)| \rightarrow 0$, and fixed L ,

$$\max_{1 \leq \ell \leq L} \|\hat{q}_{d_B,B}^{(-\ell)} - q_{d_B}\|_{2,O,q,d_B} = O_P(\rho_{q,B}).$$

Assumption 20 (ass:outcome-weak-rate-vanishes). $\delta_{h,B} = o(1)$.

Standard (Maintained estimator-level rate condition, [12]).

Assumption 21 (ass:outcome-strong-rate-vanishes). $\rho_{h,B} = o(1)$.

Standard (Maintained estimator-level rate condition, [12]).

Assumption 22 (ass:selection-weak-rate-vanishes). $\delta_{q,B} = o(1)$.

Standard (Maintained estimator-level rate condition, [12]).

Assumption 23 (ass:selection-strong-rate-vanishes). $\rho_{q,B} = o(1)$.

Standard (Maintained estimator-level rate condition, [12]).

Assumption 24 (ass:uniform-bridge-product-rate). $\min\{\delta_{h,B}\rho_{q,B}, \rho_{h,B}\delta_{q,B}\} = o(B^{-1/2})$.

Assumption 25 (ass:pilot-gradient-variance-rate). $\max_{d \in \mathcal{D}} \max\{|\hat{V}_{E,d} - V_{E,d}|, |\hat{V}_{O,d} - V_{O,d}|\} = O_P(r_{V,m})$, where probability is under the full-catalogue pilot law and the maximum is over the entire finite admissible family and both sources.

Assumption 26 (ass:pilot-variance-rate-vanishes). $r_{V,m} = o(1)$.

Standard (Maintained full-catalogue pilot estimator condition, [12]).

Assumption 27 (ass:main-gradient-variance-rate). Under the joint pilot and main-study law,

$$\max_{g \in \{E,O\}} |\tilde{V}_{g,\hat{d}_m,B} - V_{g,\hat{d}_m}| = O_P(r_{\tilde{V},B}).$$

Assumption 28 (ass:main-variance-rate-vanishes). $r_{\tilde{V},B} = o(1)$.

Standard (Maintained estimator-level rate condition, [12]).

Assumption 29 (ass:source-arm-positivity). For every $g \in \{E, O\}$ and $a \in \{0, 1\}$, $0 < p_{g,a} < 1$.

Novel. standard: source-arm positivity

Assumption 30 (ass:observed-outcome-bridge-existence). For every $d \in \mathcal{D}$, h_d is a square-integrable solution of

$$\mathbb{E}_P[Y \mid S_{2d}, S_{1d}, A, X, G = O] = \mathbb{E}_P[h_d(S_{3d}, S_{2d}, A, X) \mid S_{2d}, S_{1d}, A, X, G = O].$$

Standard (observed proximal outcome-bridge existence (ImbensKallusMaoWang2025), [1]).

Assumption 31 (ass:common-identification). For every $d \in \mathcal{D}$, $\theta_d(P) = \tau$.

Novel. novel: explicit common-identification subclass for design selection

Assumption 32 (ass:postselection-multiplier-condition). For every deterministic $d_B \in \mathcal{D}$ and $\alpha_B \in (0, 1)$ with $|\alpha_B - \alpha_{d_B}^*(P)| \rightarrow 0$, writing $\Delta h_{B,\ell} = \hat{h}_{d_B,B}^{(-\ell)} - h_{d_B}$ and $\Delta q_{B,\ell} = \hat{q}_{d_B,B}^{(-\ell)} - q_{d_B}$,

$$\begin{aligned} \max_{1 \leq \ell \leq L} & \left(\|\mathbf{1}\{G = O\} \Delta q_{B,\ell} (Y - h_{d_B})\|_{2,B}^\circ + \|\mathbf{1}\{G = O\} q_{d_B} \Delta h_{B,\ell}\|_{2,B}^\circ \right. \\ & \left. + \|\mathbf{1}\{G = O\} \Delta q_{B,\ell} \Delta h_{B,\ell}\|_{2,B}^\circ \right) = o_P(1). \end{aligned}$$

5 Definitions

Definition 1 (def:regular-law-class). Defined object. *Regular causal full-catalogue law class*

Construction. $\mathfrak{P}_{\mathcal{D}}$ is the subclass of $\mathfrak{P}_{\mathcal{D}}^{\text{obs}}$ satisfying long- and short-term consistency, observational latent exchangeability and overlap, experimental randomization and overlap, full short-term external validity, source support, the sequential-outcome restriction, and both proxy-completeness conditions for every $d \in \mathcal{D}$ and $a \in \{0, 1\}$.

Member properties. *ass:longterm-consistency; ass:shortterm-consistency; ass:observational-latent-exchangeability; ass:observational-overlap; ass:experimental-randomization; ass:experimental-overlap; ass:external-validity; ass:source-support; ass:sequential-outcomes; ass:late-proxy-completeness; ass:early-proxy-completeness; ass:source-arm-positivity; ass:observed-outcome-bridge-existence; ass:selection-bridge-existence; ass:operator-bijectivity; ass:nondegenerate-components.*

Definition 2 (def:observable-bridge-law-class). Defined object. *Observable bridge law class*

Construction.

$$\mathfrak{P}_{\mathcal{D}}^{\text{obs}} = \left\{ P : \begin{array}{l} \text{source-arm positivity, a square-integrable observed outcome bridge, a square-integrable} \\ \text{observed selection bridge, designwise bijectivity of } T_d, \text{ and finite strictly positive} \\ \text{source-score variances hold for every } d \in \mathcal{D} \end{array} \right\}.$$

Member properties. *ass:source-arm-positivity; ass:observed-outcome-bridge-existence; ass:selection-bridge-existence; ass:operator-bijectivity; ass:nondegenerate-components.*

Definition 3 (def:design-value-profile). Defined object. *Observable bridge-functional budget-normalized design-value profile*

Construction. For $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$, $d \in \mathcal{D}$, and $0 < \alpha < 1$, set

$$\mathcal{V}_d(\alpha; P) = \frac{c_{E,d} V_{E,d}}{\alpha} + \frac{c_{O,d} V_{O,d}}{1 - \alpha}.$$

Set the value to $+\infty$ outside the observable regular set.

Inputs. *lem:designwise-gradient; def:observable-bridge-law-class.*

Definition 4 (def:profiled-design-value). Defined object. *Profiled observable measurement-design value and oracle separation*

Construction. Set $\mathcal{V}_d^*(P) = \inf_{0 < \alpha < 1} \mathcal{V}_d(\alpha; P)$ and retain all population ties in $D^*(P) = \arg \min_{d \in \mathcal{D}} \mathcal{V}_d^*(P)$. Define $\gamma(P) = \min_{d \notin D^*(P)} \{\mathcal{V}_d^*(P) - \min_{d' \in D^*(P)} \mathcal{V}_{d'}^*(P)\}$, with $\gamma(P) = +\infty$ if $D^*(P) = \mathcal{D}$.

Inputs. *def:design-value-profile; def:observable-bridge-law-class.*

Definition 5 (def:design-value-gap). Defined object. *Pairwise observable design-value gap*

Construction. $\Delta_P(d, d') = \mathcal{V}_d^*(P) - \mathcal{V}_{d'}^*(P)$.

Inputs. *def:profiled-design-value.*

Definition 6 (def:predictive-benchmark). Defined object. *Unadjusted and treatment-aware population prediction-risk selectors with secondary paired-source diagnostic*

Construction. For each $d \in \mathcal{D}$, compute the unadjusted observational residual prediction risk and the treatment-aware observational residual prediction risk

$$R_d(P) = \inf_{f \in L_2(X, Z_{M_d})} \mathbb{E}_P[(Y - f(X, Z_{M_d}))^2 \mid G = O],$$

$$R_d^A(P) = \inf_{f \in L_2(A, X, Z_{M_d})} \mathbb{E}_P[(Y - f(A, X, Z_{M_d}))^2 \mid G = O].$$

The two generic population prediction-risk rules retain every tie:

$$\mathcal{D}_{\text{pred}}(P) = \arg \min_{d \in \mathcal{D}} R_d(P), \quad \mathcal{D}_{\text{pred}}^A(P) = \arg \min_{d \in \mathcal{D}} R_d^A(P).$$

Their best-tie arm-normalized source-score ratios are

$$\rho_{\text{pred}}(P) = \frac{\min_{d \in \mathcal{D}_{\text{pred}}(P)} \mathcal{V}_d^*(P)}{\min_{d' \in \mathcal{D}} \mathcal{V}_{d'}^*(P)}, \quad \rho_{\text{pred}}^A(P) = \frac{\min_{d \in \mathcal{D}_{\text{pred}}^A(P)} \mathcal{V}_d^*(P)}{\min_{d' \in \mathcal{D}} \mathcal{V}_{d'}^*(P)}.$$

For comparison only, define the continuous paired-source count and the secondary cost-weighted residual-risk diagnostic

$$n_{\text{CLW}, B}(d) = \frac{B}{c_{E,d} + c_{O,d}}, \quad \Psi_{\text{CLW}, d}(P; B) = \frac{R_d(P)}{n_{\text{CLW}, B}(d)} = \frac{(c_{E,d} + c_{O,d})R_d(P)}{B}.$$

This secondary quantity does not define either prediction selector or any of the displayed ratios. The subscript records only the residual-risk-per-budget shape and does not attribute a proxy-selection workflow or guarantee to Carneiro–Lee–Wilhelm. Neither population rule is attributed here to a named published proxy-selection workflow.

Inputs. *def:profiled-design-value; def:observable-bridge-law-class.*

Definition 7 (def:pilot-selector). Defined object. *Independent full-catalogue pilot selector*

Construction. For every $d \in \mathcal{D}$, fit the two observed bridge equations by cross-fitting on $\mathcal{P}_m$, estimate $V_{E,d}$ and $V_{O,d}$ by foldwise score second moments, form $\hat{V}_{d,m}^* = (\sqrt{c_{E,d}\hat{V}_{E,d}} + \sqrt{c_{O,d}\hat{V}_{O,d}})^2$, select a deterministic-tie-broken $\hat{d}_m \in \arg \min_{d \in \mathcal{D}} \hat{V}_{d,m}^*$, compute

$\hat{\alpha}_{d,m} = \sqrt{c_{E,d}(\hat{V}_{E,d} \vee \underline{v}_m)} / [\sqrt{c_{E,d}(\hat{V}_{E,d} \vee \underline{v}_m)} + \sqrt{c_{O,d}(\hat{V}_{O,d} \vee \underline{v}_m)}]$, set $\hat{\alpha}_m = \hat{\alpha}_{\hat{d}_m, m}$, and purchase $N_{E,B} = \lfloor \hat{\alpha}_m B / c_{E, \hat{d}_m} \rfloor$ experimental units and $N_{O,B} = \lfloor (1 - \hat{\alpha}_m) B / c_{O, \hat{d}_m} \rfloor$ follow-up units.

Inputs. *\mathcal{P}_m; def:profiled-design-value; def:observable-bridge-law-class.*

Definition 8 (def:selected-wald-variance). Defined object. *Selected-design-only Wald estimator and total variance*

Construction. After the independently funded pilot chooses $\hat{d}_m$ and $\hat{\alpha}_m$, the main study measures only $M_{\hat{d}_m}$. Let $\mathcal{I}_{E,B}$ and $\mathcal{I}_{O,B}$ have sizes $N_{E,B}$ and $N_{O,B}$, and partition $\mathcal{I}_{O,B} = \bigsqcup_{\ell=1}^L \mathcal{I}_{O,B,\ell}$ into deterministic balanced folds. Train only $\hat{h}_{\hat{d}_m,B}^{(-\ell)}$ and $\hat{q}_{\hat{d}_m,B}^{(-\ell)}$, excluding observational fold ℓ from both fits and using the experimental sample only for the selection-bridge fit. For $a \in \{0, 1\}$, define the selected-design arm counts, empirical arm probabilities, bridge means, residual corrections, and scores exactly by

$$e_{B,a} = \sum_{i \in \mathcal{I}_{E,B}} \mathbf{1}\{A_i = a\}, \quad o_{B,\ell,a} = \sum_{i \in \mathcal{I}_{O,B,\ell}} \mathbf{1}\{A_i = a\},$$

$$\hat{\mu}_{\hat{d}_m,a}^{(-\ell)} = \frac{1}{e_{B,a}} \sum_{i \in \mathcal{I}_{E,B}} \mathbf{1}\{A_i = a\} \hat{h}_{\hat{d}_m,B}^{(-\ell)}(S_{3\hat{d}_m,i}, S_{2\hat{d}_m,i}, a, X_i),$$

$$\hat{\mu}_{B,\hat{d}_m}(a) = \frac{1}{L} \sum_{\ell=1}^L \hat{\mu}_{\hat{d}_m,a}^{(-\ell)} + \frac{1}{L} \sum_{\ell=1}^L \frac{1}{o_{B,\ell,a}} \sum_{i \in \mathcal{I}_{O,B,\ell}} \mathbf{1}\{A_i = a\} \hat{q}_{\hat{d}_m,B}^{(-\ell)} \{Y_i - \hat{h}_{\hat{d}_m,B}^{(-\ell)}\},$$

with every function evaluated at its selected-design arguments, and set

$$\hat{\tau}_B(\hat{d}_m, \hat{\alpha}_m) = \hat{\mu}_{B,\hat{d}_m}(1) - \hat{\mu}_{B,\hat{d}_m}(0)$$

. For $i \in \mathcal{I}_{E,B}$, define

$$\hat{\phi}_{E,\hat{d}_m,B,i} = \frac{1}{L} \sum_{\ell=1}^L \sum_{a=0}^1 (-1)^{1-a} \frac{\mathbf{1}\{A_i = a\}}{e_{B,a}/N_{E,B}} \{ \hat{h}_{\hat{d}_m,B}^{(-\ell)}(S_{3\hat{d}_m,i}, S_{2\hat{d}_m,i}, a, X_i) - \hat{\mu}_{\hat{d}_m,a}^{(-\ell)} \}.$$

For $i \in \mathcal{I}_{O,B,\ell}$, define

$$\hat{\phi}_{O,\hat{d}_m,B,i} = \sum_{a=0}^1 (-1)^{1-a} \frac{\mathbf{1}\{A_i = a\}}{o_{B,\ell,a}/|\mathcal{I}_{O,B,\ell}|} \hat{q}_{\hat{d}_m,B}^{(-\ell)}(S_{2\hat{d}_m,i}, S_{1\hat{d}_m,i}, a, X_i) \times \{Y_i - \hat{h}_{\hat{d}_m,B}^{(-\ell)}(S_{3\hat{d}_m,i}, S_{2\hat{d}_m,i}, a, X_i)\}.$$

Set

$$\tilde{V}_{E,\hat{d}_m,B} = \frac{1}{N_{E,B}} \sum_{i \in \mathcal{I}_{E,B}} \hat{\phi}_{E,\hat{d}_m,B,i}^2, \quad \tilde{V}_{O,\hat{d}_m,B} = \frac{1}{N_{O,B}} \sum_{\ell=1}^L \sum_{i \in \mathcal{I}_{O,B,\ell}} \hat{\phi}_{O,\hat{d}_m,B,i}^2.$$

Let $\mathbb{D}_B$ require positive selected source, arm, and fold-arm counts. On $\mathbb{D}_B$, define

$$\hat{s}_B^2 = B \left\{ \frac{\tilde{V}_{E,\hat{d}_m,B}}{N_{E,B}} + \frac{\tilde{V}_{O,\hat{d}_m,B}}{N_{O,B}} \right\}, \quad s_{0,B}^2 = B \left\{ \frac{V_{E,\hat{d}_m}}{N_{E,B}} + \frac{V_{O,\hat{d}_m}}{N_{O,B}} \right\},$$

and $G_{\text{tot}} = D_B \cap \{\hat{s}_B^2 > 0\}$. On G_{tot} , use the selected estimator, statistic centered at $\theta_{\hat{d}_m}(P)$, and Wald interval; on its complement set the estimator, standard error, and statistic to zero and the interval to $[0,0]$. Every zero denominator is totalized at zero. No unselected main-study fit or score variance is part of this measurable map.

Inputs. *def:pilot-selector*; *def:observable-bridge-law-class*.

Definition 9 (def:binary-witness). Defined object. *Binary two-design arbitrary-cost-disadvantage witness family*

Construction. Fix an arbitrary prescribed $\chi > 1$. Set $K = 4$, $c_1 = c_2 = (5\chi - 3)/4$, $c_3 = c_4 = 1/2$, $b_E = 1$, $b_O = 2$, and $\kappa = (5\chi - 3)/2$, and put

$$y_\chi = \sqrt{1 - \frac{3}{10\chi}}.$$

Let X and every S_{2d} be constant. Set $P_\epsilon(G = E) = P_\epsilon(G = O) = 1/2$, and let a Rademacher U be independent of G . In source E , $P_\epsilon(T = t \mid U = u, G = E) = 1/2$; in source O , $P_\epsilon(T = t \mid U = u, G = O) = (1 + tu/4)/2$. The proxies $W_\epsilon, W_{y_\chi}, W_{1/2}^{(1)}, W_{1/2}^{(2)}$ are mutually independent conditional on U , are independent of G, A conditional on U , and form

$$Z(a) = (W_\epsilon, W_{y_\chi}, W_{1/2}^{(1)}, W_{1/2}^{(2)}),$$

while $Y(a) = U$ for both treatments. Take $\mathcal{D} = \{d_p, d_e\}$, where d_p assigns W_ϵ to role one and W_{y_χ} to role three, whereas d_e assigns $W_{1/2}^{(1)}$ to role one and $W_{1/2}^{(2)}$ to role three. Then

$$c_{E,d_p} = \frac{5\chi - 1}{2}, \quad c_{O,d_p} = \frac{5\chi + 1}{2}, \quad c_{E,d_e} = 2, \quad c_{O,d_e} = 3,$$

so the paired costs are 5χ and 5, respectively. For a design with early signal x and late signal y , the candidate bridges are

$$h_d(l, a) = \frac{l}{y}, \quad q_d(e, a) = \frac{1 - (2a - 1)e/(4x)}{1 - 1/16},$$

and the residual prediction variance is

$$R(x, y) = \frac{(1 - x^2)(1 - y^2)}{1 - x^2y^2}.$$

The primary prediction selector compares this unweighted residual risk itself; paired-cost weighting is retained only as a secondary diagnostic.

The binary constrained finite-cell model $\mathcal{M}_{\text{bin}}(\chi)$ is the family of laws P with the preceding prescribed catalogue, costs, cap, and role assignments, constant X and constant S_{2d} , latent variables $U, T, W_1, W_2, W_3, W_4 \in \{-1, 1\}$, $A = (T + 1)/2$, and potential variables

$$Z(a) = (W_1, W_2, W_3, W_4), \quad Y(a) = U, \quad a \in \{0, 1\}.$$

For strictly positive probability vectors $(\pi_g : g \in \{E, O\})$, $(\nu_u : u \in \{-1, 1\})$, $(\lambda_t : t \in \{-1, 1\})$, observational treatment kernels $e_u(t)$, and proxy kernels $k_{j,u}(w)$, its latent-cell mass factorizes as

$$P(G = g, U = u, T = t, W_1 = w_1, \dots, W_4 = w_4) = \pi_g \nu_u \{ \mathbf{1}\{g = E\} \lambda_t + \mathbf{1}\{g = O\} e_u(t) \} \prod_{j=1}^4 k_{j,u}(w_j). \quad (1)$$

Thus U is independent of G , treatment is randomized independently of U in source E , and the four proxies are mutually independent conditional on U and invariant across source and treatment. The design d_p assigns (W_1, W_2) to roles one and three, and d_e assigns (W_3, W_4) to roles one and three; role two is constant. For each $0 < \epsilon < 10^{-3}$, the preceding center P_ϵ is the point having $\pi_E = \pi_O = 1/2$, $\nu_{-1} = \nu_1 = 1/2$, $\lambda_{-1} = \lambda_1 = 1/2$, $e_u(t) = (1 + tu/4)/2$, and

$$k_{j,u}(w) = \frac{1 + s_j u w}{2}, \quad (s_1, s_2, s_3, s_4) = (\epsilon, y_\chi, 1/2, 1/2).$$

The model is parameterized by the relative interiors of these finite probability simplices. Its relative topology is equivalently the Euclidean parameter topology or the maximum-norm topology $\|\cdot\|_{\text{cell}}$ induced by the complete latent-cell vector in (1).

Inputs. `def:regular-law-class`; `def:predictive-benchmark`.

Definition 10 (`def:binary-region`). Defined object. *Relatively open binary joint-prediction-risk perturbation region*

Construction. Fix the prescribed $\chi > 1$. For every $0 < \epsilon < 10^{-3}$, choose the radius $r_\epsilon > 0$ furnished by Lemma `lem:binary-witness-stability` after the joint shrinking permitted by Lemma `lem:joint-predictive-binary-stability`, and set $r_\epsilon^{\text{opt}} := r_\epsilon$. Choose $0 < \delta_\epsilon < r_\epsilon^{\text{opt}}$ and define

$$\mathcal{U}_{\text{bin}} = \bigcup_{0 < \epsilon < 10^{-3}} \{P \in \mathcal{M}_{\text{bin}}(\chi) : \|P - P_\epsilon\|_{\text{cell}} < \delta_\epsilon\}. \quad (1)$$

Every member of every ball is regular, has both bridge contrasts equal to τ , and retains the strict unadjusted-risk, treatment-aware-risk, and arm-normalized source-score optimizer assignments furnished by the two stability lemmas. Thus $\mathcal{U}_{\text{bin}} \subseteq \mathfrak{P}_{\mathcal{D}}$ and is relatively open in $\mathcal{M}_{\text{bin}}(\chi)$. No claim of openness in the ambient unrestricted probability simplex is made, because the factorization, support, consistency, and causal equality restrictions define a lower-dimensional model.

Inputs. `def:binary-witness`; `def:regular-law-class`; `lem:binary-witness-stability`; `lem:joint-predictive-binary-stability`.

Definition 11 (`def:common-identification-class`). Defined object. *Common causal-identification subclass*

Construction.

$$\mathfrak{P}_{\mathcal{D}}^{\text{id}} = \{P \in \mathfrak{P}_{\mathcal{D}} : \theta_d(P) = \tau \text{ for every } d \in \mathcal{D}\}.$$

Member properties. `ass:longterm-consistency`; `ass:shortterm-consistency`; `ass:observational-latent-exchangeability`; `ass:observational-overlap`; `ass:experimental-randomization`; `ass:experimental-overlap`; `ass:external-validity`; `ass:source-support`; `ass:sequential-outcomes`; `ass:late-proxy-completeness`; `ass:early-proxy-completeness`; `ass:source-arm-positivity`; `ass:observed-outcome-bridge-existence`; `ass:selection-bridge-existence`; `ass:operator-bijectivity`; `ass:nondegenerate-components`; `ass:common-identification`.

Definition 12 (`def:ikmw-normalized-design-value`). Defined object. *IKM-normalized profiled value and unadjusted or treatment-aware risk-selector ratios*

Construction. For each regular design d , define the source scores on the Imbens–Kallus–Mao–Wang normalization by

$$\psi_{E,d}^{\text{IKM}} = (1 - p_{E,1})\phi_{E,d}, \quad \psi_{O,d}^{\text{IKM}} = (1 - p_{O,1})\phi_{O,d},$$

and define their source-conditional second moments by

$$V_{E,d}^{\text{IKM}} = \mathbb{E}_P[(\psi_{E,d}^{\text{IKM}})^2 \mid G = E] = (1 - p_{E,1})^2 V_{E,d},$$

$$V_{O,d}^{\text{IKM}} = \mathbb{E}_P[(\psi_{O,d}^{\text{IKM}})^2 \mid G = O] = (1 - p_{O,1})^2 V_{O,d}.$$

The profiled IKM-normalized value, its optimizer set, and the best-tie unadjusted and treatment-aware population prediction-risk ratios are

$$\mathcal{V}_{d,\text{IKM}}^*(P) = \left(\sqrt{c_{E,d} V_{E,d}^{\text{IKM}}} + \sqrt{c_{O,d} V_{O,d}^{\text{IKM}}} \right)^2, \\ D_{\text{IKM}}^*(P) = \arg \min_{d \in \mathcal{D}} \mathcal{V}_{d,\text{IKM}}^*(P),$$

$$\rho_{\text{pred}}^{\text{IKM}}(P) = \frac{\min_{d \in \mathcal{D}_{\text{pred}}(P)} \mathcal{V}_{d, \text{IKM}}^*(P)}{\min_{d' \in \mathcal{D}} \mathcal{V}_{d', \text{IKM}}^*(P)}, \quad \rho_{\text{pred}}^{A, \text{IKM}}(P) = \frac{\min_{d \in \mathcal{D}_{\text{pred}}^A(P)} \mathcal{V}_{d, \text{IKM}}^*(P)}{\min_{d' \in \mathcal{D}} \mathcal{V}_{d', \text{IKM}}^*(P)}.$$

These objects are defined only as normalization-specific score moments and profiles. Their interpretation as causal efficiency bounds is supplied design by design by Lemma `lem:ikmw-fixed-system-efficiency` after its causal-identification and regularity premises have been verified. Neither prediction ratio is attributed to a named published proxy-selection workflow.

Inputs. `lem:ikmw-fixed-system-efficiency`; `def:predictive-benchmark`; `def:common-identification-class`.

Definition 13 (`def:binary-ikmw-region`). Defined object. *Relatively open IKM-normalized binary joint-prediction-risk region*

Construction. Fix the prescribed $\chi > 1$. For every $0 < \epsilon < 10^{-3}$, choose the radius $r_{\epsilon}^{\text{IKM}} > 0$ furnished by Lemma `lem:binary-ikmw-normalized-stability` after the joint shrinking permitted by Lemma `lem:joint-predictive-binary-stability`, and let δ_{ϵ} be the radius used in Definition `def:binary-region`. Choose

$$0 < \tilde{\delta}_{\epsilon} < \min\{\delta_{\epsilon}, r_{\epsilon}^{\text{IKM}}\}$$

and define

$$\mathcal{U}_{\text{bin}}^{\text{IKM}} = \bigcup_{0 < \epsilon < 10^{-3}} \left\{ P \in \mathcal{M}_{\text{bin}}(\chi) : \|P - P_{\epsilon}\|_{\text{cell}} < \tilde{\delta}_{\epsilon} \right\}. \quad (1)$$

The inherited unadjusted-risk and treatment-aware-risk selectors and the strict IKM-normalized optimizer are stable throughout every displayed ball. Thus $\mathcal{U}_{\text{bin}}^{\text{IKM}} \subseteq \mathcal{U}_{\text{bin}}$ and is relatively open in $\mathcal{M}_{\text{bin}}(\chi)$.

Inputs. `def:binary-witness`; `def:binary-region`; `lem:binary-ikmw-normalized-stability`; `lem:joint-predictive-binary-stability`; `def:common-identification-class`.

Definition 14 (`def:bridge-differentiable-submodel`). Defined object. *Bridge-differentiable regular two-source submodel*

Construction. Fix $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$ and $d \in \mathcal{D}$, and write

$$H_d = (S_{3d}, S_{2d}, A, X), \quad L_d = (S_{2d}, S_{1d}, A, X).$$

A regular two-source parametric submodel $\{P_t : |t| < t_0\}$ through $P = P_0$, with source-conditional scores ℓ_E and ℓ_O , is called bridge-differentiable for design d at P when the following conditions hold. First, the source-conditional laws are dominated on a common support and the path is differentiable in quadratic mean in each source. Second, for all sufficiently small $|t|$, $P_t \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$ and its design- d outcome bridge $h_{d,t}$ is unique. Third, after identifying the common L_2 coordinate spaces along the dominated path, there is $\dot{h}_d \in L_2(H_d \mid G = O)$ such that

$$\left\| \frac{h_{d,t} - h_{d,0}}{t} - \dot{h}_d \right\|_{L_2(P(\cdot \mid G=O))} \longrightarrow 0,$$

and the conditional bridge equation

$$\mathbb{E}_{P_t} \{Y - h_{d,t}(H_d) \mid L_d, G = O\} = 0$$

may be differentiated at $t = 0$ in $L_2(L_d \mid G = O)$ using the conditional-score rule. This is a restriction on submodels, not a consequence of pointwise bijectivity of T_d .

Inputs. `def:observable-bridge-law-class`.

Definition 15 (`def:operator-comparison-handle`). Defined object. *Operator comparison handle for raw population prediction risk*

Construction. Use H_{comp} to compare the source-score quadratic forms underlying $\mathcal{V}_d^*(P)$ with the raw residual-risk quadratic form underlying $R_d(P)$. These source-score forms have a gradient interpretation only along the bridge-differentiable paths declared for design d . The exact target is a lawwise constant bounding the raw-risk selector ratio $\rho_{\text{pred}}(P)$; the handle does not assert that a generalized eigenvalue is sharp and does not give an unconditional gradient or causal-efficiency interpretation.

Inputs. `def:predictive-benchmark`; `def:profiled-design-value`; `def:bridge-differentiable-submodel`.

Definition 16 (def:treatment-aware-pilot-selector). Defined object. *Split-sample saturated treatment-aware pilot selector*

Construction. On an observational full-catalogue pilot from the binary witness family, order the records with $G = O$ by their original pilot indices and split them deterministically into a training half $\mathcal{I}_{m,O}^{\text{tr}}$ and a validation half $\mathcal{I}_{m,O}^{\text{va}}$, whose cardinalities differ by at most one. For each $d \in \mathcal{D}$ and each cell $v = (a, x, z_{M_d})$, define the saturated training-half regression

$$\hat{f}_{d,m}^A(v) = \frac{\sum_{i \in \mathcal{I}_{m,O}^{\text{tr}}} \mathbf{1}\{(A_i, X_i, Z_{M_d,i}) = v\} Y_i}{\sum_{i \in \mathcal{I}_{m,O}^{\text{tr}}} \mathbf{1}\{(A_i, X_i, Z_{M_d,i}) = v\}},$$

with value zero when the denominator vanishes. Define the held-out treatment-aware risk estimator

$$\hat{R}_{d,m}^A = \frac{1}{|\mathcal{I}_{m,O}^{\text{va}}|} \sum_{i \in \mathcal{I}_{m,O}^{\text{va}}} \{Y_i - \hat{f}_{d,m}^A(A_i, X_i, Z_{M_d,i})\}^2,$$

with value zero if the validation half is empty, and let $\hat{d}_m^A$ be the deterministically tie-broken element of $\arg \min_{d \in \mathcal{D}} \hat{R}_{d,m}^A$. This is a measurable finite-cell rule using only the independently funded full-catalogue pilot; it does not fit a bridge or use main-study outcomes.

Inputs. *def:predictive-benchmark; def:binary-witness.*

6 Main results

Lemma 1 (lem:designwise-gradient). For every $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$ and $d \in \mathcal{D}$, the observed-data bridge contrast $\theta_d(P) := \mu_{d,1} - \mu_{d,0}$ is identified. Along every bridge-differentiable regular two-source submodel for design d at P in Definition def:bridge-differentiable-submodel, its pathwise gradient has source components $\phi_{E,d}$ and $\phi_{O,d}$. Hence independent source sample sizes n_E, n_O have source-score variance

$$\frac{V_{E,d}}{n_E} + \frac{V_{O,d}}{n_O},$$

and this is the gradient variance on the declared bridge-differentiable submodel class. Outside that class, $V_{E,d}$ and $V_{O,d}$ remain source-score variances, but pointwise bijectivity alone does not assert pathwise differentiability. No latent causal restriction, completeness condition, or equality with τ is used or asserted.

Proof. Fix $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$ and $d \in \mathcal{D}$, and abbreviate

$$H_d = (S_{3d}, S_{2d}, A, X), \quad L_d = (S_{2d}, S_{1d}, A, X).$$

Observed outcome-bridge existence gives

$$\mathbb{E}_P\{Y - h_d(H_d) \mid L_d, G = O\} = 0. \quad (1)$$

Equivalently, $T_d h_d = \mathbb{E}_P(Y \mid L_d, G = O)$. Bijectivity of T_d makes this solution unique. Thus h_d , the two experimental means $\mu_{d,a}$, and the contrast $\theta_d(P) = \mu_{d,1} - \mu_{d,0}$ are identified from the two observed source laws.

Now take a bridge-differentiable regular two-source submodel in Definition def:bridge-differentiable-submodel, with source-conditional scores ℓ_E, ℓ_O and bridge derivative $\dot{h}_d$. Differentiability in quadratic mean, the declared L_2 bridge differentiability, and the usual product rule give

$$\begin{aligned} \dot{\mu}_{d,a} &= \mathbb{E}_P \left[\frac{\mathbf{1}\{A = a\}}{p_{E,a}} \{h_d(S_{3d}, S_{2d}, a, X) - \mu_{d,a}\} \ell_E \mid G = E \right] \\ &\quad + \mathbb{E}_P \{\dot{h}_d(S_{3d}, S_{2d}, a, X) \mid A = a, G = E\}. \end{aligned} \quad (2)$$

The selection-bridge equation states that $T_d^* q_d$ is the conditional density ratio of (S_{3d}, S_{2d}, X) in source E relative to source O , within each treatment arm. Therefore, for every square-integrable $f(H_d)$, iterated expectation gives

$$\mathbb{E}_P\{f(H_d) \mid A = a, G = E\} = \mathbb{E}_P\{q_d(L_d)f(H_d) \mid A = a, G = O\}. \quad (3)$$

The defining submodel condition permits differentiation of (1) in conditional L_2 . The conditional-score centering term is multiplied by the zero conditional residual mean in (1), and hence

$$T_d \dot{h}_d = \mathbb{E}_P[\{Y - h_d(H_d)\} \ell_O \mid L_d, G = O]. \quad (4)$$

Apply (3) with $f = \dot{h}_d$, then apply (4) and iterated expectation:

$$\begin{aligned} \mathbb{E}_P(\dot{h}_d \mid A = a, G = E) &= \mathbb{E}_P(q_d \dot{h}_d \mid A = a, G = O) \\ &= \mathbb{E}_P[q_d \{Y - h_d(H_d)\} \ell_O \mid A = a, G = O]. \end{aligned} \quad (5)$$

Subtracting (2) at $a = 0$ from (2) at $a = 1$, substituting (5), and using source-arm positivity to express arm-conditional expectations as source-conditional inner products yields

$$\dot{\theta}_d(P) = \mathbb{E}_P(\phi_{E,d} \ell_E \mid G = E) + \mathbb{E}_P(\phi_{O,d} \ell_O \mid G = O). \quad (6)$$

The definitions of $\mu_{d,a}$ imply $\mathbb{E}_P(\phi_{E,d} \mid G = E) = 0$, while (1) implies $\mathbb{E}_P(\phi_{O,d} \mid G = O) = 0$. The nondegenerate-component condition makes both components square-integrable. Thus they are the two source components of a pathwise gradient on the declared submodel class. For independent source samples of sizes n_E, n_O , the variances of their sample averages add, giving

$$\frac{V_{E,d}}{n_E} + \frac{V_{O,d}}{n_O}. \quad (7)$$

This derivation uses the explicit bridge-differentiability restriction at the differentiation step (4); it does not infer local inverse stability from pointwise bijectivity. The identification part and the definitions of the source-score variances remain meaningful without that path restriction. No latent causal restriction, completeness condition, or equality with τ was used. $\square$

Justification. The repaired lemma identifies the observed bridge contrast law-by-law and derives its two source gradient components only along the declared bridge-differentiable regular submodels. The restriction is placed exactly at the load-bearing differentiation of the conditional bridge equation. *Gap.* Pointwise L_2 bridge existence and bijectivity do not supply local inverse continuity or differentiability of the bridge solution along every regular path. *Consumer.* A downstream designer may always use $V_{E,d}$ and $V_{O,d}$ as source-score variance inputs, and may call their combination a gradient variance only after verifying bridge-differentiability for the relevant submodel class.

Theorem 1 (thm:budget-profile). *For every $P \in \mathfrak{P}_D^{\text{obs}}$, $d \in \mathcal{D}$, and fixed $0 < \alpha < 1$, as $B \rightarrow \infty$,*

$$B \left\{ \frac{V_{E,d}}{n_{E,B}(d, \alpha)} + \frac{V_{O,d}}{n_{O,B}(d, \alpha)} \right\} = \mathcal{V}_d(\alpha; P) + \mathfrak{r}_{d,B}(\alpha; P),$$

where, once $B \geq 2 \max\{c_{E,d}/\alpha, c_{O,d}/(1-\alpha)\}$,

$$0 \leq \mathfrak{r}_{d,B}(\alpha; P) \leq \frac{2}{B} \left\{ \frac{V_{E,d} c_{E,d}^2}{\alpha^2} + \frac{V_{O,d} c_{O,d}^2}{(1-\alpha)^2} \right\}.$$

Thus the floor-count source-score variance converges to $\mathcal{V}_d(\alpha; P)$. The unique minimizer and minimum are

$$\alpha_d^*(P) = \frac{\sqrt{c_{E,d} V_{E,d}}}{\sqrt{c_{E,d} V_{E,d}} + \sqrt{c_{O,d} V_{O,d}}}, \quad \mathcal{V}_d^*(P) = \left(\sqrt{c_{E,d} V_{E,d}} + \sqrt{c_{O,d} V_{O,d}} \right)^2.$$

On the bridge-differentiable submodel class of Definition def:bridge-differentiable-submodel, this profiles the design-wise gradient variance. For $P \in \mathfrak{P}_D^{\text{id}}$, it profiles the gradient variance of the common causal target only on that declared submodel class. It is a local causal semiparametric efficiency bound only when the designwise gradient is additionally canonical for τ , in particular under the fixed-system Imbens–Kallus–Mao–Wang conditions recorded separately.

Proof. Fix $P \in \mathfrak{P}_D^{\text{obs}}$, $d \in \mathcal{D}$, and $0 < \alpha < 1$, and set

$$x_E = \frac{\alpha B}{c_{E,d}}, \quad x_O = \frac{(1-\alpha)B}{c_{O,d}}.$$

The asserted lower bound on B makes $x_E, x_O \geq 2$. For every $x \geq 2$,

$$0 \leq \frac{1}{\lfloor x \rfloor} - \frac{1}{x} = \frac{x - \lfloor x \rfloor}{x \lfloor x \rfloor} \leq \frac{2}{x^2}, \quad (1)$$

because $0 \leq x - \lfloor x \rfloor < 1$ and $\lfloor x \rfloor \geq x/2$. Applying (1) to the two source counts gives the exact decomposition

$$B \left\{ \frac{V_{E,d}}{n_{E,B}(d, \alpha)} + \frac{V_{O,d}}{n_{O,B}(d, \alpha)} \right\} = \mathcal{V}_d(\alpha; P) + \mathfrak{r}_{d,B}(\alpha; P)$$

and

$$0 \leq \mathfrak{r}_{d,B}(\alpha; P) \leq \frac{2}{B} \left\{ \frac{V_{E,d} c_{E,d}^2}{\alpha^2} + \frac{V_{O,d} c_{O,d}^2}{(1-\alpha)^2} \right\}. \quad (2)$$

For fixed d and α , the right-hand side tends to zero. The left-hand side is the budget-scaled source-score variance; by Lemma lem:designwise-gradient, it is also the budget-scaled gradient variance along every bridge-differentiable submodel.

Put $a_d = c_{E,d} V_{E,d}$ and $b_d = c_{O,d} V_{O,d}$. Positive costs and Assumption ass:nondegenerate-components imply $a_d, b_d > 0$. Direct algebra gives

$$\frac{a_d}{\alpha} + \frac{b_d}{1-\alpha} - (\sqrt{a_d} + \sqrt{b_d})^2 = \frac{\{(1-\alpha)\sqrt{a_d} - \alpha\sqrt{b_d}\}^2}{\alpha(1-\alpha)} \geq 0. \quad (3)$$

Equality holds exactly when

$$\alpha = \frac{\sqrt{a_d}}{\sqrt{a_d} + \sqrt{b_d}} = \frac{\sqrt{c_{E,d} V_{E,d}}}{\sqrt{c_{E,d} V_{E,d}} + \sqrt{c_{O,d} V_{O,d}}} = \alpha_d^*(P).$$

This proves uniqueness and

$$\mathcal{V}_d^*(P) = (\sqrt{c_{E,d} V_{E,d}} + \sqrt{c_{O,d} V_{O,d}})^2.$$

On $\mathfrak{P}_D^{\text{id}}$, the observed contrast equals the common causal target, but the gradient interpretation still requires the declared bridge-differentiable submodel restriction. If that gradient is additionally canonical for τ , its variance is the local semiparametric efficiency lower bound; the separately recorded fixed-system Imbens–Kallus–Mao–Wang result supplies that interpretation under its own normalization and premises. Thus the calculation itself is a cost profile of source-score variance and does not, by algebra alone, assert a new cross-design canonical-efficiency theorem. $\square$

Justification. Profiles the exact two-source source-score variance under monetary costs and quantifies integer-count rounding. *Gap.* The anchor fixes a limiting source ratio; this algebraic profile optimizes it under design-specific costs, while the cross-design advance comes from comparing admissible measurement systems rather than from claiming a new fixed-system efficiency formula. *Consumer.* A study designer receives an explicit budget share and profiled source-score value, with causal-efficiency interpretation only under separately stated canonical-gradient premises.

Theorem 2 (thm:pilot-oracle-regret). Fix $P \in \mathfrak{P}_D^{\text{obs}}$. For the independently funded full-catalogue pilot selector,

$$0 \leq \mathcal{V}_{\hat{d}_m}^*(P) - \min_{d \in \mathcal{D}} \mathcal{V}_d^*(P) \leq 2\varepsilon_m, \quad \{\hat{d}_m \notin D^*(P)\} \subseteq \{2\varepsilon_m \geq \gamma(P)\}.$$

Conditional on the maintained full-catalogue pilot variance condition in Assumption `ass:pilot-gradient-variance-rate`, together with $r_{V,m} = o(1)$ from Assumption `ass:pilot-variance-rate-vanishes` and $\underline{v}_m = o(1)$,

$$\varepsilon_m = O_P(r_{V,m}), \quad \max_{d \in \mathcal{D}} |\hat{\alpha}_{d,m} - \alpha_d^*(P)| = O_P(r_{V,m}),$$

and $P\{\hat{d}_m \notin D^*(P)\} \rightarrow 0$, with the event empty when $D^*(P) = \mathcal{D}$. With the stipulated zero value when either floor count vanishes,

$$\max_{d \in \mathcal{D}} \left| B \left\{ \frac{V_{E,d}}{[\hat{\alpha}_{d,m} B / c_{E,d}]} + \frac{V_{O,d}}{[(1 - \hat{\alpha}_{d,m}) B / c_{O,d}]} \right\} - \mathcal{V}_d^*(P) \right| = O_P(r_{V,m}^2 + B^{-1}) = o_P(1).$$

The selector consistently chooses the complete optimizer set under these maintained pilot conditions; no causal interpretation is needed for this observable bridge-functional regret statement. The theorem does not derive the maintained variance condition from a primitive bridge learner.

Proof. Fix $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$. Write

$$v_d = \mathcal{V}_d^*(P), \quad \hat{v}_d = \hat{\mathcal{V}}_{d,m}^*, \quad v_* = \min_{d \in \mathcal{D}} v_d,$$

and choose $d^* \in D^*(P)$. By the definition of ε_m and the minimizing property of $\hat{d}_m$,

$$v_{\hat{d}_m} \leq \hat{v}_{\hat{d}_m} + \varepsilon_m \leq \hat{v}_{d^*} + \varepsilon_m \leq v_* + 2\varepsilon_m. \quad (1)$$

Since $v_* \leq v_{\hat{d}_m}$,

$$0 \leq v_{\hat{d}_m} - v_* \leq 2\varepsilon_m. \quad (2)$$

If $\hat{d}_m \notin D^*(P)$, the definition of $\gamma(P)$ gives

$$\gamma(P) \leq v_{\hat{d}_m} - v_* \leq 2\varepsilon_m. \quad (3)$$

When every design is optimal, the left event is empty; otherwise finiteness of $\mathcal{D}$ gives $\gamma(P) > 0$.

Set

$$Z_m = \max_{d \in \mathcal{D}} \max\{|\hat{V}_{E,d} - V_{E,d}|, |\hat{V}_{O,d} - V_{O,d}|\}.$$

The maintained pilot variance condition and its vanishing-rate companion give $Z_m = O_P(r_{V,m}) = o_P(1)$. By the pilot construction and Theorem `thm:budget-profile`,

$$\hat{v}_d = G_d(\hat{V}_{E,d}, \hat{V}_{O,d}), \quad v_d = G_d(V_{E,d}, V_{O,d}),$$

where $G_d(x, y) = (\sqrt{c_{E,d}x} + \sqrt{c_{O,d}y})^2$. Positive population variances, positive costs, and finiteness of $\mathcal{D}$ put all population pairs in a compact subset of $(0, \infty)^2$; with probability tending to one, the estimated pairs lie in a slightly larger such compact set. The finitely many gradients ∇G_d are uniformly bounded there. Thus

$$\varepsilon_m = \max_{d \in \mathcal{D}} |\hat{v}_d - v_d| = O_P(r_{V,m}) = o_P(1). \quad (4)$$

Equations (3)–(4) imply $P\{\hat{d}_m \notin D^*(P)\} \rightarrow 0$.

The same compact-set argument applied to

$$A_d(x, y) = \frac{\sqrt{c_{E,d}x}}{\sqrt{c_{E,d}x} + \sqrt{c_{O,d}y}}$$

shows that the variance truncation is inactive with probability tending to one and

$$\max_{d \in \mathcal{D}} |\hat{\alpha}_{d,m} - \alpha_d^*(P)| = O_P(r_{V,m}). \quad (5)$$

Finally, Lemma `lem:second-order-allocation-loss`, whose premises are exactly the same maintained pilot variance condition, its vanishing rate, nondegeneracy, and the pilot definition, gives

$$\max_{d \in \mathcal{D}} \left| B \left\{ \frac{V_{E,d}}{[\hat{\alpha}_{d,m} B / c_{E,d}]} + \frac{V_{O,d}}{[(1 - \hat{\alpha}_{d,m}) B / c_{O,d}]} \right\} - \mathcal{V}_d^*(P) \right| = O_P(r_{V,m}^2 + B^{-1}) = o_P(1).$$

The deterministic tie rule was used only to choose an empirical minimizer in (1); the conclusion is therefore membership in the complete optimizer set, not convergence to an arbitrarily selected population minimizer. Every stochastic conclusion in the proof is conditional on the maintained pilot variance condition; no primitive bridge learner or variance-rate derivation has been supplied. $\square$

Justification. The generic regret certificate is operational under the fixed-dimensional lemma, which supplies uniform pilot source-score variance error of order $m^{-1/2}$. *Gap.* Outside the fixed-dimensional regime the displayed pilot variance condition remains a hypothesis, not a consequence of the observable class. *Consumer.* A pilot designer with fixed known bridge bases obtains a feasible plug-in selector with the theorem's oracle-regret bound.

Theorem 3 (thm:postselection-wald). *Fix $P \in \mathfrak{P}_D^{\text{obs}}$. Conditional on the independent full-catalogue pilot, suppose the purchased experimental and observational samples are mutually independent and independent and identically distributed within source. Use the selected-design-only estimator, score moments, total event, variance estimator, statistic, and interval of Definition def:selected-wald-variance. Impose the maintained estimator-level conditions in Assumptions ass:uniform-outcome-weak-rate, ass:uniform-outcome-strong-rate, ass:uniform-selection-weak-rate, ass:uniform-selection-strong-rate, ass:uniform-bridge-product-rate, ass:postselection-multiplier-condition, ass:pilot-gradient-variance-rate, and ass:main-gradient-variance-rate, together with the four declared bridge-rate vanishing conditions and the pilot and main-study variance-rate vanishing conditions. Let $m \rightarrow \infty$, $B \rightarrow \infty$, and $v_m = o(1)$. Then*

$$P(\mathbf{G}_{\text{tot}}) \longrightarrow 1, \\ s_{0,B}^2 - \mathcal{V}_{\hat{d}_m}^*(P) = o_P(1), \quad \hat{s}_B^2 - \mathcal{V}_{\hat{d}_m}^*(P) = o_P(1),$$

and the selected observable contrast obeys

$$\mathbf{T}_B = \frac{\sqrt{B}\{\hat{\tau}_B - \theta_{\hat{d}_m}(P)\}}{\hat{s}_B} \rightsquigarrow \mathcal{N}(0, 1), \quad P\{\theta_{\hat{d}_m}(P) \in I_B\} \longrightarrow 1 - \eta.$$

All listed nuisance-rate, product-rate, multiplier, pilot-variance, and main-study-variance requirements are sufficient conditions of this conditional theorem. They are not conclusions of the observable law class or of the cited fixed-design theory. Primitive selected-path attainability without additional learner structure remains the separate open object oeq:primitive-selected-path-attainability.

Proof. Fix P as in the theorem. We first prove a deterministic-design-and-allocation expansion and then transfer it through the independent pilot.

Interior counts. For every $d \in \mathcal{D}$, positivity of the costs and Assumption ass:nondegenerate-components imply $0 < \alpha_d^*(P) < 1$. Since $\mathcal{D}$ is finite, there is $a_0 \in (0, 1/2)$ such that

$$2a_0 \leq \alpha_d^*(P) \leq 1 - 2a_0 \quad (d \in \mathcal{D}). \quad (1)$$

Theorem thm:pilot-oracle-regret gives

$$\max_{d \in \mathcal{D}} |\hat{\alpha}_{d,m} - \alpha_d^*(P)| = o_P(1). \quad (2)$$

Consequently, with probability tending to one, $\hat{\alpha}_m \in [a_0, 1 - a_0]$, and both purchased source counts are bounded below by a positive multiple of B . Source-arm positivity, conditional independent and identically distributed sampling, and Chebyshev's inequality give convergence of every empirical source-arm proportion to its strictly positive population value. A finite union bound over the two experimental arms and the $2L$ observational fold-arm cells therefore gives

$$P(\mathbf{D}_B) \longrightarrow 1. \quad (3)$$

Deterministic-path linearization. Take any deterministic path (d_B, α_B) such that $d_B \in \mathcal{D}$ and $|\alpha_B - \alpha_{d_B}^*(P)| \rightarrow 0$. Let e_B, o_B be its two floor counts. Equation (1) and finiteness of the cost catalogue imply

$$e_B \asymp B, \quad o_B \asymp B. \quad (4)$$

Write $h = h_{d_B}$, $q = q_{d_B}$,

$$\Delta h_\ell = \hat{h}_{d_B, B}^{(-\ell)} - h, \quad \Delta q_\ell = \hat{q}_{d_B, B}^{(-\ell)} - q.$$

We use an elementary arm-ratio expansion. If $W_1, \dots, W_N$ are independent and identically distributed, $p_a = P(A = a) > 0$, $N_a = \sum_i \mathbf{1}\{A_i = a\}$, and $f \in L_2$, then

$$\frac{1}{N_a} \sum_{i=1}^N \mathbf{1}\{A_i = a\} f(W_i) - E\{f(W) \mid A = a\} = \frac{1}{N} \sum_{i=1}^N \frac{\mathbf{1}\{A_i = a\}}{p_a} [f(W_i) - E\{f(W) \mid A = a\}] + o_P(N^{-1/2}). \quad (5)$$

Indeed, the centered numerator divided by N is $O_P(N^{-1/2})$, whereas $Np_a/N_a - 1 = O_P(N^{-1/2})$; their product is $O_P(N^{-1})$. Conditional on the training observations, the same calculation and conditional Chebyshev inequality show that a fold-external random function with conditional L_2 -norm $o_P(1)$ contributes $o_P(N^{-1/2})$. Because the observational partition is deterministic and balanced with fixed L , replacing equal fold weights by observation weights changes the resulting average by $o_P(B^{-1/2})$.

The observed outcome-bridge equation and the selection-bridge change-of-measure identity imply, for each arm and every square-integrable early-argument function u and late-argument function f ,

$$E_O\{u(S_{2dB}, S_{1dB}, a, X)(Y - h) \mid A = a\} = 0, \quad (6)$$

$$E_E\{f(S_{3dB}, S_{2dB}, a, X) \mid A = a\} = E_O\{qf(S_{3dB}, S_{2dB}, a, X) \mid A = a\}. \quad (7)$$

Thus, conditional on the training observations, the nuisance drift in one arm is exactly

$$\begin{aligned} E_E(\Delta h_\ell \mid A = a) + E_O\{(q + \Delta q_\ell)(Y - h - \Delta h_\ell) \mid A = a\} \\ = -E_O(\Delta q_\ell \Delta h_\ell \mid A = a). \end{aligned} \quad (8)$$

The two terms linear in Δh_ℓ cancel by (7), and the terms multiplying $Y - h$ vanish by (6).

This is precisely where the corrected observational norms are needed. Because Δq_ℓ is an L_d -function, conditional expectation and Cauchy–Schwarz give

$$\begin{aligned} |E_O(\Delta q_\ell \Delta h_\ell \mid A = a)| &\leq C \|\Delta q_\ell\|_{2,O,q,d_B} \|T_{dB} \Delta h_\ell\|_{2,O,q,d_B}, \\ |E_O(\Delta q_\ell \Delta h_\ell \mid A = a)| &\leq C \|T_{dB}^* \Delta q_\ell\|_{2,O,h,d_B} \|\Delta h_\ell\|_{2,O,h,d_B}. \end{aligned} \quad (9)$$

The first line uses only the observational strong-selection and weak-outcome errors. In the second line, the maintained artificial-law norms of the source-defined functions $T_{dB}^* \Delta q_\ell$ and Δh_ℓ control their observational norms because (4) keeps the observational mass of P_{B,d_B,α_B}° bounded away from zero. The four maintained rates and the product-rate assumption therefore make (8) $o_P(B^{-1/2})$, uniformly over the fixed arms and folds. No value of $T_{dB} \Delta h_\ell$ or Δq_ℓ on experimental L_d -support is used.

The centered experimental nuisance perturbation is Δh_ℓ . Its conditional empirical fluctuation is $o_P(B^{-1/2})$ by the retained artificial-law strong-outcome rate and conditional Chebyshev inequality. The centered observational perturbation is

$$\Delta q_\ell(Y - h) - q\Delta h_\ell - \Delta q_\ell \Delta h_\ell. \quad (10)$$

The multiplier condition makes the $L_2(P_{B,d_B,\alpha_B}^\circ)$ -norm of (10), after multiplication by $\mathbf{1}\{G = O\}$, equal to $o_P(1)$, so its foldwise centered empirical contribution is $o_P(B^{-1/2})$. When empirical arm probabilities replace population probabilities, their difference is $O_P(B^{-1/2})$. For a nuisance term g_B , the resulting unscaled population error is bounded by

$$C|\hat{p}_a - p_a| |P\{\mathbf{1}(A = a)g_B\}| \leq C|\hat{p}_a - p_a| \|g_B\|_2. \quad (11)$$

The strong outcome rate handles $g_B = \Delta h_\ell$; the multiplier condition handles $q\Delta h_\ell$ and $\Delta q_\ell \Delta h_\ell$; and $\Delta q_\ell(Y - h)$ has zero arm mean by (6). Hence every denominator-replacement error is also $o_P(B^{-1/2})$. The training scheme supplies the required conditional independence: experimental score evaluation uses outcome bridges trained on observational data, and each observational evaluation fold is excluded from both bridge fits.

Applying (5)–(11) and contrasting the two arms yields

$$\hat{\tau}_B(d_B, \alpha_B) - \theta_{dB}(P) = \frac{1}{e_B} \sum_{i \in E_B} \phi_{E,d_B,i} + \frac{1}{o_B} \sum_{i \in O_B} \phi_{O,d_B,i} + o_P(B^{-1/2}). \quad (12)$$

This expansion is centered at the observed bridge contrast and uses no common-identification restriction.

The two leading sums in (12) are independent and centered, with variances $V_{E,d_B}/e_B$ and $V_{O,d_B}/o_B$. To verify their normal limit, take an arbitrary subsequence and then a further subsequence on which $d_B = d$ is constant; this is possible because $\mathcal{D}$ is finite. Put

$$s_{0,B}^2(d, \alpha_B) = B \left\{ \frac{V_{E,d}}{e_B} + \frac{V_{O,d}}{o_B} \right\}.$$

Nondegeneracy and (4) bound this quantity above and away from zero. After multiplication by $\sqrt{B}/s_{0,B}$, each summand in the two independent arrays has coefficient $O(B^{-1/2})$, and the sum of their variances is one. For every $\zeta > 0$, finite second moments and these vanishing coefficients imply

$$\sum_j E\{X_{B,j}^2 \mathbf{1}(|X_{B,j}| > \zeta)\} \rightarrow 0, \quad (13)$$

because the two tail expectations are evaluated beyond thresholds tending to infinity. Expanding each characteristic function to second order, the sum of the remainders is bounded by the left side of (13) plus a constant times the maximal coefficient, while the sum of the quadratic terms tends to $-t^2/2$. Thus the product characteristic function tends to $e^{-t^2/2}$, proving

$$\frac{\sqrt{B}}{s_{0,B}(d_B, \alpha_B)} \left\{ \frac{1}{e_B} \sum_{i \in E_B} \phi_{E,d_B,i} + \frac{1}{o_B} \sum_{i \in O_B} \phi_{O,d_B,i} \right\} \rightsquigarrow \mathcal{N}(0, 1). \quad (14)$$

Since every subsequence has such a further subsequence, (14) holds along the original deterministic path. Equations (12) and (14) prove the deterministic-path studentized limit.

Transfer through pilot selection. By (2), there exists a deterministic $t_B \downarrow 0$ along the joint sequence such that

$$P\{|\hat{\alpha}_m - \alpha_{\hat{d}_m}^*(P)| > t_B\} \rightarrow 0. \quad (15)$$

The deterministic-path conclusions are uniform over

$$\{(d, \alpha) : d \in \mathcal{D}, |\alpha - \alpha_d^*(P)| \leq t_B\}. \quad (16)$$

Otherwise, one could choose a deterministic violating pair from (16) along a subsequence, contradicting the conclusion proved for every deterministic path. Independence of the pilot and main study permits conditioning on $(\hat{d}_m, \hat{\alpha}_m)$. Equations (12)–(16) therefore imply

$$\frac{\sqrt{B}\{\hat{\tau}_B(\hat{d}_m, \hat{\alpha}_m) - \theta_{\hat{d}_m}(P)\}}{s_{0,B}} \rightsquigarrow \mathcal{N}(0, 1). \quad (17)$$

The floor-count conclusion of Theorem thm:pilot-oracle-regret gives

$$s_{0,B}^2 - \mathcal{V}_{\hat{d}_m}^*(P) = o_P(1). \quad (18)$$

On D_B , direct subtraction yields

$$\begin{aligned} |\hat{s}_B^2 - s_{0,B}^2| &\leq \frac{B}{N_{E,B}} |\tilde{V}_{E,\hat{d}_m,B} - V_{E,\hat{d}_m}| \\ &\quad + \frac{B}{N_{O,B}} |\tilde{V}_{O,\hat{d}_m,B} - V_{O,\hat{d}_m}|. \end{aligned} \quad (19)$$

The two coefficients are $O_P(1)$ by (1)–(2). The maintained main-study variance condition and its vanishing-rate companion make the right side $o_P(1)$. Combining (3), (18), and (19) proves

$$\hat{s}_B^2 - \mathcal{V}_{\hat{d}_m}^*(P) = o_P(1). \quad (20)$$

Because $\min_d \mathcal{V}_d^*(P) > 0$, equations (3), (18), and (20) also yield

$$P(\mathbf{G}_{\text{tot}}) \rightarrow 1, \quad \hat{s}_B/s_{0,B} \rightarrow_P 1. \quad (21)$$

Combining (17) and (21), with an elementary product-convergence argument, proves

$$\mathbf{T}_B \rightsquigarrow \mathcal{N}(0, 1), \quad (22)$$

including the total zero extension off $\mathbf{G}_{\text{tot}}$.

On $\mathbf{G}_{\text{tot}}$, the event $\theta_{\hat{d}_m}(P) \in I_B$ is exactly $|\mathbf{T}_B| \leq z_{1-\eta/2}$. The complement of $\mathbf{G}_{\text{tot}}$ has probability tending to zero by (21), and the standard normal distribution is continuous at both critical values. Equation (22) therefore gives

$$P\{\theta_{\hat{d}_m}(P) \in I_B\} \rightarrow 1 - \eta.$$

This proves every assertion under the explicitly listed maintained conditions, with the two corrected observational norms used only on observational support. $\square$

Justification. Provides a selected-design limit law and Wald coverage for the observed bridge contrast under well-typed selected-path conditions. *Gap.* The observable law class alone does not provide learner rates; the theorem is conditional and the fixed-basis theorem supplies a concrete attainable subclass. *Consumer.* A study using an independent pilot can conduct selected-design-only inference without fitting unselected main-study models.

Proposition 1 (prop:anchor-reduction). *For every $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$, if $\mathcal{D} = \{d\}$, the main study fixes positive counts n_E, n_O , and $B = c_{E,d}n_E + c_{O,d}n_O$, then*

$$\frac{1}{B} \mathcal{V}_d\left(\frac{c_{E,d}n_E}{B}; P\right) = \frac{V_{E,d}}{n_E} + \frac{V_{O,d}}{n_O}.$$

The right-hand side is the fixed-system two-source pathwise-gradient variance for $\theta_d(P)$. If $P \in \mathfrak{P}_{\mathcal{D}}^{\text{id}}$ and the fixed-system Imbens–Kallus–Mao–Wang canonical-gradient conditions hold, it is also their causal semiparametric efficiency bound for τ ; absent those additional conditions only the observable bridge-functional identity is asserted.

Proof. Fix $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$ and the singleton design d . Put

$$\alpha = \frac{c_{E,d}n_E}{B}.$$

Because $B = c_{E,d}n_E + c_{O,d}n_O$ and both counts and costs are positive,

$$0 < \alpha < 1, \quad 1 - \alpha = \frac{c_{O,d}n_O}{B}.$$

Substitution into the design-value profile gives

$$\begin{aligned} \frac{1}{B} \mathcal{V}_d(\alpha; P) &= \frac{1}{B} \left\{ \frac{c_{E,d}V_{E,d}}{c_{E,d}n_E/B} + \frac{c_{O,d}V_{O,d}}{c_{O,d}n_O/B} \right\} \\ &= \frac{V_{E,d}}{n_E} + \frac{V_{O,d}}{n_O}. \end{aligned}$$

By Lemma lem:designwise-gradient, the last expression is the fixed-system two-source pathwise-gradient variance for $\theta_d(P)$. If $P \in \mathfrak{P}_{\mathcal{D}}^{\text{id}}$, the causal-identification premise gives $\theta_d(P) = \tau$. Under the additional fixed-system canonical-gradient premises, Lemma lem:ikmw-fixed-system-efficiency identifies the corresponding two-source canonical-gradient variance as the causal semiparametric efficiency lower bound. Without those causal-identification and canonical-gradient premises, the displayed equality remains only the observable bridge-functional identity. $\square$

Justification. Places the design calculation on the observable bridge model and reserves causal interpretation for the separate common-identification subclass. *Gap.* The anchor fixes the short-term system and source ratio and does not optimize the finite catalogue under costs. *Consumer.* Investigators can compare admissible measurement systems and allocate the main-study budget using source-specific gradient moments.

Theorem 4 (thm:predictive-impossibility). *For the generic raw population prediction-risk rule, the following impossibility holds. For every prescribed finite cost-disadvantage factor $\chi > 1$, the generalized unequal-cost binary witness has paired costs 5χ for d_p and 5 for d_e . For every $0 < \epsilon < 10^{-3}$, the center law $P_\epsilon \in \mathfrak{P}_{\mathcal{D}}$ is regular for both designs and both bridge contrasts equal $\tau = 0$. Nevertheless, d_p uniquely minimizes raw prediction risk, while d_e uniquely minimizes both the arm-normalized source-score profile and the IKM-normalized profile. More precisely,*

$$R_{d_p}(P_\epsilon) = \frac{3}{10\chi} \frac{1 - \epsilon^2}{1 - \epsilon^2(1 - 3/(10\chi))} < \frac{3}{10\chi} < \frac{3}{10} < \frac{3}{5} = R_{d_e}(P_\epsilon),$$

and the secondary paired-cost diagnostic obeys

$$B\Psi_{\text{CLW},d_p}(P_\epsilon; B) = \frac{3}{2} \frac{1 - \epsilon^2}{1 - \epsilon^2 y_\chi^2} < \frac{3}{2} < 3 = B\Psi_{\text{CLW},d_e}(P_\epsilon; B).$$

Furthermore,

$$V_{O,d_p} = \frac{64}{75(10\chi - 3)}\epsilon^{-2} + O_\chi(1), \quad V_{E,d}^{\text{IKM}} = \frac{1}{4}V_{E,d}, \quad V_{O,d}^{\text{IKM}} = \frac{1}{4}V_{O,d},$$

$$\mathcal{V}_{d,\text{IKM}}^*(P_\epsilon) = \frac{1}{4}\mathcal{V}_d^*(P_\epsilon), \quad \rho_{\text{pred}}^{\text{IKM}}(P_\epsilon) = \rho_{\text{pred}}(P_\epsilon) = \frac{5\chi + 1}{363(10\chi - 3)}\epsilon^{-2} + O_\chi(\epsilon^{-1}).$$

Across the center family, let $\chi_{\text{opt}}(\epsilon)$ be any cost-disadvantage factor minimizing the exact ratio for sufficiently small ϵ . Then

$$\chi_{\text{opt}}(\epsilon) \sim \frac{\sqrt[3]{18}}{15}\epsilon^{-2/3},$$

and the least-divergent sharp elbow is

$$\begin{aligned} \epsilon^2 \inf_{\chi > 1} \rho_{\text{pred}}(P_{\chi,\epsilon}) &= \epsilon^2 \inf_{\chi > 1} \rho_{\text{pred}}^{\text{IKM}}(P_{\chi,\epsilon}) \\ &= \frac{1}{726} + \frac{5\sqrt[3]{12}}{968}\epsilon^{2/3} + o(\epsilon^{2/3}). \end{aligned}$$

Every $P \in \mathcal{U}_{\text{bin}}$ is regular for both designs, both bridge contrasts equal τ , $\mathcal{D}_{\text{pred}}(P) = \{d_p\}$, $D^*(P) = \{d_e\}$, and

$$\sup_{P \in \mathcal{U}_{\text{bin}}} \rho_{\text{pred}}(P) = \infty.$$

On the relatively open subregion $\mathcal{U}_{\text{bin}}^{\text{IKM}} \subseteq \mathcal{U}_{\text{bin}}$, every design remains regular and identifies τ , $\mathcal{D}_{\text{pred}}(P) = \{d_p\}$, $D_{\text{IKM}}^*(P) = \{d_e\}$, and

$$\sup_{P \in \mathcal{U}_{\text{bin}}^{\text{IKM}}} \rho_{\text{pred}}^{\text{IKM}}(P) = \infty.$$

For each prescribed $\chi > 1$, the center cost vector belongs to $\mathcal{C}_\kappa$, has paired-cost ratio exactly χ , and has a relatively open neighborhood in $\mathcal{C}_\kappa$ on which d_p remains strictly more expensive and uniquely raw-risk-selected for all sufficiently small positive ϵ , while both the arm-normalized and balanced-center IKM-normalized ratios diverge along the center-law path. Thus standalone selection by raw population prediction risk has no finite uniform source-score guarantee, even when its chosen design bears any prescribed finite cost disadvantage; even the least-divergent cost-disadvantage sequence grows at order $\epsilon^{-2/3}$, and its ratio retains the leading divergence $1/(726\epsilon^2)$. The secondary cost-weighted diagnostic selects the same center design, but it is not the headline rule. No raw-risk proxy-selection workflow is attributed to a published paper, and no claim is made about every predictive method. The IKM-normalized profile has its absolute causal-efficiency interpretation only under the separately stated fixed-system premises; the arm-normalized profile is a source-score design value on the full regular class and a gradient design value only on bridge-differentiable submodels.

Proof. Fix $\chi > 1$ and write $P_{\chi,\epsilon} = P_\epsilon$. Lemma [lem:sharp-uniform-binary-elbow](#) proves center-law regularity and common identification, the paired costs 5χ and 5 , the strict sourcewise cost disadvantage of d_p , and

$$R_{d_p}(P_{\chi,\epsilon}) = \frac{3}{10\chi} \frac{1 - \epsilon^2}{1 - \epsilon^2(1 - 3/(10\chi))} < \frac{3}{10\chi} < \frac{3}{10} < \frac{3}{5} = R_{d_e}(P_{\chi,\epsilon}). \quad (1)$$

Thus d_p is uniquely selected by raw population prediction risk, whereas d_e uniquely minimizes the arm-normalized source-score profile. Multiplying (1) by the paired costs gives

$$B\Psi_{\text{CLW},d_p}(P_{\chi,\epsilon}; B) = \frac{3}{2} \frac{1 - \epsilon^2}{1 - \epsilon^2 y_\chi^2} < \frac{3}{2} < 3 = B\Psi_{\text{CLW},d_e}(P_{\chi,\epsilon}; B). \quad (2)$$

The same lemma gives

$$V_{O,d_p} = \frac{64}{75(10\chi - 3)}\epsilon^{-2} + O_\chi(1). \quad (3)$$

At every center, treatment is balanced in both sources. Lemma `lem:binary-ikmw-normalized-stability` therefore gives

$$V_{E,d}^{\text{IKM}} = \frac{1}{4}V_{E,d}, \quad V_{O,d}^{\text{IKM}} = \frac{1}{4}V_{O,d}, \quad \mathcal{V}_{d,\text{IKM}}^*(P_{\chi,\epsilon}) = \frac{1}{4}\mathcal{V}_d^*(P_{\chi,\epsilon}). \quad (4)$$

It follows that d_e also uniquely minimizes the IKM-normalized profile and that the common factor cancels from the ratios. Lemma `lem:sharp-uniform-binary-elbow` supplies

$$\rho_{\text{pred}}^{\text{IKM}}(P_{\chi,\epsilon}) = \rho_{\text{pred}}(P_{\chi,\epsilon}) = \frac{5\chi + 1}{363(10\chi - 3)}\epsilon^{-2} + O_\chi(\epsilon^{-1}), \quad (5)$$

as well as

$$\chi_{\text{opt}}(\epsilon) \sim \frac{\sqrt[3]{18}}{15}\epsilon^{-2/3} \quad (6)$$

and

$$\begin{aligned} \epsilon^2 \inf_{\chi > 1} \rho_{\text{pred}}(P_{\chi,\epsilon}) &= \epsilon^2 \inf_{\chi > 1} \rho_{\text{pred}}^{\text{IKM}}(P_{\chi,\epsilon}) \\ &= \frac{1}{726} + \frac{5\sqrt[3]{12}}{968}\epsilon^{2/3} + o(\epsilon^{2/3}). \end{aligned} \quad (7)$$

Definition `def:binary-region` chooses each ball inside the radius from Lemma `lem:binary-witness-stability`. Hence every member of $\mathcal{U}_{\text{bin}}$ is regular, both bridge contrasts equal τ , and

$$\mathcal{D}_{\text{pred}}(P) = \{d_p\}, \quad D^*(P) = \{d_e\}. \quad (8)$$

Each center belongs to its corresponding ball. Letting $\epsilon \downarrow 0$ in (5), with χ fixed, proves

$$\sup_{P \in \mathcal{U}_{\text{bin}}} \rho_{\text{pred}}(P) = \infty. \quad (9)$$

Definition `def:binary-ikmw-region` further shrinks the balls inside the normalized-stability radii without removing their centers. The normalized-stability lemma gives regularity, common identification, and the stated optimizer assignments throughout that subregion. Equation (5) then gives

$$\sup_{P \in \mathcal{U}_{\text{bin}}^{\text{IKM}}} \rho_{\text{pred}}^{\text{IKM}}(P) = \infty. \quad (10)$$

Under its separately stated premises, Lemma `lem:ikmw-fixed-system-efficiency` supplies the absolute causal-efficiency interpretation of the normalized profile.

It remains to verify the cost-manifold clause. The center primitive cost vector and cap are

$$\left(1, 2, \frac{5\chi - 3}{4}, \frac{5\chi - 3}{4}, \frac{1}{2}, \frac{1}{2}\right), \quad \kappa = \frac{5\chi - 3}{2}. \quad (11)$$

All entries are positive, $b_E < b_O$, the measurement cost of d_p equals the cap, and that of d_e is one. Hence (11) belongs to $\mathcal{C}_\kappa$, and its paired-cost ratio is exactly χ . At $\epsilon = 0$, the continuous extensions of the raw risks are $3/(10\chi)$ and $3/5$, so their gap is strict. The raw-risk gap, the two sourcewise cost gaps, and the cap inequalities are continuous in the relative topology of $\mathcal{C}_\kappa$. Therefore a relatively open cost neighborhood, together with all sufficiently small positive ϵ , preserves strict cost disadvantage and unique raw-risk selection. The secondary diagnostic gap in (2) is also strict and continuous, so it persists after a further shrinking.

On this neighborhood, c_{O,d_p} stays bounded below and both costs of d_e stay bounded above. By (3),

$$\mathcal{V}_{d_p}^*(P_{\chi,\epsilon}) \geq c_{O,d_p} V_{O,d_p} \asymp_\chi \epsilon^{-2},$$

whereas $\mathcal{V}_{d_e}^*$ stays finite and positive. Thus the arm-normalized ratio diverges. The balanced-center factorization in (4) holds for every positive primitive cost vector, so the IKM-normalized ratio diverges as well. Equations (1)–(11) establish every formula, optimizer, neighborhood, and divergence in the theorem. $\square$

Justification. The theorem establishes unbounded inefficiency for the unadjusted population risk on a regular common-identification region; the added two-rule theorem extends the same witness and constants to the treatment-aware risk. *Gap.* Prediction accuracy, even after treatment adjustment, does not control bridge-score sensitivity or cost-normalized efficiency. *Consumer.* It supplies the negative frontier result against which design-value-gap selection is justified.

Lemma 2 (lem:ikmw-fixed-design-limit). *For a fixed short-term-outcome system and source-conditional law P , let $n = n_E + n_O$ and define the artificial superpopulation law P_n° by*

$$P_n^\circ(G = E) = \frac{n_E}{n}, \quad P_n^\circ(G = O) = \frac{n_O}{n}, \quad P_n^\circ(\cdot \mid G = g) = P(\cdot \mid G = g).$$

Every norm in the statement below is $\|\cdot\|_2 = \|\cdot\|_{L_2(P_n^\circ)}$. Use a fixed number $L_0 \geq 2$ of observational-only cross-fitting folds, independent i.i.d. observations within the two mutually independent sources, $n_E, n_O \rightarrow \infty$, and $n_E/n_O \rightarrow \lambda \in (0, \infty)$. Suppose Imbens–Kallus–Mao–Wang Theorem 4 statements 1 and 2, their square-integrable latent outcome and selection bridge premises, the corresponding observed moment equations, and both parts of their Assumption 8 hold with nuisance limits equal to the true bridges h_0 and q_0 . Thus, for every $1 \leq \ell \leq L_0$,

$$\begin{aligned} \|\mathbf{T}_0(\hat{h}^{(-\ell)} - h_0)\|_2 &= O_P(\delta_{h,n}), & \|\hat{h}^{(-\ell)} - h_0\|_2 &= O_P(\rho_{h,n}), \\ \|\mathbf{T}_0^*(\hat{q}^{(-\ell)} - q_0)\|_2 &= O_P(\delta_{q,n}), & \|\hat{q}^{(-\ell)} - q_0\|_2 &= O_P(\rho_{q,n}), \end{aligned}$$

all four rates tend to zero, and

$$\min\{\delta_{h,n}\rho_{q,n}, \rho_{h,n}\delta_{q,n}\} = o(n^{-1/2}).$$

Define the two score moments printed in their Theorem 5, kept distinct from this paper’s arm-normalized $V_{E,d}$ and $V_{O,d}$, by

$$V_E^{\text{IKM}} = \mathbb{E}_P \left[\left\{ \frac{A - P(A = 1 \mid G = E)}{P(A = 1 \mid G = E)} (h_0(S_3, S_2, A, X) - \mu_0(A)) \right\}^2 \mid G = E \right]$$

and

$$V_O^{\text{IKM}} = \mathbb{E}_P \left[\left\{ \frac{A - P(A = 1 \mid G = O)}{P(A = 1 \mid G = O)} q_0(S_2, S_1, A, X) (Y - h_0(S_3, S_2, A, X)) \right\}^2 \mid G = O \right].$$

If these two moments are finite and strictly positive, their Theorem 5 states that

$$\sqrt{n} (\hat{\tau}_{\text{DR}} - \tau_0) \rightsquigarrow \mathcal{N} \left(0, \frac{1 + \lambda}{\lambda} V_E^{\text{IKM}} + (1 + \lambda) V_O^{\text{IKM}} \right).$$

This cited statement records the source’s printed normalization only. It neither identifies V_g^{IKM} with this paper’s $V_{g,d}$ nor supplies the arm-normalized fixed-path expansion used by the post-selection theorem.

Justification. This is a source-faithful comparison node that binds the artificial-superpopulation norm and preserves the source’s printed score normalization. *Gap.* It does not prove the arm-normalized expansion used by this note and is not a substitute for that derivation. *Consumer.* Readers can audit exactly what is imported from the fixed-system paper and what is newly proved here.

Lemma 3 (lem:ikmw-fixed-system-efficiency). *For a fixed short-term-outcome system and a design d , let $V_{E,d}^{\text{IKM}}$ and $V_{O,d}^{\text{IKM}}$ denote the two source-score second moments printed in Imbens–Kallus–Mao–Wang Theorem 5 for the causal target τ . Suppose the conditions of their Theorem 7 hold: their Assumptions 1–4 and 6 define the local semiparametric model; at the law under evaluation, their Assumptions 6 and 7 admit unique square-integrable outcome and selection bridges; the corresponding conditional-expectation operator is bijective; and $V_{E,d}^{\text{IKM}}$ and $V_{O,d}^{\text{IKM}}$ are finite. Suppose also that $n_E, n_O \rightarrow \infty$ and $n_E/n_O \rightarrow \lambda \in (0, \infty)$. Then their Theorem 7 gives the local efficiency lower bound*

$$\sigma_{d,\text{IKM}}^2 = \frac{1 + \lambda}{\lambda} V_{E,d}^{\text{IKM}} + (1 + \lambda) V_{O,d}^{\text{IKM}} = \lim(n_E + n_O) \left(\frac{V_{E,d}^{\text{IKM}}}{n_E} + \frac{V_{O,d}^{\text{IKM}}}{n_O} \right).$$

The proof of their Theorem 7 identifies the full two-source influence function assembled from the experimental and observational source terms as the efficient influence function, equivalently the canonical gradient, for τ ; the two conditional source-score terms are components of that influence function rather than separate canonical gradients.

For attainment, additionally impose the remaining sampling and estimation premises of their Theorem 5: independent identically distributed observations within the two sources, both identification-and-consistency premises in statements 1 and 2 of their Theorem 4, cross-fitting, and both parts of their Assumption 8 with nuisance limits equal to the true bridges, all weak- and strong-metric errors vanishing, and the minimum of the outcome-weak-selection-strong and outcome-strong-selection-weak error products equal to $o((n_E + n_O)^{-1/2})$. Under these premises and the same fixed-ratio regime, Theorem 5 gives asymptotic normality with variance $\sigma_{d,\text{IKM}}^2$, while Theorem 7 identifies that variance as the efficiency lower bound; hence their doubly robust estimator attains the bound for τ . Equivalently, on the original estimator scale, the leading variance is $V_{E,d}^{\text{IKM}}/n_E + V_{O,d}^{\text{IKM}}/n_O$.

Justification. This cited leaf records the published IKM-normalized fixed-ratio efficiency and attainment result, including its explicit λ -dependent variance. *Gap.* The source establishes efficiency for the assembled two-source influence function; it does not make the two conditional source terms separate canonical gradients or justify arbitrary source-ratio sequences. *Consumer.* The binary calculation and every absolute causal-efficiency interpretation use this leaf only after proving the normalization and all causal-identification premises.

Lemma 4 (lem:second-order-allocation-loss). *Let $\mathcal{D}$ be finite and nonempty. Suppose the costs and population variance components obey*

$$0 < \min_{d \in \mathcal{D}} \min\{c_{E,d}, c_{O,d}, V_{E,d}, V_{O,d}\} \leq \max_{d \in \mathcal{D}} \max\{c_{E,d}, c_{O,d}, V_{E,d}, V_{O,d}\} < \infty.$$

Suppose

$$\max_{d \in \mathcal{D}} \max\{|\widehat{V}_{E,d} - V_{E,d}|, |\widehat{V}_{O,d} - V_{O,d}|\} = O_P(r_{V,m}), \quad r_{V,m} = o(1), \quad \underline{v}_m \downarrow 0,$$

and define $\widehat{\alpha}_{d,m}$ by the full-catalogue pilot selector. For every d inside the following maximum, assign the entire floor-count expression the value zero whenever either floor count is zero. Then, along every joint sequence $m \rightarrow \infty$ and $B \rightarrow \infty$,

$$\max_{d \in \mathcal{D}} \left| B \left\{ \frac{V_{E,d}}{[\widehat{\alpha}_{d,m} B / c_{E,d}]} + \frac{V_{O,d}}{[(1 - \widehat{\alpha}_{d,m}) B / c_{O,d}]} \right\} - \mathcal{V}_d^*(P) \right| = O_P(r_{V,m}^2 + B^{-1}).$$

In particular, allocation estimation enters the profiled value only at second order, and the asymptotically inactive truncation contributes no term to the sharp rate.

Proof. Write

$$Z_m = \max_{d \in \mathcal{D}} \max\{|\widehat{V}_{E,d} - V_{E,d}|, |\widehat{V}_{O,d} - V_{O,d}|\}.$$

By hypothesis, $Z_m = O_P(r_{V,m}) = o_P(1)$. Finiteness of $\mathcal{D}$ and the displayed uniform bounds furnish constants $0 < \nu_- < \nu_+ < \infty$ containing every population variance and every cost. On the event

$$\mathcal{E}_m = \{Z_m < \nu_-/2\} \cap \{\underline{v}_m < \nu_-/2\}, \tag{1}$$

whose probability tends to one, every estimated variance is positive and exceeds $\underline{v}_m$. Hence the truncation in $\widehat{\alpha}_{d,m}$ is inactive.

For

$$A_d(x, y) = \frac{\sqrt{c_{E,d}x}}{\sqrt{c_{E,d}x} + \sqrt{c_{O,d}y}},$$

the gradients are uniformly bounded on a common compact rectangle around all population pairs. The mean-value theorem and (1) therefore yield

$$\max_{d \in \mathcal{D}} |\widehat{\alpha}_{d,m} - \alpha_d^*(P)| = O_P(r_{V,m}). \tag{2}$$

The population optimal shares form a finite subset of $(0, 1)$; hence there is $a_0 > 0$ such that, with probability tending to one,

$$a_0 \leq \hat{\alpha}_{d,m} \leq 1 - a_0 \quad (d \in \mathcal{D}). \quad (3)$$

For fixed d , set $a = c_{E,d}V_{E,d}$, $b = c_{O,d}V_{O,d}$, and $F_d(u) = a/u + b/(1-u)$. The budget-profile theorem gives $u_d^* = \alpha_d^*(P) = \sqrt{a}/(\sqrt{a} + \sqrt{b})$ and $\mathcal{V}_d^*(P) = (\sqrt{a} + \sqrt{b})^2$. A direct calculation gives the exact loss identity

$$F_d(u) - \mathcal{V}_d^*(P) = \mathcal{V}_d^*(P) \frac{(u - u_d^*)^2}{u(1-u)}. \quad (4)$$

Combining (2)–(4), uniformly over the finite design family, gives

$$\max_{d \in \mathcal{D}} |F_d(\hat{\alpha}_{d,m}) - \mathcal{V}_d^*(P)| = O_P(r_{V,m}^2). \quad (5)$$

On the event (3), both continuous source counts are at least a positive constant times B . For all sufficiently large B , the floor inequality used in Theorem thm:budget-profile applies uniformly and gives

$$\begin{aligned} 0 &\leq B \left\{ \frac{V_{E,d}}{[\hat{\alpha}_{d,m}B/c_{E,d}]} + \frac{V_{O,d}}{[(1 - \hat{\alpha}_{d,m})B/c_{O,d}]} \right\} - F_d(\hat{\alpha}_{d,m}) \\ &\leq \frac{C}{B} \end{aligned} \quad (6)$$

with one finite constant C for all d . Equations (5)–(6) prove the asserted $O_P(r_{V,m}^2 + B^{-1})$ bound. The stipulated zero convention matters only off an event whose probability tends to zero, so it does not alter this stochastic order. Equation (1) also shows that truncation is asymptotically inactive, which proves the final assertion. $\square$

Justification. The explicit finite-family expansion shows that estimating the square-root allocation changes the profiled value only quadratically, apart from integer rounding. *Gap.* The fixed-ratio anchor does not quantify the cost of learning the source allocation from a pilot. *Consumer.* The pilot designer can separate first-order measurement-system regret from second-order allocation error.

Proposition 2 (prop:postselection-wald-ate-corollary). *For every $P \in \mathfrak{P}_{\mathcal{D}}^{\text{id}}$, under the sampling and maintained estimator-level conditions of Theorem thm:postselection-wald,*

$$\frac{\sqrt{B}(\hat{\tau}_B - \tau)}{\hat{s}_B} \rightsquigarrow \mathcal{N}(0, 1), \quad P\{\tau \in I_B\} \longrightarrow 1 - \eta,$$

with the same total-event zero extension as in that theorem.

Proof. Let $P \in \mathfrak{P}_{\mathcal{D}}^{\text{id}}$. By Definition def:common-identification-class,

$$\theta_d(P) = \tau \quad \text{for every } d \in \mathcal{D}.$$

In particular, $\theta_{\hat{d}_m}(P) = \tau$ for every pilot realization. Therefore the studentized statistic in Theorem thm:postselection-wald is exactly

$$\frac{\sqrt{B}(\hat{\tau}_B - \tau)}{\hat{s}_B}$$

on $\mathbf{G}_{\text{tot}}$, with the same zero extension on its complement. The theorem's normal limit consequently gives the first conclusion. Likewise, on $\mathbf{G}_{\text{tot}}$, its coverage event $\{\theta_{\hat{d}_m}(P) \in I_B\}$ equals $\{\tau \in I_B\}$; the complement of $\mathbf{G}_{\text{tot}}$ has probability tending to zero. Hence the coverage conclusion of that theorem gives

$$P\{\tau \in I_B\} \longrightarrow 1 - \eta.$$

$\square$

Justification. The corollary translates selected-contrast inference to ATE inference exactly on the common-identification class and inherits every maintained condition of the parent Wald theorem. *Gap.* The equality $\theta_d(P) = \tau$ is not available on the whole observable or causal regular class. *Consumer.* Causal analysts obtain the exact additional identification restriction required for ATE coverage, without changing the selected estimator or studentizer.

Theorem 5 (thm:finite-dimensional-selected-path-attainability). *Fix $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$. For each $d \in \mathcal{D}$, write $H_d = (S_{3d}, S_{2d}, A, X)$ and $L_d = (S_{2d}, S_{1d}, A, X)$. Suppose the following fixed-dimensional conditions hold.*

For each (d, a) , there are known, bounded, finite-dimensional bridge bases $b_{d,a}^h(H_d)$ and $b_{d,a}^q(L_d)$, and known bounded test bases $z_{d,a}^h(L_d)$ and $z_{d,a}^q(H_d)$, whose dimensions do not depend on m or B , such that

$$h_d(H_d) = b_{d,a}^h(H_d)^\top \beta_{d,a}, \quad q_d(L_d) = b_{d,a}^q(L_d)^\top \gamma_{d,a} \quad \text{on } \{A = a\}. \quad (1)$$

The square population moment matrices

$$M_{d,a}^h = E_P\{z_{d,a}^h(L_d)b_{d,a}^h(H_d)^\top \mid A = a, G = O\}, \quad M_{d,a}^q = E_P\{z_{d,a}^q(H_d)b_{d,a}^q(L_d)^\top \mid A = a, G = O\} \quad (2)$$

have smallest singular values bounded below by a constant $c_0 > 0$, uniformly over the finite set of (d, a) ; the corresponding basis Gram matrices have eigenvalues in $[c_0, c_0^{-1}]$. The exact moment equations are

$$M_{d,a}^h \beta_{d,a} = E_P\{z_{d,a}^h(L_d)Y \mid A = a, G = O\}, \quad M_{d,a}^q \gamma_{d,a} = E_P\{z_{d,a}^q(H_d) \mid A = a, G = E\}. \quad (3)$$

There are $\delta > 0$ and $C_0 < \infty$ such that all four bases are bounded by C_0 , $E_P(|Y|^{4+\delta} \mid G = O) \leq C_0$, and every source and source-arm probability is at least c_0 . Pilot source-arm sizes, main-study source-arm sizes, and every training and evaluation fold size are proportional to m or B , as appropriate, with fixed L .

On each training sample, form the empirical conditional analogues of (2)–(3) and set

$$\hat{\beta}_{d,a} = \widehat{M}_{d,a}^{h,-1} \hat{g}_{d,a}^h, \quad \hat{\gamma}_{d,a} = \widehat{M}_{d,a}^{q,-1} \hat{g}_{d,a}^q, \quad (4)$$

with value zero when a displayed empirical matrix is singular. Use the resulting fold-external linear bridge fits in the pilot and main study, and use the empirical plug-in source-score second moments specified in Definitions def:pilot-selector and def:selected-wald-variance.

Then, uniformly over the finite catalogue and the fixed folds,

$$\|\hat{h}_{d,B}^{(-\ell)} - h_d\|_{2,B}^\circ + \|T_d(\hat{h}_{d,B}^{(-\ell)} - h_d)\|_{2,O,q,d} = O_P(B^{-1/2}), \quad (5)$$

$$\|\hat{q}_{d,B}^{(-\ell)} - q_d\|_{2,O,q,d} + \|T_d^*(\hat{q}_{d,B}^{(-\ell)} - q_d)\|_{2,B}^\circ = O_P(B^{-1/2}). \quad (6)$$

Both weak-strong products are $O_P(B^{-1}) = o_P(B^{-1/2})$, the multiplier expression in Assumption ass:postselection-multiplier-condition is $O_P(B^{-1/2}) = o_P(1)$, and

$$\max_{d \in \mathcal{D}} \max\{|\widehat{V}_{E,d} - V_{E,d}|, |\widehat{V}_{O,d} - V_{O,d}|\} = O_P(m^{-1/2}), \quad \max\{|\widetilde{V}_{E,\hat{d}_m,B} - V_{E,\hat{d}_m}|, |\widetilde{V}_{O,\hat{d}_m,B} - V_{O,\hat{d}_m}|\} = O_P(B^{-1/2}). \quad (7)$$

Consequently the observational-norm weak-outcome and strong-selection conditions, the retained artificial-law strong-outcome and weak-selection conditions, and all product, multiplier, pilot-variance, and main-study-variance conditions used by Theorems thm:pilot-oracle-regret and thm:postselection-wald are verified on this fixed-dimensional subclass, with $r_{V,m} = m^{-1/2}$, $r_{\widetilde{V},B} = B^{-1/2}$, and all four bridge rates of order $B^{-1/2}$. Hence the oracle-regret, selected-design limit law, and Wald-coverage conclusions of those theorems hold without separately maintaining those rate conditions on this subclass. No such conclusion is asserted for all of $\mathfrak{P}_{\mathcal{D}}^{\text{obs}}$.

Proof. We prove the support-respecting statement. Uniformity below follows because the design set, treatment set, basis dimensions, and fold count are all finite and fixed.

Fix (d, a) , and let n denote a relevant training-sample size. Bounded bases, the $(4 + \delta)$ -moment bound for Y , source-arm positivity, and the proportional cell-size condition imply, entry by entry,

$$\widehat{M}_{d,a}^h - M_{d,a}^h = O_P(n^{-1/2}), \quad \widehat{g}_{d,a}^h - g_{d,a}^h = O_P(n^{-1/2}), \quad (8)$$

and the same bounds hold for $\widehat{M}_{d,a}^q - M_{d,a}^q$ and $\widehat{g}_{d,a}^q - g_{d,a}^q$. Each coordinate has finite variance, so this follows from Chebyshev's inequality after conditioning on the relevant source-arm cell. A finite union bound makes (8) simultaneous over designs, arms, and folds.

If $\|\widehat{M}_{d,a}^h - M_{d,a}^h\|_{\text{op}} < c_0/2$, then, for every vector x ,

$$\|\widehat{M}_{d,a}^h x\| \geq \|M_{d,a}^h x\| - \|(\widehat{M}_{d,a}^h - M_{d,a}^h)x\| \geq \frac{c_0}{2} \|x\|.$$

Thus $\widehat{M}_{d,a}^h$ is nonsingular and its inverse has operator norm at most $2/c_0$. The same argument applies to $\widehat{M}_{d,a}^q$. Equation (8) shows that these events hold simultaneously with probability tending to one. On them, the exact population moment equations give

$$\widehat{\beta}_{d,a} - \beta_{d,a} = (\widehat{M}_{d,a}^h)^{-1} \left\{ (\widehat{g}_{d,a}^h - g_{d,a}^h) - (\widehat{M}_{d,a}^h - M_{d,a}^h) \beta_{d,a} \right\}, \quad (9)$$

and the analogous identity holds for $\widehat{\gamma}_{d,a} - \gamma_{d,a}$. The population coefficient vectors are uniformly bounded: the population inverses have norm at most c_0^{-1} , the outcome right-hand sides are bounded by the moment condition, and the selection right-hand sides are bounded by the basis envelope. Consequently,

$$\max_{d,a,\ell} \left\{ \|\widehat{\beta}_{d,a}^{(-\ell)} - \beta_{d,a}\| + \|\widehat{\gamma}_{d,a}^{(-\ell)} - \gamma_{d,a}\| \right\} = O_P(n^{-1/2}). \quad (10)$$

The stipulated zero convention is asymptotically irrelevant because the joint singularity event has probability tending to zero.

We now translate coefficient rates into the four correctly typed bridge rates. The bounded bases give the pointwise bounds

$$|\widehat{h}_d - h_d| \leq C \|\widehat{\beta}_d - \beta_d\|, \quad |\widehat{q}_d - q_d| \leq C \|\widehat{\gamma}_d - \gamma_d\|. \quad (11)$$

Thus the first bound in (11) gives the retained source-defined strong-outcome rate

$$\|\widehat{h}_{d,B}^{(-\ell)} - h_{d,B}^\circ\|_{2,B} = O_P(B^{-1/2}). \quad (12)$$

On observational support, conditional expectation is an L_2 -contraction, and hence

$$\|T_d(\widehat{h}_{d,B}^{(-\ell)} - h_d)\|_{2,O,q,d} \leq \|\widehat{h}_{d,B}^{(-\ell)} - h_d\|_{2,O,h,d} = O_P(B^{-1/2}). \quad (13)$$

The second bound in (11) directly gives the corrected strong-selection rate

$$\|\widehat{q}_{d,B}^{(-\ell)} - q_d\|_{2,O,q,d} = O_P(B^{-1/2}). \quad (14)$$

Finally, conditional expectation preserves the pointwise envelope in (11), so every source-defined version of $T_d^*(\widehat{q}_d - q_d)$ is bounded in absolute value by $C\|\widehat{\gamma}_d - \gamma_d\|$. It follows under the artificial two-source law that

$$\|T_d^*(\widehat{q}_{d,B}^{(-\ell)} - q_d)\|_{2,B}^\circ = O_P(B^{-1/2}). \quad (15)$$

Equations (12)–(15) prove the proposed displays (5)–(6). In particular, either weak–strong product is $O_P(B^{-1}) = o_P(B^{-1/2})$. Notice that (13)–(14), unlike the rejected claims, never evaluate an observational L_d -object on unrestricted experimental L_d -support.

It remains to verify the multiplier and variance conditions. Bounded bridge bases and (10)–(11) imply, uniformly in d, a, ℓ ,

$$|\Delta h_{B,\ell}| = O_P(B^{-1/2}), \quad |\Delta q_{B,\ell}| = O_P(B^{-1/2}) \quad (16)$$

with the stochastic orders multiplied by fixed deterministic envelopes. The population selection bridge is bounded because its basis and coefficient are bounded, while the outcome moment assumption makes $Y - h_d$ square-integrable. Therefore

$$\|\mathbf{1}\{G = O\} \Delta q_{B,\ell} (Y - h_d)\|_{2,B}^\circ + \|\mathbf{1}\{G = O\} q_d \Delta h_{B,\ell}\|_{2,B}^\circ = O_P(B^{-1/2}), \quad (17)$$

and

$$\|\mathbf{1}\{G = O\} \Delta q_{B,\ell} \Delta h_{B,\ell}\|_{2,B}^\circ = O_P(B^{-1}). \quad (18)$$

Thus the multiplier expression is $O_P(B^{-1/2}) = o_P(1)$.

For either source, let $\phi_{g,d}$ be the population score and $\widehat{\phi}_{g,d}$ its fold-external plug-in version. Empirical arm probabilities are root- n consistent and bounded away from zero with probability tending to one. On that event, the score is locally Lipschitz in the finite coefficient vectors and arm probabilities. Equations (10)–(11), together with $E_P(|Y|^{4+\delta} | G = O) < \infty$, give

$$\|\widehat{\phi}_{g,d} - \phi_{g,d}\|_2 = O_P(n^{-1/2}), \quad \|\widehat{\phi}_{g,d} + \phi_{g,d}\|_2 = O_P(1), \quad E_P(\phi_{g,d}^4 | G = g) < \infty. \quad (19)$$

Conditional on the training data for an evaluation fold,

$$|\mathbb{P}_n \hat{\phi}_{g,d}^2 - P \phi_{g,d}^2| \leq |(\mathbb{P}_n - P) \phi_{g,d}^2| + \mathbb{P}_n \left[|\hat{\phi}_{g,d} - \phi_{g,d}| |\hat{\phi}_{g,d} + \phi_{g,d}| \right]. \quad (20)$$

The first term is $O_P(n^{-1/2})$ by Chebyshev's inequality applied to $\phi_{g,d}^2$. Conditional Cauchy-Schwarz, (19), and the law of large numbers make the second term $O_P(n^{-1/2})$. Fixed-fold averaging and a finite union bound preserve the rate. Taking $n \asymp m$ in the full-catalogue pilot and $n \asymp B$ in the main study yields

$$\max_{d \in \mathcal{D}} \max \{ |\hat{V}_{E,d} - V_{E,d}|, |\hat{V}_{O,d} - V_{O,d}| \} = O_P(m^{-1/2}) \quad (21)$$

and

$$\max_{g \in \{E, O\}} |\tilde{V}_{g, \hat{d}_m, B} - V_{g, \hat{d}_m}| = O_P(B^{-1/2}). \quad (22)$$

Equations (12)–(18) and (21)–(22) verify, on this fixed-dimensional subclass, the corrected learner-rate conditions and every product, multiplier, pilot-variance, and main-study-variance condition named in Theorems thm:pilot-oracle-regret and thm:postselection-wald, with $r_{V,m} = m^{-1/2}$, $r_{\tilde{V},B} = B^{-1/2}$, and all four bridge rates of order $B^{-1/2}$. Applying those theorems gives the asserted oracle regret, selected-design normal limit, and Wald coverage. The proof uses the additional fixed-dimensional conditions stated here and therefore makes no claim over all of $\mathfrak{P}_{\mathcal{D}}^{\text{obs}}$. $\square$

Justification. Derives every maintained selected-path rate from fixed-dimensional coefficient root rates, using observational norms for the weak outcome and strong selection errors. *Gap.* This closes attainability only for exact fixed bases and makes no growing-sieve or unrestricted-learner claim. *Consumer.* Supplies a concrete primitive subclass on which pilot regret and post-selection Wald conclusions are operational.

Lemma 5 (lem:binary-witness-stability). *Fix any prescribed $\chi > 1$ and use Definition def:binary-witness. For every $0 < \epsilon < 10^{-3}$, $P_\epsilon \in \mathfrak{P}_{\mathcal{D}}$, both admissible bridge contrasts equal $\tau = 0$, the unique raw population prediction-risk optimizer is d_p , and the unique arm-normalized design-value optimizer is d_e . More precisely,*

$$R_{d_p}(P_\epsilon) = \frac{3}{10\chi} \frac{1 - \epsilon^2}{1 - \epsilon^2(1 - 3/(10\chi))} < \frac{3}{10\chi} < \frac{3}{10} < \frac{3}{5} = R_{d_e}(P_\epsilon).$$

The paired costs and the secondary cost-weighted prediction objectives satisfy

$$c_{E,d_p} + c_{O,d_p} = 5\chi, \quad c_{E,d_e} + c_{O,d_e} = 5,$$

$$B\Psi_{\text{CLW},d_p}(P_\epsilon; B) = \frac{3}{2} \frac{1 - \epsilon^2}{1 - \epsilon^2 y_\chi^2} < \frac{3}{2} < 3 = B\Psi_{\text{CLW},d_e}(P_\epsilon; B).$$

Moreover,

$$V_{O,d_p} = \frac{64}{75(10\chi - 3)} \epsilon^{-2} + \frac{896}{75(10\chi - 3)}, \quad \mathcal{V}_{d_e}^*(P_\epsilon) = \frac{3872}{25},$$

and, as $\epsilon \downarrow 0$ with χ fixed,

$$\rho_{\text{pred}}(P_\epsilon) = \frac{5\chi + 1}{363(10\chi - 3)} \epsilon^{-2} + O_\chi(\epsilon^{-1}).$$

For each fixed pair (χ, ϵ) , there is a radius $r_\epsilon > 0$ such that every law $P \in \mathcal{M}_{\text{bin}}(\chi)$ with $\|P - P_\epsilon\|_{\text{cell}} < r_\epsilon$ is regular, has the same strict raw-risk and arm-normalized optimizer assignments, has both bridge contrasts equal to τ , and satisfies $\rho_{\text{pred}}(P) > \rho_{\text{pred}}(P_\epsilon)/2$.

Proof. Fix $\chi > 1$ and $0 < \epsilon < 10^{-3}$. Lemma `lem:sharp-uniform-binary-elbow` proves that $P_\epsilon \in \mathfrak{P}_{\mathcal{D}}$, both bridge contrasts equal $\tau = 0$, and the raw-risk and arm-normalized optimizers are uniquely d_p and d_e , respectively. It also proves the displayed raw-risk chain and

$$V_{O,d_p} = \frac{64}{75(10\chi - 3)}(\epsilon^{-2} + 14) = \frac{64}{75(10\chi - 3)}\epsilon^{-2} + \frac{896}{75(10\chi - 3)}, \quad (1)$$

$$\mathcal{V}_{d_e}^*(P_\epsilon) = \frac{3872}{25}, \quad \rho_{\text{pred}}(P_\epsilon) = \frac{5\chi + 1}{363(10\chi - 3)}\epsilon^{-2} + O_\chi(\epsilon^{-1}). \quad (2)$$

The cost identities in Definition `def:binary-witness` give

$$c_{E,d_p} + c_{O,d_p} = 5\chi, \quad c_{E,d_e} + c_{O,d_e} = 5. \quad (3)$$

Multiplying the exact residual risks by these paired costs and using $y_\chi^2 = 1 - 3/(10\chi)$ yields

$$B\Psi_{\text{CLW},d_p}(P_\epsilon; B) = \frac{3}{2} \frac{1 - \epsilon^2}{1 - \epsilon^2 y_\chi^2} < \frac{3}{2} < 3 = B\Psi_{\text{CLW},d_e}(P_\epsilon; B), \quad (4)$$

because $1 - \epsilon^2 < 1 - \epsilon^2 y_\chi^2$.

We now prove the neighborhood assertion. The center is in the relative interior of the finite-cell parameterization of $\mathcal{M}_{\text{bin}}(\chi)$. Positivity of cells, source and treatment overlap, and nonzero determinants of the finite conditional-expectation matrices are open conditions. Matrix inversion is continuous where the determinant is nonzero, so the two bridges, their contrasts, and their source-score variances are continuous on a sufficiently small ball. The variances remain finite and strictly positive there. The constrained model keeps $Y(1) = Y(0) = U$, randomized treatment in source E , source-invariant proxy kernels, and conditional proxy independence. Thus its causal restrictions persist, and the bridge argument using completeness gives both bridge contrasts equal to the common $\tau = 0$ throughout the ball.

The two raw risks are continuous finite-cell conditional-variance functionals. The profiled values are continuous compositions of the positive costs and source-score variances. Since both optimizer gaps at P_ϵ are strict, a sufficiently small radius preserves the two unique optimizer assignments. Finally, the denominator defining ρ_{pred} stays positive and both numerator and denominator are continuous. Because $\rho_{\text{pred}}(P_\epsilon) > 0$, shrinking the radius once more ensures

$$\rho_{\text{pred}}(P) > \frac{1}{2}\rho_{\text{pred}}(P_\epsilon).$$

Taking the minimum of these finitely many positive radii gives the asserted r_ϵ . $\square$

Justification. Transfers the strict raw-risk and source-score optimizer separation from each center to a relatively open regular common-identification neighborhood. *Gap.* A center-only calculation would not establish a nondegenerate law region or rule out identification failure as the source of inefficiency. *Consumer.* Supplies the stable witness regions used by the raw-risk impossibility theorem.

Lemma 6 (`lem:sharp-uniform-binary-elbow`). *Fix $\chi > 1$ and $0 < \epsilon < 10^{-3}$, and write $P_{\chi,\epsilon} = P_\epsilon$ for the binary center in Definition `def:binary-witness`, with the dependence on χ made explicit. Then $P_{\chi,\epsilon} \in \mathfrak{P}_{\mathcal{D}}$, both designwise bridge contrasts equal $\tau = 0$, and*

$$R_{d_p}(P_{\chi,\epsilon}) = \frac{3}{10\chi} \frac{1 - \epsilon^2}{1 - \epsilon^2(1 - 3/(10\chi))} < \frac{3}{10\chi} < \frac{3}{10} < \frac{3}{5} = R_{d_e}(P_{\chi,\epsilon}).$$

Consequently $\mathcal{D}_{\text{pred}}(P_{\chi,\epsilon}) = \{d_p\}$ and $D^(P_{\chi,\epsilon}) = \{d_e\}$. The raw-risk-selected design is costlier in each source:*

$$c_{E,d_p} - c_{E,d_e} = c_{O,d_p} - c_{O,d_e} = \frac{5(\chi - 1)}{2} > 0.$$

Moreover,

$$V_{E,d_p} = \frac{40\chi}{10\chi - 3}, \quad V_{O,d_p} = \frac{64}{75(10\chi - 3)}(\epsilon^{-2} + 14), \quad \mathcal{V}_{d_e}^*(P_{\chi,\epsilon}) = \frac{3872}{25},$$

and the exact center ratio is

$$\begin{aligned}\rho_{\text{pred}}(P_{\chi,\epsilon}) &= \rho_{\text{pred}}^{\text{IKM}}(P_{\chi,\epsilon}) \\ &= \frac{25}{3872} \left\{ \sqrt{\frac{20\chi(5\chi-1)}{10\chi-3}} + \sqrt{\frac{32(5\chi+1)}{75(10\chi-3)} (\epsilon^{-2} + 14)} \right\}^2.\end{aligned}\quad (1)$$

Consequently,

$$\rho_{\text{pred}}(P_{\chi,\epsilon}) \geq \frac{5\chi+1}{363(10\chi-3)} (\epsilon^{-2} + 14) > \frac{\epsilon^{-2} + 14}{726}, \quad (2)$$

with the identical inequalities for $\rho_{\text{pred}}^{\text{IKM}}$, and

$$\inf_{\chi>1} \rho_{\text{pred}}(P_{\chi,\epsilon}) = \inf_{\chi>1} \rho_{\text{pred}}^{\text{IKM}}(P_{\chi,\epsilon}) > \frac{\epsilon^{-2} + 14}{726}. \quad (3)$$

For all sufficiently small ϵ , the infimum in (3) is attained. If $\chi_{\text{opt}}(\epsilon)$ is any minimizer and

$$k_{\star} = \frac{\sqrt[3]{18}}{15},$$

then

$$\chi_{\text{opt}}(\epsilon) \sim k_{\star} \epsilon^{-2/3}. \quad (4)$$

Equivalently, on writing $\chi = k\epsilon^{-2/3}$, the coefficient of $\epsilon^{2/3}$ inside the braces-squared expression before multiplication by $25/3872$ is

$$q(k) = 2\sqrt{\frac{32k}{15}} + \frac{8}{75k}, \quad (5)$$

and q has the unique minimizer $k_{\star}$. The sharp second-order uniform elbow is

$$\begin{aligned}\epsilon^2 \inf_{\chi>1} \rho_{\text{pred}}(P_{\chi,\epsilon}) &= \epsilon^2 \inf_{\chi>1} \rho_{\text{pred}}^{\text{IKM}}(P_{\chi,\epsilon}) \\ &= \frac{1}{726} + \frac{5\sqrt[3]{12}}{968} \epsilon^{2/3} + o(\epsilon^{2/3}).\end{aligned}\quad (6)$$

In particular, the first-order limit in (6) is $1/726$.

Proof. Fix $\chi > 1$, $0 < \epsilon < 10^{-3}$, and write $x = \epsilon$ and $y = y_{\chi}$. The center construction has positive finite cells, consistency by construction, randomized treatment in source E , latent exchangeability and overlap in source O , source-invariant conditional proxy kernels, and conditional independence of the early and late proxies given U . Hence external validity and the sequential-outcome restriction hold. For a binary proxy W_s , the two conditional laws given $U = 1$ and $U = -1$ differ whenever $s \neq 0$; the associated two-by-two conditional-expectation matrix therefore has nonzero determinant. Since $x, y, 1/2 \neq 0$ and every arm has positive mass, both completeness conditions and both designwise operator-bijectivity conditions hold. The displayed bridges solve the observed equations, and their finite scores have strictly positive variances. Thus $P_{\chi,\epsilon} \in \mathfrak{P}_{\mathcal{D}}$.

Because $Y(1) = Y(0) = U$, $\tau = 0$. Completeness and the sequential restriction give

$$\mathbb{E}_P\{h_d(S_{3d}, S_{2d}, a, X) \mid U, A = a, G = O\} = U.$$

External validity and experimental randomization then imply $\mu_{d,a} = \mathbb{E}_P(U) = 0$, so both bridge contrasts equal τ .

The residual-risk formula from the binary construction and $y_{\chi}^2 = 1 - 3/(10\chi)$ give

$$R_{d_p}(P_{\chi,\epsilon}) = \frac{3}{10\chi} \frac{1 - \epsilon^2}{1 - \epsilon^2(1 - 3/(10\chi))}. \quad (1)$$

The denominator in the last fraction minus its numerator is $3\epsilon^2/(10\chi) > 0$, so the fraction is strictly below one. For d_e , direct substitution gives

$$R_{d_e}(P_{\chi,\epsilon}) = \frac{(1 - 1/4)^2}{1 - 1/16} = \frac{3}{5}. \quad (2)$$

Since $\chi > 1$, (1)–(2) prove the displayed strict raw-risk chain and $\mathcal{D}_{\text{pred}}(P_{\chi,\epsilon}) = \{d_p\}$.

For a design with early signal x and late signal y , the outcome bridge is $h(W_y, a) = W_y/y$. At the balanced experimental center,

$$\phi_{E,d} = 2T \frac{W_y}{y}, \quad V_{E,d} = \frac{4}{y^2}. \quad (3)$$

In source O ,

$$q_d(W_x, a) = \frac{16}{15} \left(1 - \frac{TW_x}{4x} \right).$$

Conditional on U , the residual $U - W_y/y$ has mean zero and variance $(1 - y^2)/y^2$. Also $\mathbb{E}_P(TW_x | G = O) = x/4$, whence direct expansion gives

$$\mathbb{E}_P(q_d^2 | G = O) = \frac{16(1 + 14x^2)}{225x^2}, \quad V_{O,d} = \frac{64(1 - y^2)(1 + 14x^2)}{225x^2y^2}. \quad (4)$$

For d_p , (3)–(4) yield

$$V_{E,d_p} = \frac{40\chi}{10\chi - 3}, \quad V_{O,d_p} = \frac{64}{75(10\chi - 3)}(\epsilon^{-2} + 14). \quad (5)$$

For d_e , they yield $V_{E,d_e} = 16$ and $V_{O,d_e} = 384/25$. Theorem thm:budget-profile therefore gives

$$\mathcal{V}_{d_e}^*(P_{\chi, \epsilonpsilon}) = \left(\sqrt{2 \cdot 16} + \sqrt{3 \cdot 384/25} \right)^2 = \frac{3872}{25}. \quad (6)$$

The cost formulas give the two stated sourcewise gaps. Combining (5)–(6) with the profiled formula gives exactly equation (1) in the statement. At the centers, both treatment probabilities equal $1/2$, so every IKM-normalized source moment is one quarter of its arm-normalized counterpart. Every profiled value is multiplied by one quarter and the ratio is unchanged.

Dropping the first nonnegative square-root term from the exact ratio gives

$$\rho_{\text{pred}}(P_{\chi, \epsilonpsilon}) \geq \frac{5\chi + 1}{363(10\chi - 3)}(\epsilon^{-2} + 14) > \frac{\epsilon^{-2} + 14}{726}, \quad (7)$$

because $(5\chi + 1)/(10\chi - 3) > 1/2$. Since $\epsilon < 10^{-3}$, the lower bound exceeds one; hence d_e uniquely minimizes both profiles. This proves the ratio identities and inequalities.

For the sharp optimization, set

$$A(\chi) = \frac{20\chi(5\chi - 1)}{10\chi - 3}, \quad D(\chi) = \frac{32(5\chi + 1)}{75(10\chi - 3)}, \quad C_0 = \frac{25}{3872}.$$

The exact ratio implies

$$\epsilon^2 \rho_{\text{pred}}(P_{\chi, \epsilonpsilon}) = C_0 \left[\epsilon^2 A(\chi) + D(\chi)(1 + 14\epsilon^2) + 2\epsilon \sqrt{1 + 14\epsilon^2} \sqrt{A(\chi)D(\chi)} \right], \quad (8)$$

where

$$A(\chi) = 10\chi + 1 + \frac{3}{10\chi - 3}, \quad D(\chi) = \frac{16}{75} + \frac{16}{15(10\chi - 3)}. \quad (9)$$

The objective is continuous on $(1, \infty)$ and diverges as $\chi \rightarrow \infty$ for fixed $\epsilon > 0$. A candidate $\chi = k\epsilon^{-2/3}$ has leading limit $C_0(16/75) = 1/726$, whereas every bounded sequence of χ 's has a strictly larger leading constant $C_0 D(\chi)$. Hence, for all sufficiently small ϵ , the infimum is attained at an interior point and every minimizing sequence diverges.

Writing $t = \epsilon^{2/3}$ and $\chi = k/t$, (8)–(9) give, locally uniformly for $k \in (0, \infty)$,

$$\epsilon^2 \rho_{\text{pred}}(P_{k/t, \epsilonpsilon}) = \frac{1}{726} + C_0 t \left\{ 2\sqrt{\frac{32k}{15}} + \frac{8}{75k} \right\} + o(t). \quad (10)$$

The rescaled minimizers stay in a compact subset of $(0, \infty)$: as $k \downarrow 0$, the positive excess from $D(k/t)$, divided by t , diverges like $8/(75k)$, while as $k \rightarrow \infty$, the cross term divided by t diverges like $2\sqrt{32k/15}$. Let

$$q(k) = 2\sqrt{\frac{32k}{15}} + \frac{8}{75k}.$$

Then

$$q'(k) = \sqrt{\frac{32}{15}k^{-1/2} - \frac{8}{75}k^{-2}},$$

which changes sign once, at $k_\star = \sqrt[3]{18}/15$. Local uniformity and compactness yield $\chi_{\text{opt}}(\epsilon) \sim k_\star \epsilon^{-2/3}$. Finally,

$$C_0 q(k_\star) = \frac{5\sqrt[3]{12}}{968},$$

which proves the sharp second-order expansion. The common one-quarter IKM normalization proves the same expansion for $\rho_{\text{pred}}^{\text{IKM}}$. $\square$

Justification. This lemma supplies the exact two-design score calculations and the sharp cost-disadvantage optimization. Its treatment-aware companion proves that adding treatment to the population prediction regression leaves every ratio and sharp constant unchanged. *Gap.* The exact elbow shows that even optimizing the witness cost disadvantage cannot remove the leading ϵ^{-2} divergence. *Consumer.* It feeds both population impossibility theorems, the joint stability construction, and the feasible triangular-pilot result.

Lemma 7 (lem:binary-ikmw-normalized-stability). *Fix any prescribed $\chi > 1$. Under the IKM score and profile normalization, every binary center law satisfies, for $d \in \mathcal{D}$,*

$$\psi_{E,d}^{\text{IKM}} = \frac{1}{2}\phi_{E,d}, \quad \psi_{O,d}^{\text{IKM}} = \frac{1}{2}\phi_{O,d}, \quad V_{E,d}^{\text{IKM}} = \frac{1}{4}V_{E,d}, \quad V_{O,d}^{\text{IKM}} = \frac{1}{4}V_{O,d},$$

$$\mathcal{V}_{d,\text{IKM}}^*(P_\epsilon) = \frac{1}{4}\mathcal{V}_d^*(P_\epsilon).$$

Consequently $D_{\text{IKM}}^*(P_\epsilon) = \{d_e\}$ and $\rho_{\text{pred}}^{\text{IKM}}(P_\epsilon) = \rho_{\text{pred}}(P_\epsilon)$. For each fixed $0 < \epsilon < 10^{-3}$, there is $r_\epsilon^{\text{IKM}} > 0$ such that every law $P \in \mathcal{M}_{\text{bin}}(\chi)$ with $\|P - P_\epsilon\|_{\text{cell}} < r_\epsilon^{\text{IKM}}$ is regular, both admissible bridge contrasts equal τ , $\mathcal{D}_{\text{pred}}(P) = \{d_p\}$, $D_{\text{IKM}}^*(P) = \{d_e\}$, and

$$\rho_{\text{pred}}^{\text{IKM}}(P) > \frac{1}{2}\rho_{\text{pred}}^{\text{IKM}}(P_\epsilon).$$

On this neighborhood, the IKM-normalized profile is the profiled local causal semiparametric efficiency bound under the fixed-system conditions in Lemma lem:ikmw-fixed-system-efficiency.

Proof. At every binary center, $P(A = 1 \mid G = E) = P(A = 1 \mid G = O) = 1/2$. The definitions of the IKM-normalized scores therefore give, design by design,

$$\psi_{E,d}^{\text{IKM}} = \frac{1}{2}\phi_{E,d}, \quad \psi_{O,d}^{\text{IKM}} = \frac{1}{2}\phi_{O,d}.$$

Taking conditional second moments gives

$$V_{E,d}^{\text{IKM}} = \frac{1}{4}V_{E,d}, \quad V_{O,d}^{\text{IKM}} = \frac{1}{4}V_{O,d}. \quad (1)$$

The positive homogeneity of the profiled formula, or direct substitution into Definition def:ikmw-normalized-design-value, now yields

$$\mathcal{V}_{d,\text{IKM}}^*(P_\epsilon) = \left\{ \sqrt{c_{E,d}V_{E,d}/4} + \sqrt{c_{O,d}V_{O,d}/4} \right\}^2 = \frac{1}{4}\mathcal{V}_d^*(P_\epsilon). \quad (2)$$

Lemma lem:sharp-uniform-binary-elbow shows that d_e is the unique arm-normalized minimizer and d_p the unique raw-risk minimizer. Equation (2) preserves the first ordering, and its common factor cancels from the best-tie ratio. Hence

$$D_{\text{IKM}}^*(P_\epsilon) = \{d_e\}, \quad \rho_{\text{pred}}^{\text{IKM}}(P_\epsilon) = \rho_{\text{pred}}(P_\epsilon). \quad (3)$$

For stability, start with the regularity and common-identification ball furnished by Lemma lem:binary-witness-stability. Within this finite-cell ball, source-arm probabilities, bridge coefficients, score second moments, and therefore all IKM-normalized profiled values are continuous functions of the

positive cell vector. At the center, the raw-risk gap, the IKM-profile gap, and the positive ratio in (3) are strict. Shrinking the ball preserves regularity, both bridge contrasts equal to τ , the two unique optimizer assignments, and

$$\rho_{\text{pred}}^{\text{IKM}}(P) > \frac{1}{2} \rho_{\text{pred}}^{\text{IKM}}(P_\epsilon).$$

This supplies $r_\epsilon^{\text{IKM}} > 0$. Finally, when the fixed-system premises stated in Lemma `lem:ikmw-fixed-system-efficiency` hold, that cited result identifies the IKM score as the canonical gradient for τ . Profiling its two source moments over the budget share therefore gives the asserted local causal semiparametric efficiency bound on this neighborhood. $\square$

Justification. Balanced treatment transfers the raw-risk failure to the IKM score normalization and preserves it on a relatively open neighborhood. *Gap.* The arm-normalized source-score profile and the published IKM normalization must not be conflated. *Consumer.* Supplies the conditional fixed-system causal-efficiency interpretation of the binary failure.

Theorem 6 (`thm:sharp-predictive-operator-certificate`). *Fix $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$, and abbreviate*

$$p_d = R_d(P), \quad v_d = \mathcal{V}_d^*(P).$$

Define the raw-prediction-order relation and its order-restricted design-value constant by

$$\mathcal{O}_P = \{(d, d') \in \mathcal{D}^2 : p_d \leq p_{d'}\}, \quad C_{\text{ord}}(P) = \max_{(d, d') \in \mathcal{O}_P} \frac{v_d}{v_{d'}}. \quad (1)$$

Then $1 \leq C_{\text{ord}}(P) < \infty$, and, for every $C \geq 0$,

$$[p_d \leq p_{d'} \implies v_d \leq C v_{d'} \text{ for all } d, d' \in \mathcal{D}] \iff C \geq C_{\text{ord}}(P). \quad (2)$$

The smallest constant for the actual best-tie raw prediction-risk rule is

$$C_{\text{sel}}(P) = \frac{\min_{d \in \mathcal{D}_{\text{pred}}(P)} v_d}{\min_{d' \in \mathcal{D}} v_{d'}} = \rho_{\text{pred}}(P), \quad 1 \leq C_{\text{sel}}(P) \leq C_{\text{ord}}(P) < \infty. \quad (3)$$

Consequently, for any nonempty subclass $\mathcal{Q} \subseteq \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$, a finite constant uniform over $\mathcal{Q}$ exists for the best-tie rule if and only if $\sup_{P \in \mathcal{Q}} \rho_{\text{pred}}(P) < \infty$; for all pairwise raw-risk-order implications it exists if and only if $\sup_{P \in \mathcal{Q}} C_{\text{ord}}(P) < \infty$. Thus (1), rather than an operator norm, is the weakest exact lawwise comparison for selection by raw population prediction risk. The design-specific operators T_d enter (1) through the bridges and hence through v_d , but no further comparison of the operators is necessary for lawwise finiteness.

More specifically, consider any numerical realization of the handle $\mathbf{H}_{\text{comp}}$ with the following properties. For each d , it supplies a real vector space $\mathcal{H}_d$, a cone $\mathcal{X}_d(P) \subseteq \mathcal{H}_d \setminus \{0\}$ containing a transported law-specific direction $x_d(P)$, a positive prediction quadratic form $Q_{R,d}$ on that cone, and two nonnegative source-score quadratic forms $Q_{E,d}$ and $Q_{O,d}$, with

$$p_d = Q_{R,d}\{x_d(P)\}, \quad v_d = F_d\{x_d(P)\}, \quad (4)$$

where

$$F_d(x) = \left\{ \sqrt{c_{E,d} Q_{E,d}(x)} + \sqrt{c_{O,d} Q_{O,d}(x)} \right\}^2. \quad (5)$$

On the bridge-differentiable submodel class of Definition `def:bridge-differentiable-submodel`, these source scores are gradient components; outside that class, only the source-score interpretation is asserted. Set

$$\Lambda_-(P) = \min_{d \in \mathcal{D}} \inf_{x \in \mathcal{X}_d(P)} \frac{F_d(x)}{Q_{R,d}(x)}, \quad \Lambda_+(P) = \max_{d \in \mathcal{D}} \sup_{x \in \mathcal{X}_d(P)} \frac{F_d(x)}{Q_{R,d}(x)}. \quad (6)$$

The weakest two-sided quadratic-form envelope on the chosen cones is precisely

$$0 < \Lambda_-(P) \leq \Lambda_+(P) < \infty, \quad (7)$$

and it implies the valid, generally conservative certificate

$$\rho_{\text{pred}}(P) \leq C_{\text{ord}}(P) \leq \frac{\Lambda_+(P)}{\Lambda_-(P)}. \quad (8)$$

The sharp constant in (2) or (3) is not, in general, a largest generalized eigenvalue. Indeed,

$$F_d(x) = \inf_{0 < \alpha < 1} \left\{ \frac{c_{E,d} Q_{E,d}(x)}{\alpha} + \frac{c_{O,d} Q_{O,d}(x)}{1 - \alpha} \right\}, \quad (9)$$

so the profiled numerator is generally not a quadratic form. If $\mathcal{H}_d$ is finite dimensional, $Q_{R,d}$ is positive definite, and $\mathcal{X}_d(P) = \mathcal{H}_d \setminus \{0\}$, a largest generalized eigenvalue at a fixed α gives only a one-sided full-space envelope. Selection requires both the upper and lower envelopes in (6), and equality in (8) additionally requires saturation by law-realized, raw-risk-ordered directions. Hence equality with a largest generalized eigenvalue can hold only after adding a common-space representation, a normalization of the positive lower envelope, and the required saturation; none follows from observable-law operator bijectivity or from the raw-risk ordering alone.

Proof. Fix $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$. Positive unit costs, Assumption `ass:nondegenerate-components`, and Theorem `thm:budget-profile` give, for every $d \in \mathcal{D}$,

$$0 < v_d = \left\{ \sqrt{c_{E,d} V_{E,d}} + \sqrt{c_{O,d} V_{O,d}} \right\}^2 < \infty. \quad (10)$$

This is a source-score statement on $\mathfrak{P}_{\mathcal{D}}^{\text{obs}}$. By the gradient interpretation recorded in Definition `def:bridge-differentiable-submodel`, it is a gradient-variance statement only along the declared bridge-differentiable submodels.

Because $\mathcal{D}$ is finite and nonempty, $\mathcal{O}_P$ is finite and nonempty; indeed, it contains (d, d) for every d . By (10), every ratio in (1) is well-defined, positive, and finite. Thus its maximum exists, is finite, and is at least one. If the implication on the left of (2) holds, then for every $(d, d') \in \mathcal{O}_P$, division by $v_{d'} > 0$ yields

$$\frac{v_d}{v_{d'}} \leq C.$$

Taking the maximum gives $C_{\text{ord}}(P) \leq C$. Conversely, if $C \geq C_{\text{ord}}(P)$ and $p_d \leq p_{d'}$, then $(d, d') \in \mathcal{O}_P$, so

$$v_d \leq C_{\text{ord}}(P) v_{d'} \leq C v_{d'}.$$

This proves (2), including the sharpness of $C_{\text{ord}}(P)$.

Choose

$$d_p^* \in \arg \min_{d \in \mathcal{D}_{\text{pred}}(P)} v_d, \quad d^* \in \arg \min_{d \in \mathcal{D}} v_d.$$

Both choices exist because the relevant sets are nonempty subsets of the finite family. Since d_p^* minimizes p_d ,

$$p_{d_p^*} \leq p_{d^*},$$

and hence $(d_p^*, d^*) \in \mathcal{O}_P$. The definitions and (10) give

$$1 \leq \frac{v_{d_p^*}}{v_{d^*}} \leq C_{\text{ord}}(P). \quad (11)$$

The middle quantity in (11) is exactly the displayed $C_{\text{sel}}(P) = \rho_{\text{pred}}(P)$. Any constant bounding the best-tie rule at this law must be at least this ratio after division by $v_{d^*} > 0$, while this ratio itself gives equality. It is therefore the least best-tie constant. Taking suprema over any nonempty $\mathcal{Q}$ proves both uniform characterizations: a finite uniform constant exists exactly when the supremum of the corresponding pointwise least constants is finite. This also proves that the finite preorder maximum in (1), without any additional operator comparison, is the weakest exact lawwise pairwise certificate. The dependence on T_d is already carried by the designwise bridges and the resulting values v_d .

Now impose the stated numerical realization of $\mathbf{H}_{\text{comp}}$. Positivity of $Q_{R,d}$ on $\mathcal{X}_d(P)$ makes every quotient in (6) well-defined. By the definitions of infimum and supremum, (7) is equivalent to the simultaneous inequalities

$$\Lambda_-(P) Q_{R,d}(x) \leq F_d(x) \leq \Lambda_+(P) Q_{R,d}(x) \quad (12)$$

for every $d \in \mathcal{D}$ and $x \in \mathcal{X}_d(P)$, with common strictly positive finite endpoints. Conversely, any common two-sided envelope with constants $0 < L \leq U < \infty$ implies $L \leq \Lambda_-(P)$ and $\Lambda_+(P) \leq U$. Thus no larger common lower

constant than $\Lambda_-(P)$, and no smaller common upper constant than $\Lambda_+(P)$, can work. Since every cone is nonempty, its quotient range is nonempty, and hence $\Lambda_-(P) \leq \Lambda_+(P)$. This proves that (7) is precisely the weakest existence condition for the asserted two-sided envelope and that (6) gives its sharp endpoints.

For any $(d, d') \in \mathcal{O}_P$, apply (12) at the transported law-specific directions in (4). The raw-risk ordering gives

$$\begin{aligned} v_d &\leq \Lambda_+(P)Q_{R,d}\{x_d(P)\} = \Lambda_+(P)p_d \\ &\leq \Lambda_+(P)p_{d'} = \Lambda_+(P)Q_{R,d'}\{x_{d'}(P)\} \\ &\leq \frac{\Lambda_+(P)}{\Lambda_-(P)}v_{d'}. \end{aligned} \tag{13}$$

Maximizing (13) over $\mathcal{O}_P$, and then using (11), proves (8). The exact finite-order argument above did not divide by p_d , so prediction risks may vanish there. Strict positivity is required only for the conditional quotient representation in (6).

It remains to establish the profiling and generalized-eigenvalue assertions. Put

$$u = c_{E,d}Q_{E,d}(x), \quad w = c_{O,d}Q_{O,d}(x).$$

When $u, w > 0$, direct algebra gives

$$\frac{u}{\alpha} + \frac{w}{1-\alpha} - (\sqrt{u} + \sqrt{w})^2 = \frac{\{(1-\alpha)\sqrt{u} - \alpha\sqrt{w}\}^2}{\alpha(1-\alpha)} \geq 0. \tag{14}$$

Equality holds at $\alpha = \sqrt{u}/(\sqrt{u} + \sqrt{w})$. If exactly one of u, w is zero, the same infimum is obtained by taking the appropriate endpoint limit; if both are zero, both sides vanish. Therefore (14) proves (9) for all nonnegative u, w .

The infimum in (9) need not be a quadratic form. For an explicit witness, take $\mathcal{H}_d = \mathbb{R}^2$, unit costs, and

$$Q_{E,d}(x) = x_1^2, \quad Q_{O,d}(x) = x_2^2.$$

Then $F_d(x) = (|x_1| + |x_2|)^2$. With $e_1 = (1, 0)$ and $e_2 = (0, 1)$,

$$F_d(e_1 + e_2) + F_d(e_1 - e_2) = 8, \quad 2F_d(e_1) + 2F_d(e_2) = 4,$$

so the parallelogram identity fails and F_d is not quadratic. A largest generalized eigenvalue for a fixed α can therefore control only the upper quotient associated with that fixed-share quadratic form. The selection comparison in (13) also needs a strictly positive lower envelope. Equality further requires transported, law-realized directions that attain the relevant upper and lower envelopes while obeying the raw-risk order. Comparing design-specific spaces by a single eigenvalue additionally requires a common-space representation and a normalization of the lower envelope. Observable-law operator bijectivity and raw-risk ordering imply none of those extra properties. This proves all claims. $\square$

Justification. Gives exact finite-catalogue constants for raw-risk ordering and separates them from conservative quadratic-form envelopes. *Gap.* A generalized eigenvalue is neither automatically sharp nor sufficient after budget profiling. *Consumer.* Analysts can compute exact lawwise constants by finite enumeration and audit any proposed operator envelope.

Theorem 7 (thm:finite-catalogue-predictive-guarantee). *For every $P \in \mathfrak{P}_D^{\text{obs}}$, the exact fixed-catalogue raw-risk-selection identity is*

$$1 \leq \rho_{\text{pred}}(P) = \frac{\min_{d \in \mathcal{D}_{\text{pred}}(P)} (\sqrt{c_{E,d}}V_{E,d} + \sqrt{c_{O,d}}V_{O,d})^2}{\min_{d' \in \mathcal{D}} (\sqrt{c_{E,d'}}V_{E,d'} + \sqrt{c_{O,d'}}V_{O,d'})^2} < \infty.$$

The smallest constant in the actual best-tie inequality is exactly $\rho_{\text{pred}}(P)$. The least constant that makes every pairwise implication $R_d(P) \leq R_{d'}(P) \Rightarrow \mathcal{V}_d^(P) \leq C(P)\mathcal{V}_{d'}^*(P)$ valid is*

$$\max_{d, d' \in \mathcal{D}: R_d(P) \leq R_{d'}(P)} \frac{\mathcal{V}_d^*(P)}{\mathcal{V}_{d'}^*(P)}.$$

Every displayed constant is finite at a fixed P . Nevertheless, for every prescribed finite $\chi > 1$, the χ -indexed binary region satisfies $\sup_{P \in \mathcal{U}_{\text{bin}}} \rho_{\text{pred}}(P) = \infty$, even though the uniquely raw-risk-selected design has paired cost χ times

that of the arm-normalized source-score-optimal design at the center laws. Across all center cost-disadvantage factors, every minimizing sequence satisfies

$$\chi_{\text{opt}}(\epsilon) \sim \frac{\sqrt[3]{18}}{15} \epsilon^{-2/3},$$

and the exact least-divergent envelope obeys

$$\begin{aligned} \epsilon^2 \inf_{\chi > 1} \rho_{\text{pred}}(P_{\chi, \epsilon}) &= \epsilon^2 \inf_{\chi > 1} \rho_{\text{pred}}^{\text{IKM}}(P_{\chi, \epsilon}) \\ &= \frac{1}{726} + \frac{5\sqrt[3]{12}}{968} \epsilon^{2/3} + o(\epsilon^{2/3}). \end{aligned}$$

Thus finiteness of $\mathcal{D}$ yields an exact law-specific enumeration, not a primitive uniform raw-risk-to-source-score guarantee; optimizing the witness's cost disadvantage does not remove the leading $1/(726\epsilon^2)$ divergence.

Proof. Fix $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$. The family $\mathcal{D}$ is finite and nonempty. Positive unit costs and Assumption `ass:nondegenerate-components` imply, for every d ,

$$0 < c_{E,d} V_{E,d} < \infty, \quad 0 < c_{O,d} V_{O,d} < \infty.$$

Theorem `thm:budget-profile` therefore gives

$$v_d := \mathcal{V}_d^*(P) = (\sqrt{c_{E,d} V_{E,d}} + \sqrt{c_{O,d} V_{O,d}})^2 \in (0, \infty). \quad (1)$$

Definition `def:predictive-benchmark` gives

$$\rho_{\text{pred}}(P) = \frac{\min_{d \in \mathcal{D}_{\text{pred}}(P)} v_d}{\min_{d' \in \mathcal{D}} v_{d'}}. \quad (2)$$

Substitution of (1) proves the displayed identity. The numerator minimizes over a nonempty subset of the denominator's index set, so the ratio is at least one; finiteness and positivity give a finite ratio. Dividing the best-tie inequality by the positive denominator in (2) shows that its least constant is exactly $\rho_{\text{pred}}(P)$.

For the pairwise comparison, set

$$C_{\text{pair}}(P) = \max_{d, d' \in \mathcal{D}: R_d(P) \leq R_{d'}(P)} \frac{v_d}{v_{d'}}. \quad (3)$$

The index set contains all diagonal pairs and is finite; (1) makes every ratio finite. Any constant valid for all raw-risk-order implications must dominate every ratio in (3), hence their maximum. Conversely, the maximum itself makes all implications valid. This proves exactness of (3).

Theorem `thm:predictive-impossibility` proves, for every prescribed $\chi > 1$,

$$\sup_{P \in \mathcal{U}_{\text{bin}}} \rho_{\text{pred}}(P) = \infty, \quad (4)$$

while the uniquely raw-risk-selected center design has paired cost 5χ and the source-score-optimal design has paired cost 5. Their ratio is χ . Lemma `lem:sharp-uniform-binary-elbow` further proves

$$\chi_{\text{opt}}(\epsilon) \sim \frac{\sqrt[3]{18}}{15} \epsilon^{-2/3} \quad (5)$$

and

$$\begin{aligned} \epsilon^2 \inf_{\chi > 1} \rho_{\text{pred}}(P_{\chi, \epsilon}) &= \epsilon^2 \inf_{\chi > 1} \rho_{\text{pred}}^{\text{IKM}}(P_{\chi, \epsilon}) \\ &= \frac{1}{726} + \frac{5\sqrt[3]{12}}{968} \epsilon^{2/3} + o(\epsilon^{2/3}). \end{aligned} \quad (6)$$

Equations (1)–(3) establish exact lawwise enumeration. Equations (4)–(6) show that neither finiteness of the catalogue nor optimization of the witness's cost disadvantage gives a uniform raw-risk-to-source-score approximation guarantee. $\square$

Justification. The theorem distinguishes exact fixed-law enumeration from a uniform guarantee. Its two-rule companion supplies the treatment-aware and IKM-normalized enumerations. *Gap.* Finiteness of the catalogue guarantees evaluability at a fixed law but no law-uniform approximation factor for either prediction ordering. *Consumer.* It gives downstream readers the exact constants to compute and the condition whose uniform boundedness would be needed on a restricted law class.

Lemma 8 (lem:treatment-aware-sharp-binary-elbow). *Fix $\chi > 1$ and $0 < \epsilon < 10^{-3}$. For a binary center design having early and late proxy signals $x, y \in (0, 1)$, its treatment-aware observational prediction risk is*

$$\mathcal{R}_A(x, y) = \frac{15(1-x^2)(1-y^2)(16-x^2-y^2-14x^2y^2)}{\{16(1+xy)^2 - (x+y)^2\}\{16(1-xy)^2 - (x-y)^2\}}. \quad (1)$$

Consequently,

$$R_{d_e}^A(P_{\chi, \epsilon}) = \frac{585}{1024}, \quad R_{d_p}^A(P_{\chi, \epsilon}) \leq R_{d_p}(P_{\chi, \epsilon}) < \frac{3}{10\chi} < \frac{3}{10} < \frac{585}{1024}. \quad (2)$$

Thus

$$\mathcal{D}_{\text{pred}}(P_{\chi, \epsilon}) = \mathcal{D}_{\text{pred}}^A(P_{\chi, \epsilon}) = \{d_p\},$$

and, at every center,

$$\rho_{\text{pred}}^A(P_{\chi, \epsilon}) = \rho_{\text{pred}}(P_{\chi, \epsilon}), \quad \rho_{\text{pred}}^{A, \text{IKM}}(P_{\chi, \epsilon}) = \rho_{\text{pred}}^{\text{IKM}}(P_{\chi, \epsilon}). \quad (3)$$

Hence the exact ratio, the lower bound $(\epsilon^{-2} + 14)/726$, the optimizing scale $\chi_{\text{opt}}(\epsilon) \sim (\sqrt[3]{18}/15)\epsilon^{-2/3}$, and the sharp second-order expansion in Lemma lem:sharp-uniform-binary-elbow hold unchanged for both treatment-aware ratios.

Proof. Because X is constant and $Y = U$ on the center family, R_d^A is the Bayes mean-square error for predicting U from (T, W_x, W_y) . In the observational source, conditional independence given U gives, for $u, t, e, l \in \{-1, 1\}$,

$$L_u(t, e, l) := P(T = t, W_x = e, W_y = l \mid U = u, G = O) = \frac{1}{8} \left(1 + \frac{tu}{4}\right) (1 + xue)(1 + yul). \quad (1)$$

All eight observable cells have positive mass because $x, y \in (0, 1)$. With the Rademacher prior on U , Bayes' rule gives

$$E(U \mid t, e, l, G = O) = \frac{L_1(t, e, l) - L_{-1}(t, e, l)}{L_1(t, e, l) + L_{-1}(t, e, l)}.$$

The contribution of cell (t, e, l) to the Bayes risk is therefore

$$\frac{L_1 + L_{-1}}{2} \left\{ 1 - \left(\frac{L_1 - L_{-1}}{L_1 + L_{-1}} \right)^2 \right\} = \frac{2L_1L_{-1}}{L_1 + L_{-1}}. \quad (2)$$

Summing (2) over the eight triples, pairing opposite sign triples, and placing the resulting four rational terms over their common denominator gives

$$\sum_{t, e, l} \frac{2L_1(t, e, l)L_{-1}(t, e, l)}{L_1(t, e, l) + L_{-1}(t, e, l)} = \frac{15(1-x^2)(1-y^2)(16-x^2-y^2-14x^2y^2)}{\{16(1+xy)^2 - (x+y)^2\}\{16(1-xy)^2 - (x-y)^2\}}. \quad (3)$$

This proves the first display in the statement. The two denominator factors are positive because the summands in (2) have positive likelihood denominators.

For d_e , substituting $x = y = 1/2$ into (3) gives

$$R_{d_e}^A(P_{\chi, \epsilon}) = \frac{15(3/4)^2(117/8)}{24 \cdot 9} = \frac{585}{1024}. \quad (4)$$

For any square-integrable response, enlarging the predictor sigma-field cannot increase optimal squared-error risk, because conditional expectation is the orthogonal projection in L_2 . Since

$$\sigma(X, Z_{M_d}) \subseteq \sigma(A, X, Z_{M_d}),$$

we have $R_d^A(P) \leq R_d(P)$. Applying this to d_p and using the strict raw-risk chain in Lemma `lem:sharp-uniform-binary-elbow` proves the second display. In particular, the treatment-aware risk gap is bounded below uniformly in ϵ by

$$\frac{585}{1024} - \frac{3}{10} = \frac{1389}{5120} > 0. \quad (5)$$

Thus both prediction selectors have the same unique center optimizer d_p . Their best-tie numerators are consequently identical for each of the arm-normalized and IKM-normalized profiles, proving the two ratio identities. Every exact ratio and every optimization over χ then agrees pointwise with the corresponding quantity in Lemma `lem:sharp-uniform-binary-elbow`, so its lower bound, optimizing scale, and sharp expansion carry over unchanged. $\square$

Lemma 9 (`lem:joint-predictive-binary-stability`). *For each fixed $\chi > 1$ and $0 < \epsilon < 10^{-3}$, the radii r_ϵ and r_ϵ^{IKM} in Lemmas `lem:binary-witness-stability` and `lem:binary-ikmw-normalized-stability` may be jointly decreased, without losing any conclusion of those lemmas, so that every law in the corresponding smaller finite-cell ball also satisfies*

$$\mathcal{D}_{\text{pred}}^A(P) = \{d_p\}, \quad \rho_{\text{pred}}^A(P) > \frac{1}{2} \rho_{\text{pred}}^A(P_\epsilon),$$

and, in the IKM-normalized ball,

$$\rho_{\text{pred}}^{A,\text{IKM}}(P) > \frac{1}{2} \rho_{\text{pred}}^{A,\text{IKM}}(P_\epsilon).$$

Thus the radii used to form $\mathcal{U}_{\text{bin}}$ and $\mathcal{U}_{\text{bin}}^{\text{IKM}}$ can preserve both strict prediction-risk orderings simultaneously with the arm-normalized and IKM-normalized optimizer orderings.

Proof. Inside the relative interior of $\mathcal{M}_{\text{bin}}(\chi)$, every conditioning cell of (A, X, Z_{M_d}) has positive probability in a sufficiently small neighborhood of P_ϵ . Since $Y = U \in \{-1, 1\}$, the treatment-aware risk has the finite-cell representation

$$R_d^A(P) = E_P(Y^2 \mid G = O) - \sum_v P\{(A, X, Z_{M_d}) = v \mid G = O\} [E_P\{Y \mid (A, X, Z_{M_d}) = v, G = O\}]^2. \quad (1)$$

Every term in (1) is a continuous rational function of the positive latent-cell vector. Hence $R_d^A(P)$ is continuous for each of the two designs. The source-score and IKM-normalized profiled values are continuous on the regularity neighborhoods already constructed in the two cited stability lemmas.

Lemma `lem:treatment-aware-sharp-binary-elbow` gives the strict treatment-aware center gap

$$R_{d_\epsilon}^A(P_\epsilon) - R_{d_p}^A(P_\epsilon) > \frac{1389}{5120}. \quad (2)$$

It also identifies both treatment-aware center ratios with their positive unadjusted counterparts. By continuity, a sufficiently small ball preserves (2), the unique treatment-aware optimizer, and more than one half of each center ratio. Intersect that ball with the radii already furnished by Lemmas `lem:binary-witness-stability` and `lem:binary-ikmw-normalized-stability`. The intersection contains a positive-radius ball and preserves every earlier regularity, common-identification, unadjusted-risk, and profile-ordering conclusion. This is exactly the asserted joint decrease of the radii. $\square$

Theorem 8 (`thm:treatment-aware-predictive-operator-certificate`). *Fix $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$. For either positive design-value profile*

$$w_d = \mathcal{V}_d^*(P) \quad \text{or} \quad w_d = \mathcal{V}_{d,\text{IKM}}^*(P),$$

the least constant making every treatment-aware prediction-order implication

$$R_d^A(P) \leq R_{d'}^A(P) \implies w_d \leq C w_{d'} \quad (d, d' \in \mathcal{D})$$

valid is

$$\max_{d, d' \in \mathcal{D}: R_d^A(P) \leq R_{d'}^A(P)} \frac{w_d}{w_{d'}}. \quad (1)$$

For the arm-normalized profile, the smallest constant for the actual best-tie selector is $\rho_{\text{pred}}^A(P)$; for the IKM-normalized profile it is $\rho_{\text{pred}}^{A,\text{IKM}}(P)$. Each is at most (1) for its profile and is finite at a fixed regular law. Consequently,

a uniform constant over a nonempty law class exists exactly when the supremum of the corresponding best-tie ratio is finite.

If a numerical realization of $\mathbf{H}_{\text{comp}}$ represents $R_d^A(P)$ by a positive quadratic form $Q_{R,d}^A\{x_d(P)\}$ and represents the arm-normalized value by the profiled source-score form $F_d\{x_d(P)\}$, then the two-sided envelope

$$0 < \Lambda_-^A(P) := \min_d \inf_x \frac{F_d(x)}{Q_{R,d}^A(x)} \leq \max_d \sup_x \frac{F_d(x)}{Q_{R,d}^A(x)} =: \Lambda_+^A(P) < \infty$$

implies

$$\rho_{\text{pred}}^A(P) \leq \max_{d,d': R_d^A(P) \leq R_{d'}^A(P)} \frac{\mathcal{V}_d^*(P)}{\mathcal{V}_{d'}^*(P)} \leq \frac{\Lambda_+^A(P)}{\Lambda_-^A(P)}. \quad (2)$$

As for the unadjusted rule, this envelope is generally conservative and the sharp constant is not generally a largest generalized eigenvalue, because profiling the two source terms produces a nonquadratic numerator and equality additionally requires saturation by law-realized, treatment-aware-risk-ordered directions.

Proof. Fix $P \in \mathfrak{P}_{\mathcal{D}}^{\text{obs}}$. For the arm-normalized profile, positivity and finiteness follow from positive unit costs, Assumption `ass:nondegenerate-components`, and Theorem `thm:budget-profile`. For the IKM-normalized profile, Assumption `ass:source-arm-positivity` and Definition `def:ikmw-normalized-design-value` give

$$0 < (1 - p_{g,1})^2 V_{g,d} < \infty \quad (g \in \{E, O\}, d \in \mathcal{D}),$$

so positive costs again imply $0 < w_d < \infty$ for every d . Thus the following argument applies to either displayed profile.

Define the finite treatment-aware order relation

$$\mathcal{O}_P^A = \{(d, d') \in \mathcal{D}^2 : R_d^A(P) \leq R_{d'}^A(P)\}.$$

It is nonempty because it contains every diagonal pair. If C satisfies all treatment-aware order implications, then division by $w_{d'} > 0$ gives

$$\frac{w_d}{w_{d'}} \leq C \quad \text{for every } (d, d') \in \mathcal{O}_P^A.$$

Consequently C is at least the maximum in (1). Conversely, that finite maximum dominates every displayed ratio, so it satisfies every implication. This proves (1) and its sharpness.

Choose a design minimizing w_d among $\mathcal{D}_{\text{pred}}^A(P)$ and a design minimizing w_d over all of $\mathcal{D}$. The first design minimizes $R_d^A(P)$, so the pair consisting of these two choices belongs to $\mathcal{O}_P^A$. Hence the ratio of their profile values is at most (1). By Definitions `def:predictive-benchmark` and `def:ikmw-normalized-design-value`, that ratio is respectively $\rho_{\text{pred}}^A(P)$ or $\rho_{\text{pred}}^{A,\text{IKM}}(P)$. Division by the strictly positive global minimum also shows that each ratio is the least constant for its actual best-tie selector and belongs to $[1, \infty)$. Taking suprema over a nonempty law class proves that a finite uniform best-tie constant exists if and only if the supremum of the corresponding pointwise ratio is finite. This is the treatment-aware analogue of the finite-order argument in Theorem `thm:sharp-predictive-operator-certificate`.

Now impose the stated treatment-aware quadratic-form realization, with all infima and suprema taken over the cones supplied by $\mathbf{H}_{\text{comp}}$. The envelope definitions give, for every admissible d and direction x ,

$$\Lambda_-^A(P) Q_{R,d}^A(x) \leq F_d(x) \leq \Lambda_+^A(P) Q_{R,d}^A(x). \quad (10)$$

For every $(d, d') \in \mathcal{O}_P^A$, evaluation at the transported law-specific directions yields

$$\begin{aligned} \mathcal{V}_d^*(P) &= F_d\{x_d(P)\} \\ &\leq \Lambda_+^A(P) R_d^A(P) \leq \Lambda_+^A(P) R_{d'}^A(P) \\ &\leq \frac{\Lambda_+^A(P)}{\Lambda_-^A(P)} F_{d'}\{x_{d'}(P)\} \\ &= \frac{\Lambda_+^A(P)}{\Lambda_-^A(P)} \mathcal{V}_{d'}^*(P). \end{aligned} \quad (11)$$

Maximizing (11) over $\mathcal{O}_P^A$, and using the best-tie bound just proved, gives (2).

Finally, Theorem `thm:sharp-predictive-operator-certificate` proves the profile identity

$$F_d(x) = \inf_{0 < \alpha < 1} \left\{ \frac{c_{E,d} Q_{E,d}(x)}{\alpha} + \frac{c_{O,d} Q_{O,d}(x)}{1 - \alpha} \right\}$$

and gives an explicit two-dimensional example in which F_d violates the parallelogram identity. Thus the profiled numerator is not generally quadratic. A largest generalized eigenvalue at a fixed allocation supplies at most a one-sided full-space upper envelope; it does not provide the positive lower envelope in (10). Even when both envelopes exist, equality in (11) additionally requires law-realized treatment-aware-risk-ordered directions to attain the relevant extrema. Therefore the envelope is generally conservative and the sharp treatment-aware selection constant is not generally a largest generalized eigenvalue. $\square$

Justification. This theorem gives the exact order-restricted constant for treatment-aware prediction and the valid two-sided quadratic-form certificate for the arm-normalized profile. *Gap.* A largest generalized eigenvalue is insufficient because design profiling is nonquadratic and selection requires a positive lower envelope plus saturation. *Consumer.* It characterizes precisely what additional law-class comparison would be required to rescue a finite treatment-aware prediction guarantee.

Theorem 9 (thm:two-rule-predictive-impossibility). *Choose the radii defining $\mathcal{U}_{\text{bin}}$ and $\mathcal{U}_{\text{bin}}^{\text{IKM}}$ after the joint shrinking in Lemma lem:joint-predictive-binary-stability. Then every $P \in \mathcal{U}_{\text{bin}}$ is regular, identifies the common contrast τ , and satisfies*

$$\mathcal{D}_{\text{pred}}(P) = \mathcal{D}_{\text{pred}}^A(P) = \{d_p\}, \quad D^*(P) = \{d_e\},$$

$$\sup_{P \in \mathcal{U}_{\text{bin}}} \rho_{\text{pred}}(P) = \sup_{P \in \mathcal{U}_{\text{bin}}} \rho_{\text{pred}}^A(P) = \infty. \quad (1)$$

Every $P \in \mathcal{U}_{\text{bin}}^{\text{IKM}}$ additionally satisfies $D_{\text{IKM}}^*(P) = \{d_e\}$ and

$$\sup_{P \in \mathcal{U}_{\text{bin}}^{\text{IKM}}} \rho_{\text{pred}}^{\text{IKM}}(P) = \sup_{P \in \mathcal{U}_{\text{bin}}^{\text{IKM}}} \rho_{\text{pred}}^{A, \text{IKM}}(P) = \infty. \quad (2)$$

At every center, all four ratios agree, so the exact ratio, prescribed finite cost disadvantage, optimizer scale $(\sqrt[3]{18/15})\epsilon^{-2/3}$, leading constant $1/726$, and second-order constant $5\sqrt[3]{12}/968$ are unchanged. For each fixed $\chi > 1$, the primitive cost vector has a relatively open cost-manifold neighborhood on which d_p remains strictly costlier and is uniquely selected by both population prediction rules for all sufficiently small positive ϵ , while the arm-normalized and balanced-center IKM-normalized ratios diverge. Neither rule is attributed to a named published proxy-selection workflow.

Proof. Theorem thm:predictive-impossibility supplies regularity, common identification, the unadjusted selector, both profile optimizers, the exact center ratio, both original-region divergences, the sharp cost optimization, and the relatively open cost-neighborhood conclusion. Lemma lem:treatment-aware-sharp-binary-elbow proves at every center that the treatment-aware selector is the same singleton $\{d_p\}$ and that its arm-normalized and IKM-normalized ratios equal the two original ratios. Lemma lem:joint-predictive-binary-stability permits the defining radii to be chosen so that both strict prediction orderings and both profile orderings persist simultaneously throughout the displayed regions.

Every center remains in its own open ball. Therefore the center identity and the divergence from Lemma lem:sharp-uniform-binary-elbow imply

$$\sup_{P \in \mathcal{U}_{\text{bin}}} \rho_{\text{pred}}(P) = \sup_{P \in \mathcal{U}_{\text{bin}}} \rho_{\text{pred}}^A(P) = \infty$$

and

$$\sup_{P \in \mathcal{U}_{\text{bin}}^{\text{IKM}}} \rho_{\text{pred}}^{\text{IKM}}(P) = \sup_{P \in \mathcal{U}_{\text{bin}}^{\text{IKM}}} \rho_{\text{pred}}^{A, \text{IKM}}(P) = \infty$$

by letting $\epsilon \downarrow 0$ at fixed χ . The pointwise equality of all four center ratios transports the exact formula and both sharp-elbow constants without modification.

For the cost-neighborhood clause, at $\epsilon = 0$ the continuously extended treatment-aware risk of d_p is no larger than its unadjusted risk $3/(10\chi) < 3/10$, while the treatment-aware risk of d_e is $585/1024$. Thus the treatment-aware gap is strict at the center cost vector. Finite-cell conditional risks and primitive costs vary continuously. Intersecting the cost neighborhood from the original impossibility theorem with a neighborhood preserving this additional strict gap proves simultaneous unique selection by both prediction rules. The source-score divergence and balanced-center one-quarter IKM scaling are unchanged because neither depends on which of the two coincident center selectors is used. This proves every assertion. $\square$

Theorem 10 (thm:two-rule-finite-catalogue-predictive-guarantee). *For every $P \in \mathfrak{P}_D^{\text{obs}}$, both the unadjusted and treatment-aware population squared-risk rules have exact finite-catalogue constants. In addition to the identity in Theorem thm:finite-catalogue-predictive-guarantee,*

$$\rho_{\text{pred}}^A(P) = \frac{\min_{d \in \mathcal{D}_{\text{pred}}^A(P)} (\sqrt{c_{E,d} V_{E,d}} + \sqrt{c_{O,d} V_{O,d}})^2}{\min_{d' \in \mathcal{D}} (\sqrt{c_{E,d'} V_{E,d'}} + \sqrt{c_{O,d'} V_{O,d'}})^2} \in [1, \infty), \quad (1)$$

and

$$\rho_{\text{pred}}^{A, \text{IKM}}(P) = \frac{\min_{d \in \mathcal{D}_{\text{pred}}^A(P)} \mathcal{V}_{d, \text{IKM}}^*(P)}{\min_{d' \in \mathcal{D}} \mathcal{V}_{d', \text{IKM}}^*(P)} \in [1, \infty). \quad (2)$$

Each ratio is the smallest constant for its actual best-tie selector. For either positive profile w_d , the least constant for every treatment-aware pairwise implication is

$$\max_{d, d' \in \mathcal{D}: R_d^A(P) \leq R_{d'}^A(P)} \frac{w_d}{w_{d'}}. \quad (3)$$

These are exact lawwise enumerations, not primitive uniform guarantees. For every fixed $\chi > 1$, the jointly shrunk binary regions have unbounded best-tie ratios for both prediction rules and both profiles in their respective regions. Across the center family, all four ratios agree and obey the same sharp expansion

$$\epsilon^2 \inf_{\chi > 1} \rho(P_{\chi, \epsilon}) = \frac{1}{726} + \frac{5\sqrt[3]{12}}{968} \epsilon^{2/3} + o(\epsilon^{2/3}), \quad (4)$$

where ρ may be any of the four center ratios. Thus treatment adjustment in the prediction objective does not produce a finite uniform source-score or IKM-normalized guarantee.

Proof. Theorem thm:budget-profile and Assumption ass:nondegenerate-components imply that every arm-normalized profiled value is finite and strictly positive. Source-arm positivity gives $0 < 1 - p_{g,1} < 1$, so every IKM-normalized source component and profiled value is likewise finite and strictly positive. Since $\mathcal{D}$ is finite and nonempty, both treatment-aware selector sets are nonempty. Substituting the profiled arm-normalized formula into the definition of $\rho_{\text{pred}}^A(P)$ proves (1), and the IKM definition gives (2). In each case, the selected index set is a subset of $\mathcal{D}$, so the ratio lies in $[1, \infty)$, and division by the positive global minimum proves that it is the smallest best-tie constant.

For either positive profile w_d , let

$$C_A(P) = \max_{d, d' \in \mathcal{D}: R_d^A(P) \leq R_{d'}^A(P)} \frac{w_d}{w_{d'}}. \quad (1)$$

The index set is finite and contains every diagonal pair. Any valid pairwise constant must dominate every ratio indexed by this set, while $C_A(P)$ plainly makes every implication valid. This proves exactness of the third display.

Theorem thm:two-rule-predictive-impossibility proves the two-rule regional divergences and the equality of all four ratios at every center. Lemma lem:sharp-uniform-binary-elbow supplies the expansion for the two original ratios, and Lemma lem:treatment-aware-sharp-binary-elbow identifies the other two with them pointwise. Hence

$$\epsilon^2 \inf_{\chi > 1} \rho(P_{\chi, \epsilonpsilon}) = \frac{1}{726} + \frac{5\sqrt[3]{12}}{968} \epsilon^{2/3} + o(\epsilon^{2/3})$$

for every one of the four center ratios. A finite decision set therefore gives only an evaluable lawwise enumeration; it cannot turn either prediction ordering into a uniform efficiency approximation. $\square$

Theorem 11 (thm:empirical-treatment-aware-predictive-selector). *Fix $\chi > 1$, let $\epsilon_m \in (0, 10^{-3})$ satisfy $\epsilon_m \downarrow 0$, and suppose the independently funded full-catalogue pilot consists of independent and identically distributed draws from P_{χ, ϵ_m} , independently of the main study. Apply the split-sample saturated finite-cell rule in Definition def:treatment-aware-pilot-selector. Then*

$$\max_{d \in \mathcal{D}} |\hat{R}_{d,m}^A - R_d^A(P_{\chi, \epsilon_m})| = O_P(m^{-1/2}), \quad (1)$$

$$P_{\chi, \epsilon_m} \{\hat{d}_m^A = d_p\} \longrightarrow 1. \quad (2)$$

Moreover, the achieved arm-normalized and IKM-normalized profile ratios diverge in probability:

$$\frac{\mathcal{V}_{\hat{d}_m^A}^*(P_{\chi, \epsilon_m})}{\min_{d \in \mathcal{D}} \mathcal{V}_d^*(P_{\chi, \epsilon_m})} \longrightarrow_P \infty, \quad \frac{\mathcal{V}_{\hat{d}_m^{A, \text{IKM}}}^*(P_{\chi, \epsilon_m})}{\min_{d \in \mathcal{D}} \mathcal{V}_{d, \text{IKM}}^*(P_{\chi, \epsilon_m})} \longrightarrow_P \infty. \quad (3)$$

Thus the treatment-aware population impossibility is attained by a feasible pilot selector using only bounded binary outcomes and saturated finite-cell regression; no inverse-problem learner or growing sieve is used.

Proof. Fix $\chi > 1$. Along $0 \leq \epsilon \leq 10^{-3}$, the source probability is $1/2$, the observational treatment kernel is bounded below by $3/8$, and every proxy kernel appearing in either design is bounded away from zero by a positive constant depending only on χ . The two designs have finitely many cells. Hence there is $c_\chi > 0$ such that every observational cell of (A, X, Z_{M_d}) , for both d , has probability at least c_χ , uniformly along the triangular sequence.

The training and validation observational counts are both proportional to m in probability. Binomial concentration followed by a finite union bound therefore shows that every training cell count is proportional to m with probability tending to one. Since $|Y| = 1$, each saturated training-cell mean satisfies, uniformly over the finitely many cells and designs,

$$\hat{f}_{d,m}^A(v) - E(Y \mid (A, X, Z_{M_d}) = v, G = O) = O_P(m^{-1/2}). \quad (1)$$

Condition on the training half. The held-out squared losses are independent, bounded by four, and their number is proportional to m . Their empirical mean differs from its conditional expectation by $O_P(m^{-1/2})$, uniformly over the two designs. If $f_d^A = E(Y \mid A, X, Z_{M_d}, G = O)$, the conditional-expectation projection identity gives

$$E\{(Y - \hat{f}_{d,m}^A)^2 \mid G = O, \mathcal{I}_{m,O}^{\text{tr}}\} - R_d^A(P_{\chi, \epsilonpsilon_m}) = E\{(\hat{f}_{d,m}^A - f_d^A)^2 \mid G = O, \mathcal{I}_{m,O}^{\text{tr}}\} = O_P(m^{-1}). \quad (2)$$

Combining the held-out fluctuation with (2) and taking the finite maximum proves

$$\max_{d \in \mathcal{D}} |\hat{R}_{d,m}^A - R_d^A(P_{\chi, \epsilonpsilon_m})| = O_P(m^{-1/2}). \quad (3)$$

The zero-denominator conventions are irrelevant with probability tending to one by the cell-count bound.

Lemma lem:treatment-aware-sharp-binary-elbow gives uniformly along the triangular sequence

$$R_{d_e}^A(P_{\chi, \epsilonpsilon_m}) - R_{d_p}^A(P_{\chi, \epsilonpsilon_m}) > \frac{1389}{5120}. \quad (4)$$

On the event that the maximum error in (3) is less than half the right-hand side of (4), the empirical minimizer is uniquely d_p . The probability of this event tends to one, proving

$$P_{\chi, \epsilonpsilon_m} \{\hat{d}_m^A = d_p\} \longrightarrow 1. \quad (5)$$

On $\{\hat{d}_m^A = d_p\}$, each achieved ratio in the theorem equals its corresponding center population ratio. Lemma lem:treatment-aware-sharp-binary-elbow identifies these ratios with the original ones, and Lemma lem:sharp-uniform-binary-elbow gives

$$\rho_{\text{pred}}^A(P_{\chi, \epsilonpsilon_m}) = \rho_{\text{pred}}^{A, \text{IKM}}(P_{\chi, \epsilonpsilon_m}) > \frac{\epsilon_m^{-2} + 14}{726} \longrightarrow \infty. \quad (6)$$

Since the event on which (6) is the achieved ratio has probability tending to one, both achieved ratios diverge in probability. Independence of the pilot and main study, Assumption `ass:independent-pilot`, ensures that this pilot selection is a feasible *ex ante* measurement rule and does not use main-study outcomes. This proves the theorem. $\square$

Justification. This theorem turns the treatment-aware population counterexample into a feasible independent-pilot procedure using a split-sample saturated finite-cell regression. *Gap.* The population failure is not caused by an infeasible oracle criterion: the empirical selector consistently makes the same inefficient choice. *Consumer.* It provides an implementable diagnostic counterexample and clarifies why the pilot should estimate design values rather than prediction risks.

7 Interpretation

Predictive sufficiency and bridge efficiency are different geometric objects. Treatment adjustment improves or preserves prediction risk because it enlarges the projection space, but it does not control the inverse sensitivity of the early proxy inside the selection bridge. In the witness, the late proxy makes d_p highly predictive, even after conditioning on treatment, while the early signal ϵ drives its observational score variance at order ϵ^{-2} . The exact design-value gap therefore answers the measurement question that prediction risk cannot. The IKM-normalized profile is interpreted as an absolute causal efficiency bound only under the separately stated fixed-system canonical-gradient premises.

Technical internal limitation (diagnostic only)

The empirical treatment-aware selector result is deliberately finite-cell: it uses bounded binary outcomes, uniformly positive cells along a fixed- χ triangular path, and a split-sample saturated regression. The general post-selection Wald theorem remains conditional on the declared selected-path bridge-learning, multiplier, and variance conditions outside the separate fixed-dimensional attainability subclass. No uniform claim is made for arbitrary predictive algorithms, continuous high-dimensional catalogues, or laws approaching empty treatment-proxy cells.

8 Honest scope

The positive result is exact design-value selection and inference for the observed bridge contrast on the observable regular class, with causal ATE coverage only on the common-identification subclass. The negative result concerns two explicit population squared-risk rules, $R_d(P)$ and $R_d^A(P)$, plus a feasible saturated finite-cell implementation of the latter. It does not claim that every predictive method fails, does not attribute either rule to a named published workflow, and does not elevate the arm-normalized profile to an unconditional causal efficiency bound. The relatively open witness regions are open only within the constrained finite-cell model.

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